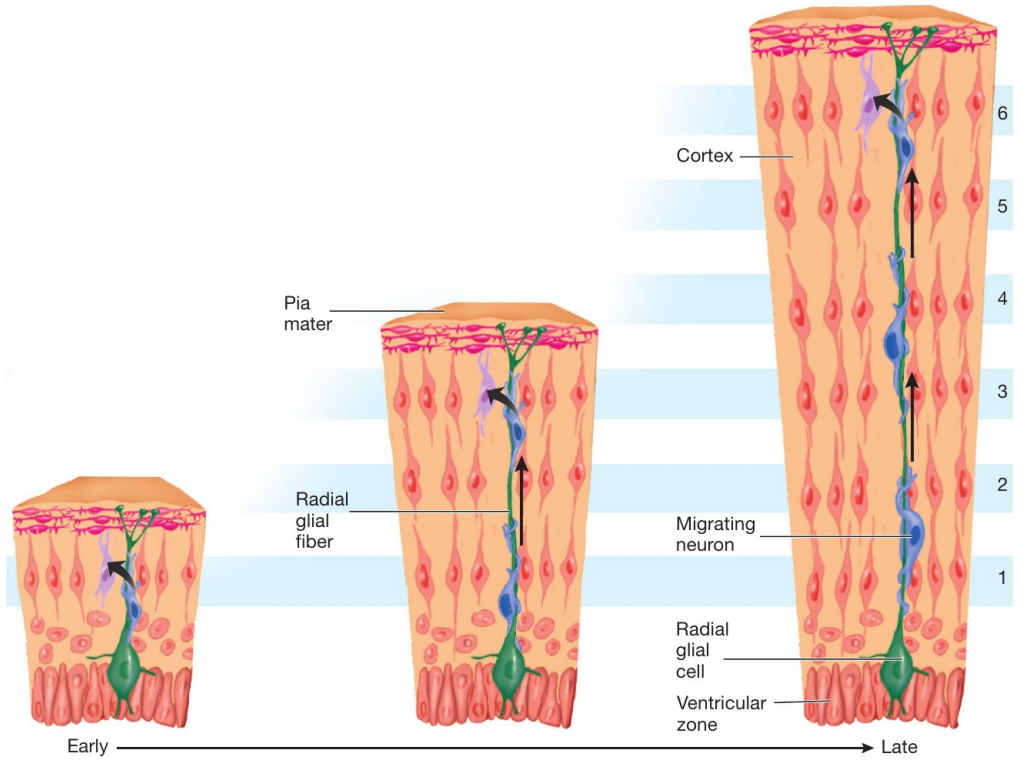


Carlson and Birkett (2022), Fig. 15.3, p. 495.

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From Neural Progenitors to Developing Circuits and Behavior

Building the Brain: Development, Experience, and Neurodiversity

Fall 2026

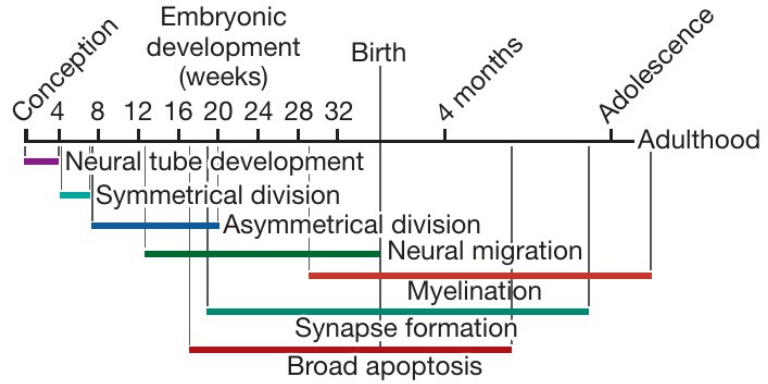
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Development builds cells pathways and circuits

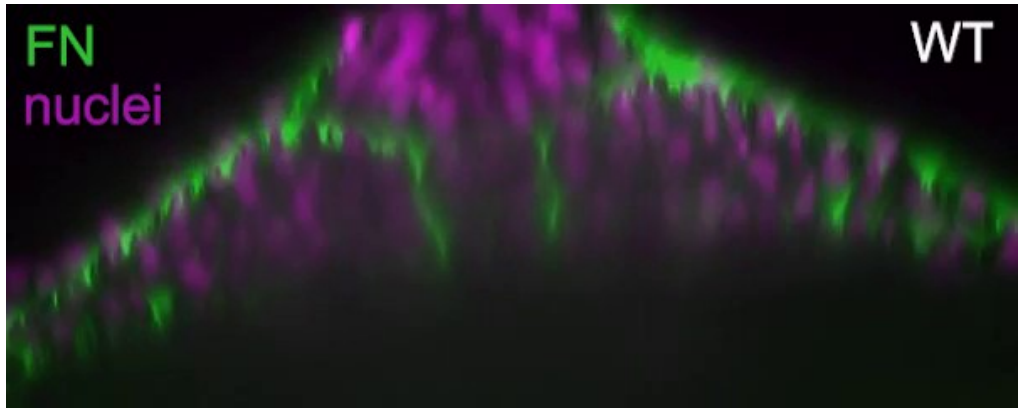
The nervous system develops through a coordinated sequence of cell production, movement, differentiation, and connection. Differentiation means that a cell acquires a more specialized identity and set of functions. A developing neuron must reach an appropriate location, extend processes toward other cells, establish synapses, and become incorporated into circuits that can process information and control behavior.

These processes overlap rather than forming a single assembly line that finishes before birth. Neuron production, migration, synapse formation, myelination, and elimination occur on different schedules across regions. Some pathways begin operating while others are still being constructed. The developing brain therefore changes both its cellular composition and the patterns of activity through which its cells influence one another.

Genes provide proteins and regulatory instructions, while local chemical signals, mechanical forces, and experience influence how those instructions are expressed. A change at one stage can alter the conditions encountered at later stages. Developmental disorders can consequently arise through different biological routes, and similar behavioral difficulties do not necessarily identify the same underlying cellular change.



Carlson and Birkett (2022), Fig. 15.5, p. 497.



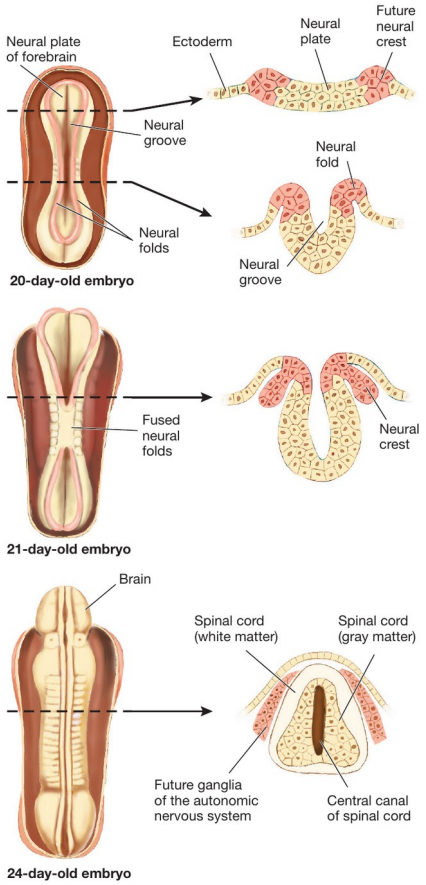
Guillon et al. (2020), Video 1.

Neurulation creates the neural tube

Early in embryonic development, part of the outer cell layer becomes a thickened neural plate. The plate bends and its edges approach one another, forming neural folds. Fusion of those folds creates the neural tube, the embryonic structure from which the brain and spinal cord develop. Its central cavity persists as the ventricular system and spinal central canal.

Cells must change shape, maintain adhesion, and coordinate forces with neighboring tissues for closure to proceed. The tissue is not simply rolled up by an external hand. Molecular signals regulate cell behavior, while the surrounding extracellular matrix transmits mechanical influences. Different positions along the body close through a spatially organized process rather than all sealing at exactly the same instant.

Neural crest cells arise near the border of the neural plate and contribute to many structures, including peripheral sensory and autonomic ganglia. Their departure separates a migratory cell population from the central neural tube. Defects in closure can affect brain or spinal development, and the early timing explains why conditions present before pregnancy is recognized can already influence this stage.



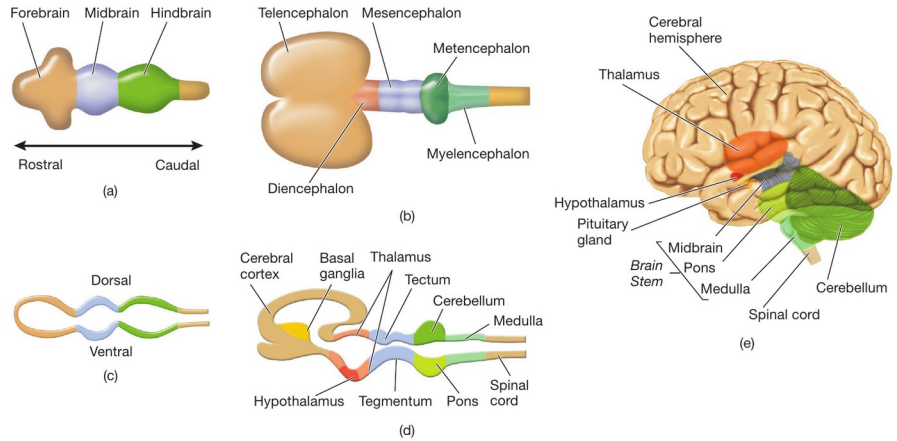
Carlson and Birkett (2022), Fig. 15.1, p. 493.

Brain vesicles preserve a central cavity

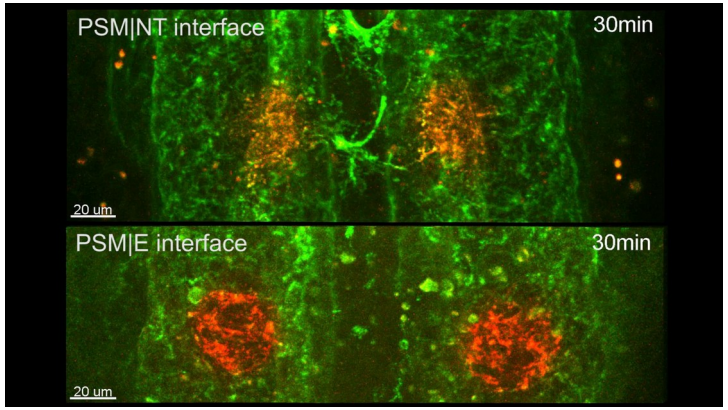
The rostral end of the neural tube expands into early brain vesicles that give rise to the forebrain, midbrain, and hindbrain. Their walls produce neural tissue, while their cavities become connected fluid spaces. Following the changing cavity helps explain the relationship between adult brain structures and the ventricles that remain inside them.

The forebrain later includes the telencephalon and diencephalon. The telencephalon forms the cerebral hemispheres and related structures, while the diencephalon includes the thalamus and hypothalamus. The midbrain surrounds the narrow cerebral aqueduct. Hindbrain territories form structures including the pons, cerebellum, and medulla as growth and folding alter the original simple tubular arrangement.

Regional identity depends on developmental signals that regulate which genes cells express. Cells in different positions therefore do not all follow an identical program, even though they originate from a continuous tube. Subsequent proliferation and migration expand those regional populations, producing the distinct tissue architecture and pathways that later support sensory, motor, and regulatory functions.



Carlson and Birkett (2022), Fig. 15.2, p. 494.



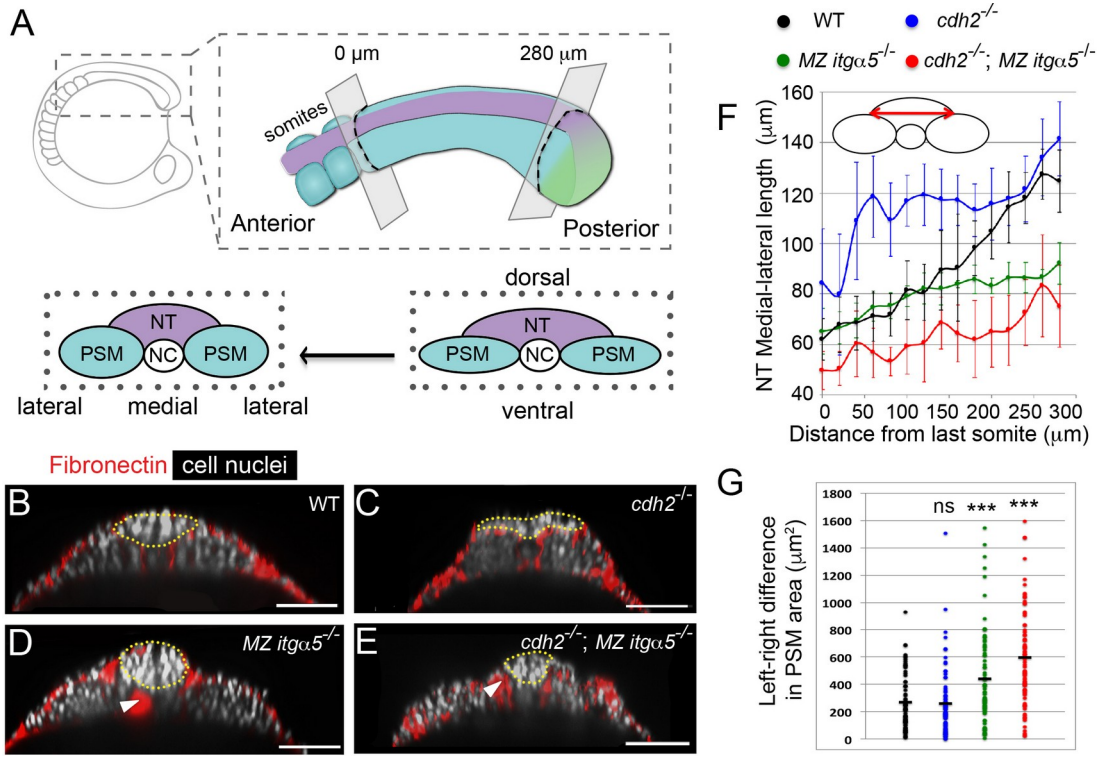
Guillon et al. (2020), Video 2.

Fibronectin couples neighboring tissues

A zebrafish study examined fibronectin, an extracellular protein that forms a matrix between tissues. The neural tube develops alongside columns of paraxial mesoderm, tissue that contributes to structures of the trunk. Fibronectin helps these neighboring tissues adhere, allowing forces generated during changes in one tissue to influence the shape and movement of the other.

The investigators imaged the interfaces and found that fibronectin was distributed in a mechanically organized pattern rather than as a uniform coating. It accumulated toward lateral edges where shearing forces were greater. Shear occurs when neighboring surfaces tend to slide relative to one another, so the location of the matrix affected how the tissues resisted that movement.

The original microscopy reveals a physical aspect of neurulation that a gene list alone cannot explain. Adhesion transmits force across the boundary between tissues. Changing that adhesion can alter convergence of the neural tissue even when the affected protein is located outside the neural cells rather than inside a developing neuron's electrical signaling machinery.



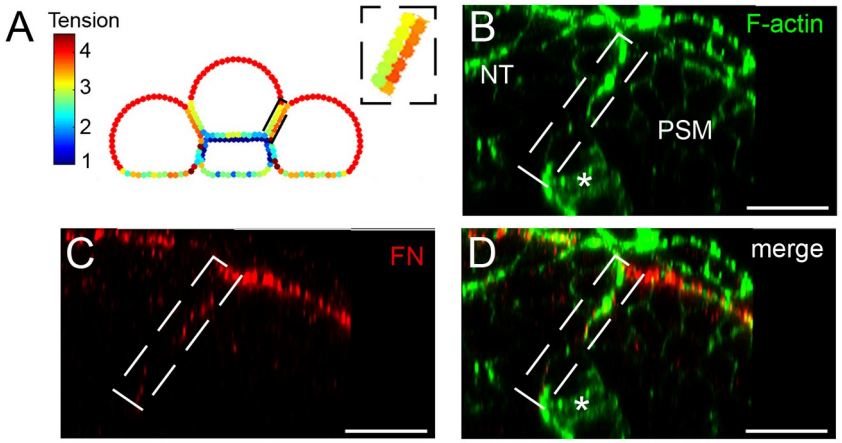
Guillon et al. (2020), Fig. 1.

Adhesion changes neural tube convergence

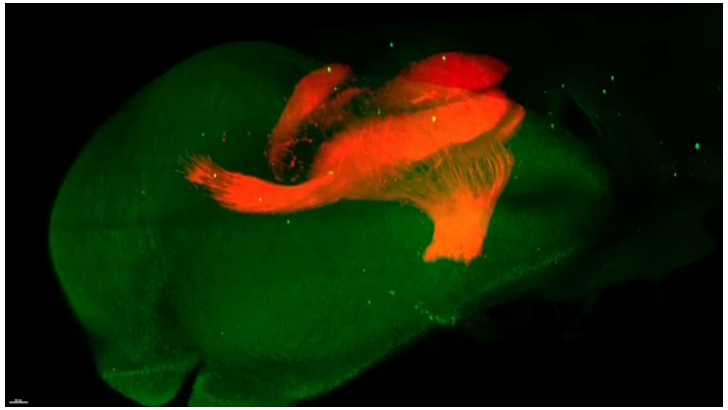
The fibronectin study tested what happened when matrix mediated adhesion was disturbed. Fibronectin was required for normal bilateral organization of neighboring mesoderm, showing that the two sides of the developing trunk were mechanically coordinated. The neural tube and adjacent tissues therefore could not be treated as completely independent structures undergoing unrelated shape changes.

In embryos with altered cadherin dependent adhesion, perturbing fibronectin unexpectedly improved a neural tube convergence defect. Cadherins are cell adhesion proteins that help cells attach to one another. Reducing one adhesive interaction could relieve a problem produced by another defect, because the final tissue shape depended on the combined forces and attachments across the developing region.

This interaction does not mean that less adhesion is generally beneficial. The same extracellular matrix supports normal morphogenesis in one condition and contributes to a convergence problem in another. The experiment identifies a specific mechanical relationship: the consequences of changing fibronectin depend on the state of the cell adhesion system with which it operates.



Guillon et al. (2020), Fig. 3.



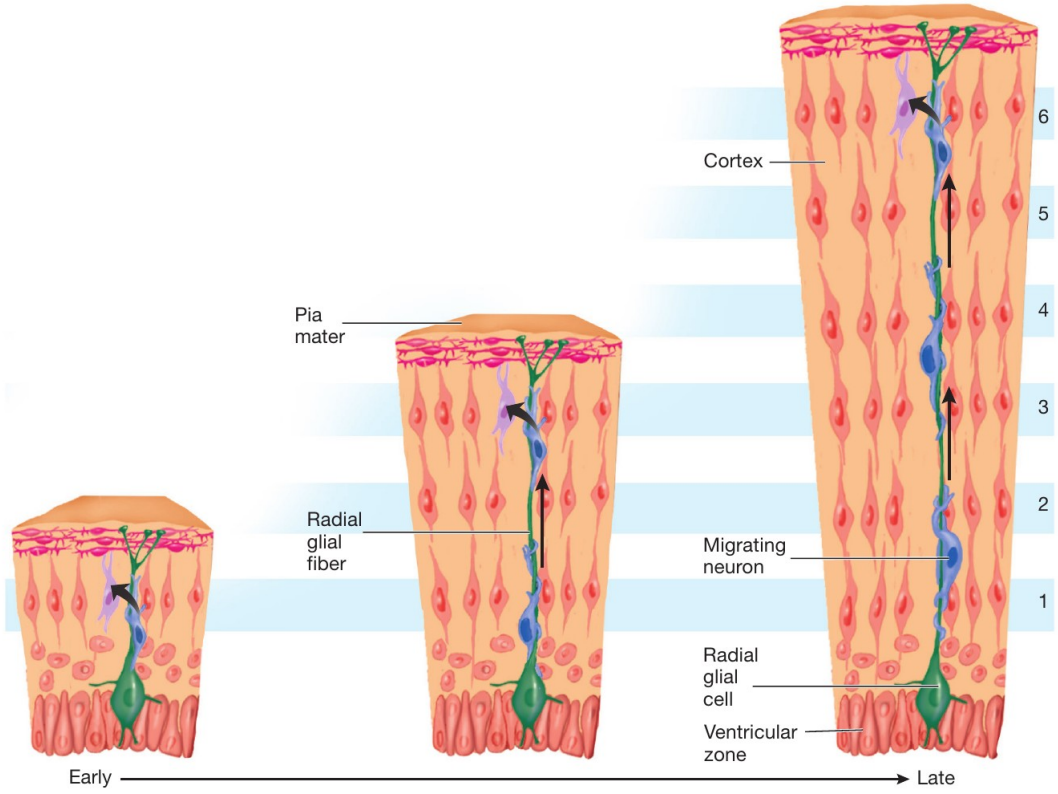
Pedraza et al. (2025), fig4-Video 1.

Progenitors balance renewal and differentiation

A neural progenitor is a dividing cell that can produce descendants destined for neural tissue. Early divisions expand the progenitor population, while later divisions also generate neurons and glia. Self renewal preserves a source of additional cells, whereas differentiation directs descendants toward specialized functions. The balance between these outcomes determines how many cells remain available for subsequent development.

A symmetric division produces daughters with similar developmental potential, while an asymmetric division produces daughters with different outcomes. These terms describe cell fate and are not determined only by whether a microscope image looks geometrically even. The orientation of division, inherited cellular components, local signals, and the daughters' later behavior can all contribute to their developmental trajectories.

Progenitors near the ventricles occupy organized epithelial regions with connections to the tissue surface. As the cortex expands, additional progenitor populations appear farther from the ventricle. Altering the timing of division or the ability of daughter cells to separate can change the size and organization of these populations before mature synaptic communication has even begun.



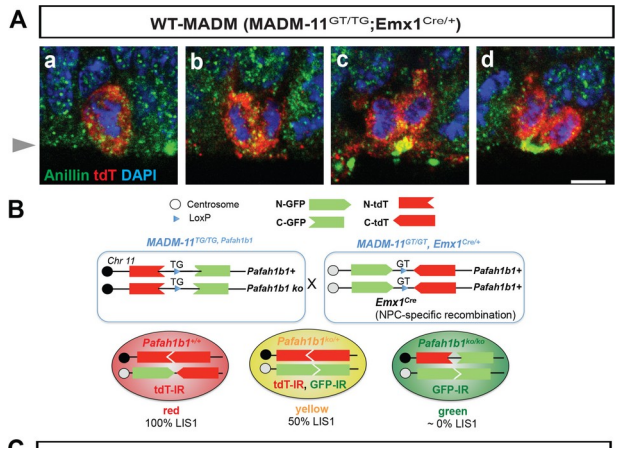
Carlson and Birkett (2022), Fig. 15.3, p. 495.

LIS1 regulates progenitor cell division

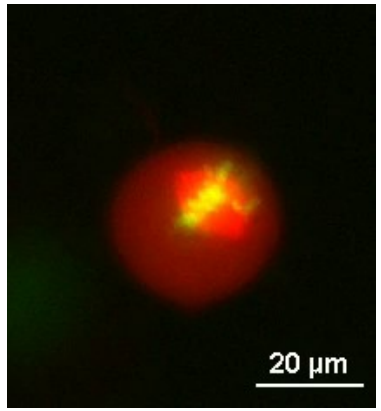
A study examined LIS1, a protein encoded by PAFAH1B1, whose reduced function is associated with disrupted cortical development. The investigators studied dividing neural progenitors in mice and complementary cultured mouse embryonic fibroblasts. They asked how LIS1 influences the physical separation of daughter cells, extending its known roles in microtubules, motor proteins, and nuclear movement.

Before a cell divides, chromosomes separate and the cell membrane constricts between the future daughters. This constriction depends on actin filaments and myosin, a motor protein that uses ATP to generate force. The location and timing of the constriction must be coordinated with the chromosome separating machinery so that each daughter receives an appropriate cellular complement.

LIS1 deficient progenitors showed displacement of the cleavage plane and abnormal localization of proteins associated with the division furrow. Related abnormalities appeared in the fibroblast preparation used for live imaging. The shared result implicated a cellular mechanical process, while the neural progenitor measurements established its relevance to the developing cortex rather than only to cultured nonneural cells.



Moon et al. (2020), Fig. 1.



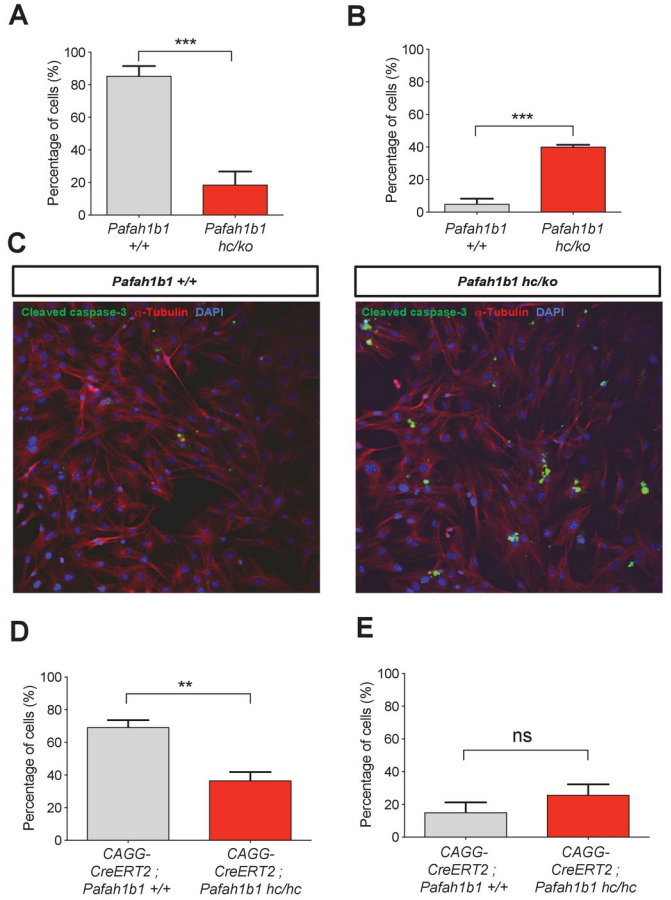
Moon et al. (2020), Video 1.

Excess contraction disrupts cell separation

The LIS1 study found abnormal actomyosin organization and excessive membrane contractility in deficient cells. Actomyosin is the interacting system of actin filaments and myosin motors that produces cellular force. When that force is misplaced or poorly regulated, a cell can constrict incorrectly rather than dividing into two normally separated daughters.

Live recordings in mouse embryonic fibroblasts showed failures of cytokinesis, the final physical separation of the daughters. Some cells became binucleated because chromosome separation was not followed by successful division of the cell body. The source movies identify this preparation explicitly; they are not recordings of mature neurons splitting during normal brain activity.

The experiments connect LIS1 to coordination between microtubule related processes and the contractile membrane apparatus. Cortical development depends on producing the correct descendants at the correct times, so a mechanical division defect can influence later neural organization. The finding specifies a route from a protein change to abnormal progenitor behavior before considering its eventual effects on brain function.



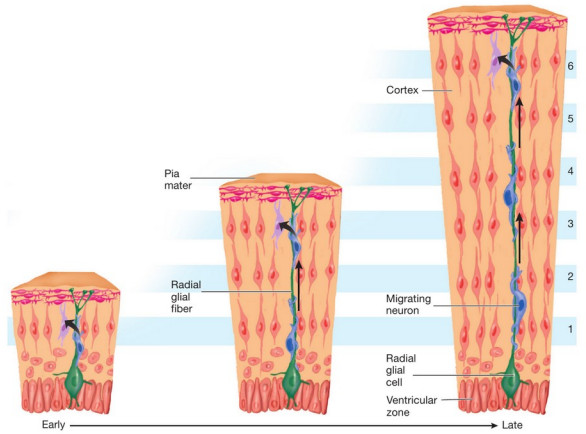
Moon et al. (2020), Fig. 4.

Radial glia generate cells and guide migration

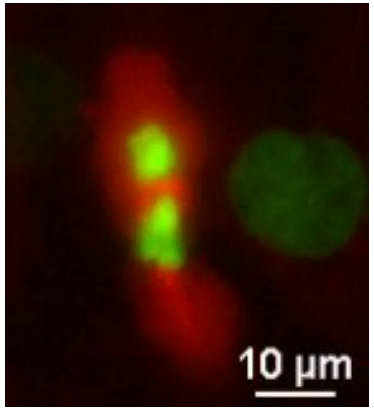
Radial glia are elongated developmental cells with processes extending across the thickness of the forming cortex. They can generate neural descendants and also provide a scaffold along which many newborn neurons move. Their name describes this radial orientation, from regions near the ventricle toward the outer surface rather than along the adult horizontal cortical layers.

Apical radial glia retain a connection near the ventricular surface, while outer radial glia occupy more superficial progenitor zones and have a different arrangement of processes. Outer radial glia are especially prominent in the expanded developing primate cortex. The shape and attachments of these cells influence both their own division behavior and the paths encountered by migrating neurons.

The scaffold is dynamic because the tissue thickens while cells continue dividing and moving. Maintaining a long process requires organization of the cytoskeleton, the internal framework that supports cellular shape. A signaling change that alters that framework can therefore affect migration routes without necessarily changing every progenitor's identity or the total rate at which cells divide.



Carlson and Birkett (2022), Fig. 15.3, p. 495.



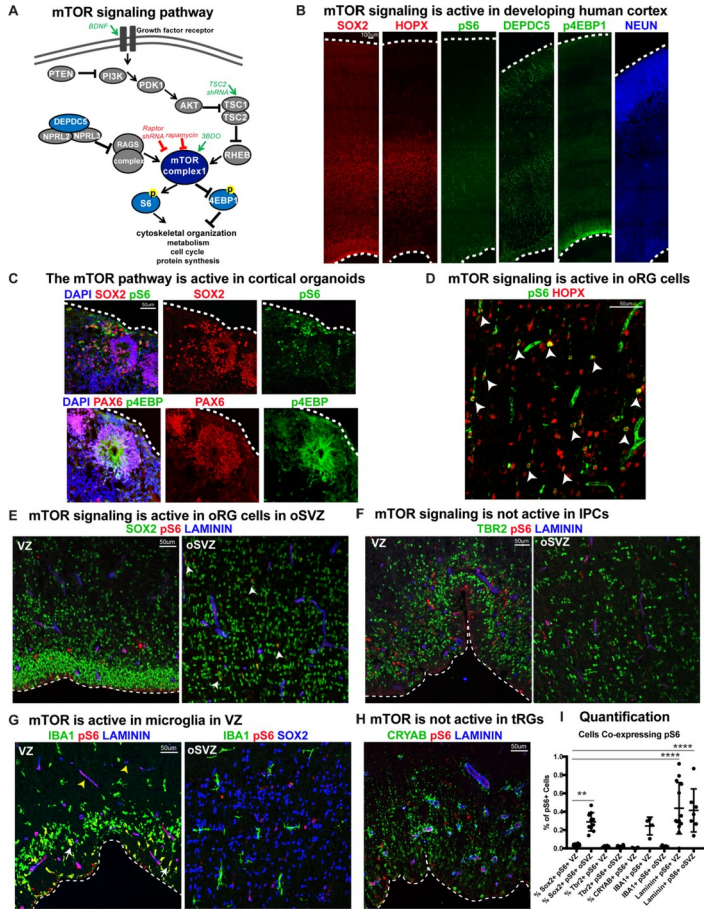
Moon et al. (2020), Video 2.

mTOR regulates outer radial glial shape

A study of developing human cortex examined mTOR signaling in outer radial glia. mTOR is a protein kinase, an enzyme that adds phosphate groups to other proteins and helps regulate cellular growth related processes. The investigators asked whether changing this pathway altered the behavior of particular progenitor types rather than assuming that every developing cell responded in the same way.

Disrupting mTOR signaling changed the morphology, migration, and division associated movements of outer radial glia. Other tested progenitor types were less affected in these respects. The findings identified a selective influence on the cells that help construct and organize the expanding radial scaffold, rather than a uniform suppression of all neural tissue development.

The study related these changes to the actin cytoskeleton and CDC42, a molecular switch that regulates cellular organization. Actin supports the cell's processes and movement. Altered regulation can therefore shorten or destabilize the structures through which a radial glial cell maintains its shape, changing the physical architecture available to neighboring developing neurons.



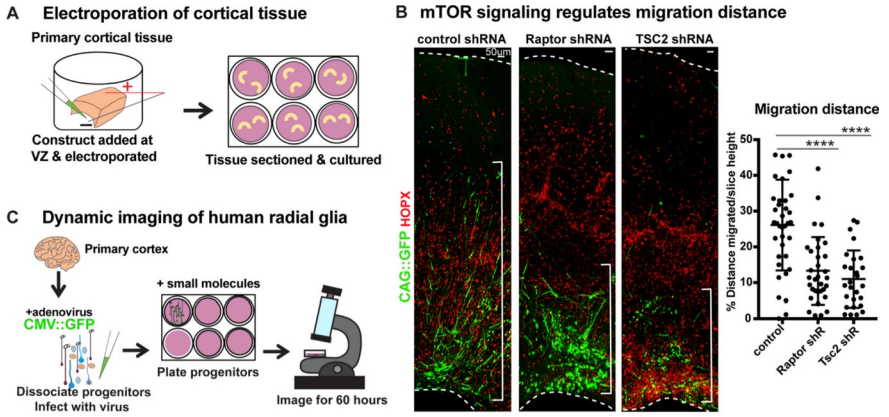
Andrews et al. (2020), Fig. 1.

Scaffold changes can spare cell identity

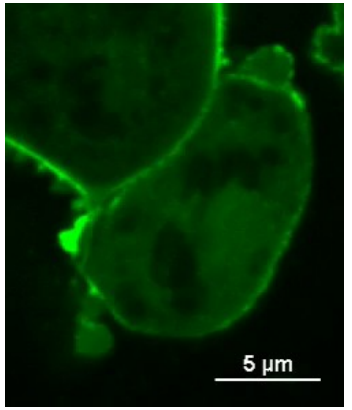
The outer radial glia experiments separated cell shape and migration from proliferation and identity. Under the tested manipulations, mTOR dysregulation altered morphology and movement without a corresponding change in proliferation or cell fate. A progenitor could therefore retain its broad identity while behaving differently within the developing tissue and supplying a less typical structural scaffold.

This distinction explains why counting cells alone can miss a developmental defect. Two samples can contain similar numbers of a cell type but differ in process length, attachment, orientation, or movement. Those properties determine how the cells occupy space and how other cells interact with them, influencing the organization of the forming cortex.

The study connected pathway activity to a concrete cellular sequence: altered regulation of the actin system changed outer radial glial architecture and migration behavior. It did not establish that every disorder involving mTOR has the same cellular phenotype. The consequences depend on which population is affected, the developmental stage, and the direction of the signaling disturbance.



Andrews et al. (2020), Fig. 3.



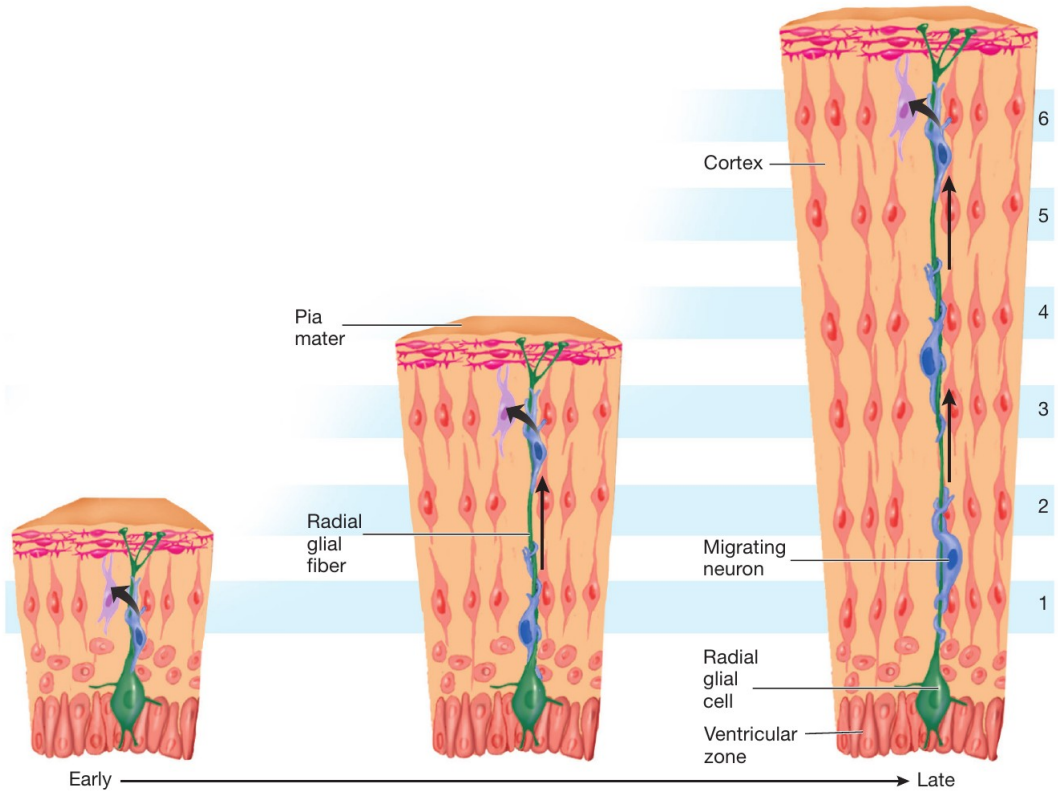
Moon et al. (2020), Video 3.

Cortical neurons migrate into organized layers

Many excitatory cortical neurons are generated in progenitor zones near the ventricle and migrate outward. Earlier generated neurons generally settle in deeper layers, while later generated neurons pass them to reach more superficial positions. This inside out organization is a developmental pattern of the cortex, not a statement that every neural cell follows the same route or schedule.

Migration requires the cell to extend a leading process, move its nucleus and cell body, and release or adjust attachments behind it. Cytoskeletal proteins and motor proteins generate the required forces. Chemical signals and interactions with radial glia help organize movement, while the neuron's developing identity influences when it stops and which connections it later forms.

Inhibitory interneurons often originate in ventral forebrain regions and first migrate tangentially into the cortex before settling within it. Tangential movement runs across the radial scaffold rather than simply outward from the nearest ventricular zone. Different developmental routes therefore bring excitatory and inhibitory populations together, making circuit assembly sensitive to disruption at more than one anatomical origin.



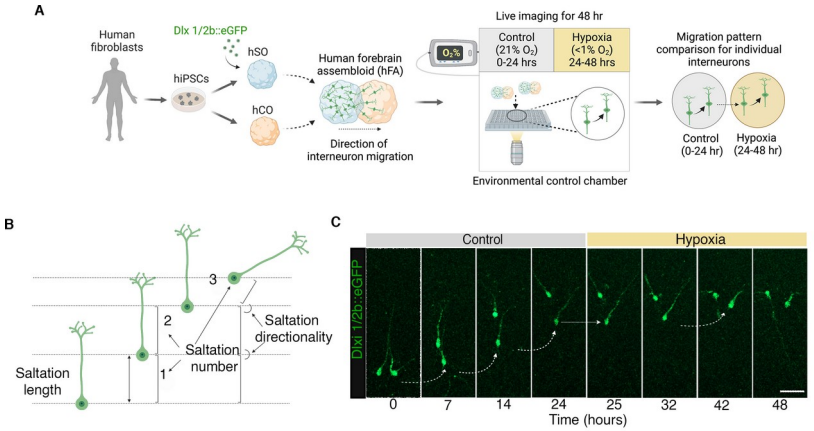
Carlson and Birkett (2022), Fig. 15.3, p. 495.

Hypoxia slows human interneuron migration

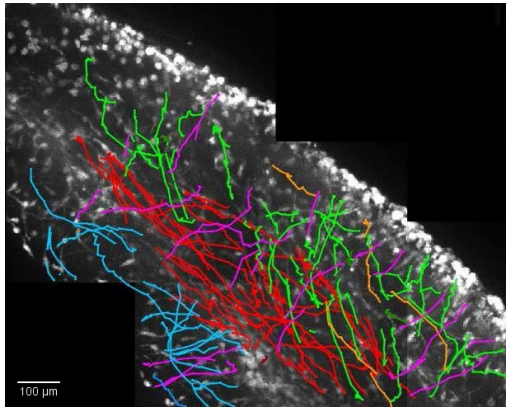
A study examined how hypoxia, a reduction in available oxygen, affected human cortical interneuron migration. The researchers used forebrain assembloids made from induced pluripotent stem cells and developing human tissue studied outside the body. Assembloids combine related tissue models so that cells can migrate between regions resembling their developmental origin and destination.

Multiday live imaging showed that interneurons migrated less effectively under hypoxic conditions. The researchers measured movement directly rather than inferring migration from the final number of cells in a region. This distinction matters because fewer cells at a destination could also result from altered production, cell death, or a change in the identity markers used to recognize them.

Gene expression analysis identified a strong increase in adrenomedullin, a signaling peptide. A peptide is a short chain of amino acids that can act on receptors to change cellular activity. The investigators then tested whether adding this molecule could alter the migration response, turning an observed molecular change into an experimentally examined candidate protective mechanism.



Puno et al. (2026), Fig. 1.



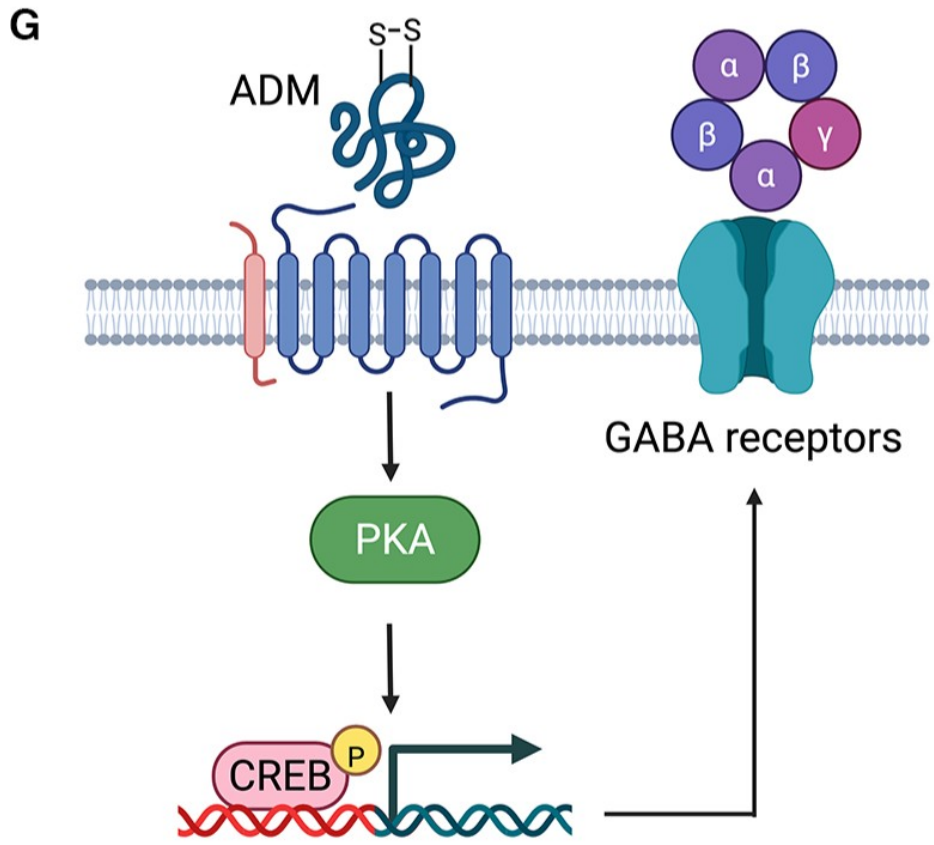
Pedraza et al. (2025), fig6-Video 1.

Adrenomedullin restores migration signaling

Adding adrenomedullin to the hypoxic culture conditions restored interneuron migration in the tested human preparations. The experiment therefore compared movement under normal oxygen, reduced oxygen, and reduced oxygen with an added signaling molecule. The result concerned a developmental cellular behavior, not a completed clinical treatment for premature infants with brain injury.

The study identified participation of the cyclic AMP and protein kinase A pathway. Cyclic AMP is an intracellular messenger, and protein kinase A changes the activity of target proteins by phosphorylation. This signaling increased phosphorylation of CREB, a protein that regulates gene expression, and helped restore expression of a subset of GABA receptors associated with promoting migration.

GABA has developmental effects beyond the inhibition it commonly produces in mature circuits. Receptor signaling can influence how an immature interneuron moves before it becomes integrated into its final cortical network. Restoring that signaling under hypoxia therefore supplied a plausible intermediate step between the added peptide and the measured improvement in migration rather than leaving the rescue as an unexplained association.



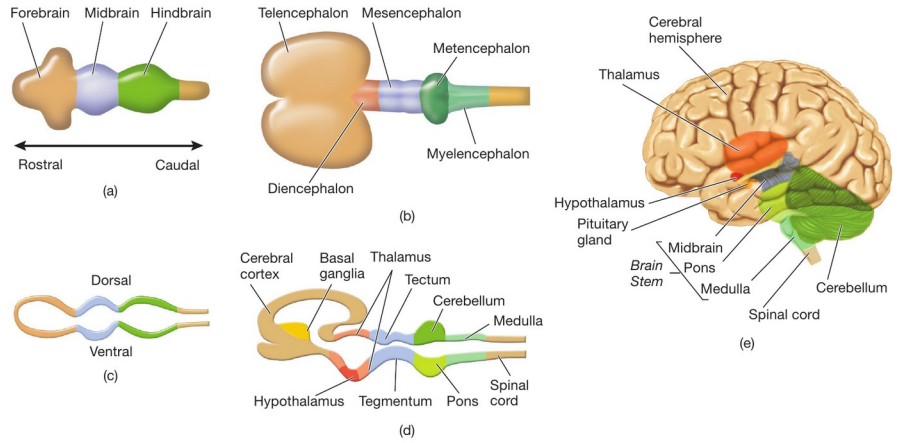
Puno et al. (2026), Fig. 6.

Progenitor expansion enlarges cortical surfaces

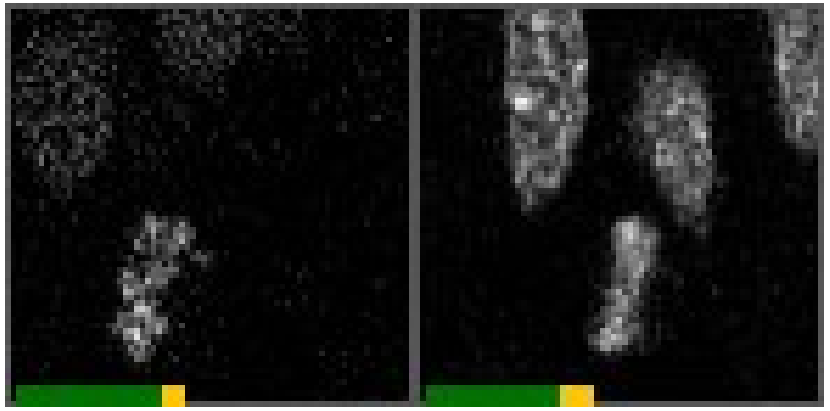
The size and folding of a developing cortex depend partly on how progenitor populations expand and produce descendants. More self renewing divisions can increase the pool available for later neuron generation. The timing of differentiation also matters because an early shift away from proliferation reduces the number of progenitors available to contribute additional cells afterward.

Primate cortices include expanded outer progenitor zones and abundant outer radial glia. Their organization differs from the smaller and smoother cortex of many laboratory rodents. Comparing species can identify developmental processes associated with expansion, but a larger cortical surface does not follow from a single universal extra division that explains every difference in adult brain organization.

Genetic changes can alter the timing or regulation of existing developmental processes rather than inventing an entirely new cell type for every species. Organoids provide one way to compare human and other primate progenitors under related culture conditions. They reproduce selected developmental features while lacking much of the complete embryo's anatomy, sensory experience, and systemic environment.



Carlson and Birkett (2022), Fig. 15.2, p. 494.



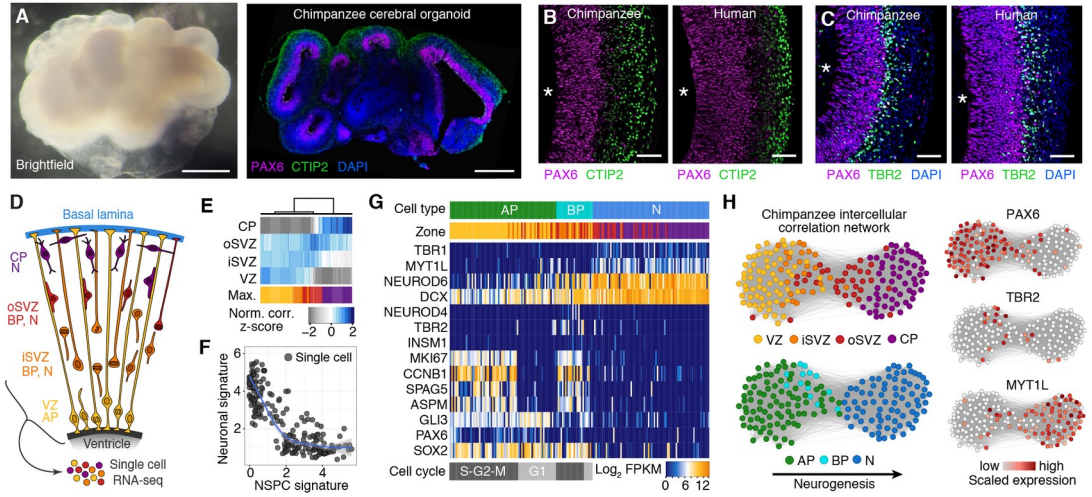
Mora-Bermúdez et al. (2016), media1.

Primate organoids share developmental programs

A comparative study generated cerebral organoids from human and chimpanzee cells and examined their developing cortical tissue. The researchers combined tissue staining, live imaging, and single cell gene expression analysis. This allowed them to compare both cellular organization and dynamic behavior rather than interpreting every species difference from the size of a finished organoid.

The organoids showed substantial similarities in tissue architecture, major cell types, and neurogenic gene expression programs. Both species used related developmental machinery to generate cortical progenitors and neurons. These shared features provided a background against which more specific differences could be identified, rather than assuming that human and chimpanzee brains arise through entirely unrelated developmental sequences.

The investigators then focused on the timing of progenitor division and the balance of proliferative and neurogenic populations. A subtle change in a recurring cell cycle event can accumulate across many descendants. Measuring that event directly provides a more concrete developmental hypothesis than attributing cortical expansion to a general increase in cellular complexity.



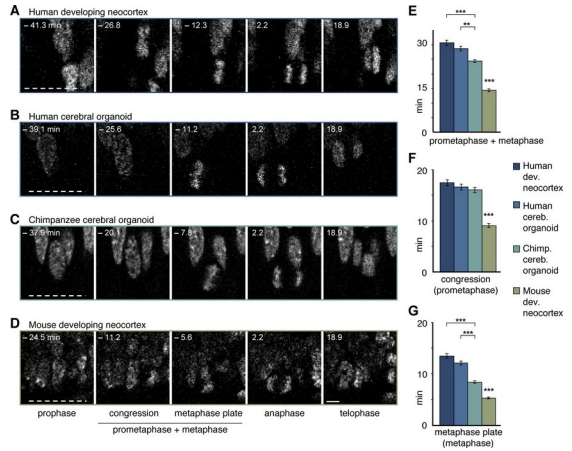
Mora-Bermúdez et al. (2016), Fig. 1.

Human progenitors differ in mitotic timing

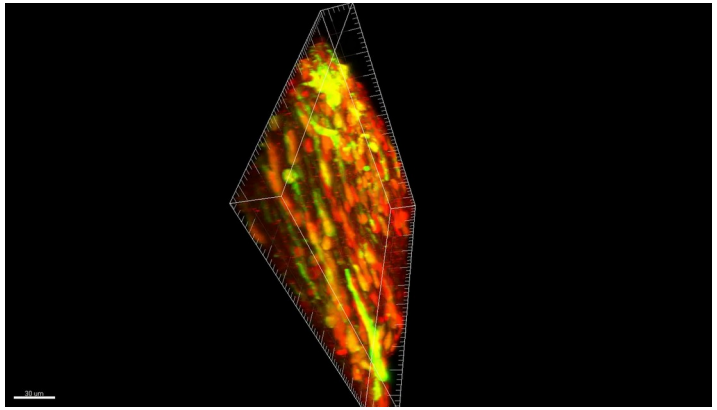
Live imaging in the human and chimpanzee organoids revealed a longer prometaphase to metaphase interval in human apical progenitors. These are stages when chromosomes and the division spindle are being organized before chromosome separation. The difference was specific to proliferating progenitors in the reported comparisons and was not a universal timing difference across all tested nonneural cells.

The study also found differences consistent with a greater proliferative capacity in the human progenitor population and a lower proportion of neurogenic basal progenitors. These observations concerned how the population was maintained and allocated during development. They did not show that a longer mitotic stage directly creates a particular adult cognitive ability.

The source movie records dividing cells in organoid tissue, allowing the relevant phases to be followed over time. The organoid preparation gives experimental access to an early developmental process that is otherwise difficult to observe. Its strongest conclusion is about measurable progenitor behavior under the tested conditions, which can contribute to hypotheses about the evolution of cortical growth.



Mora-Bermúdez et al. (2016), Fig. 5.



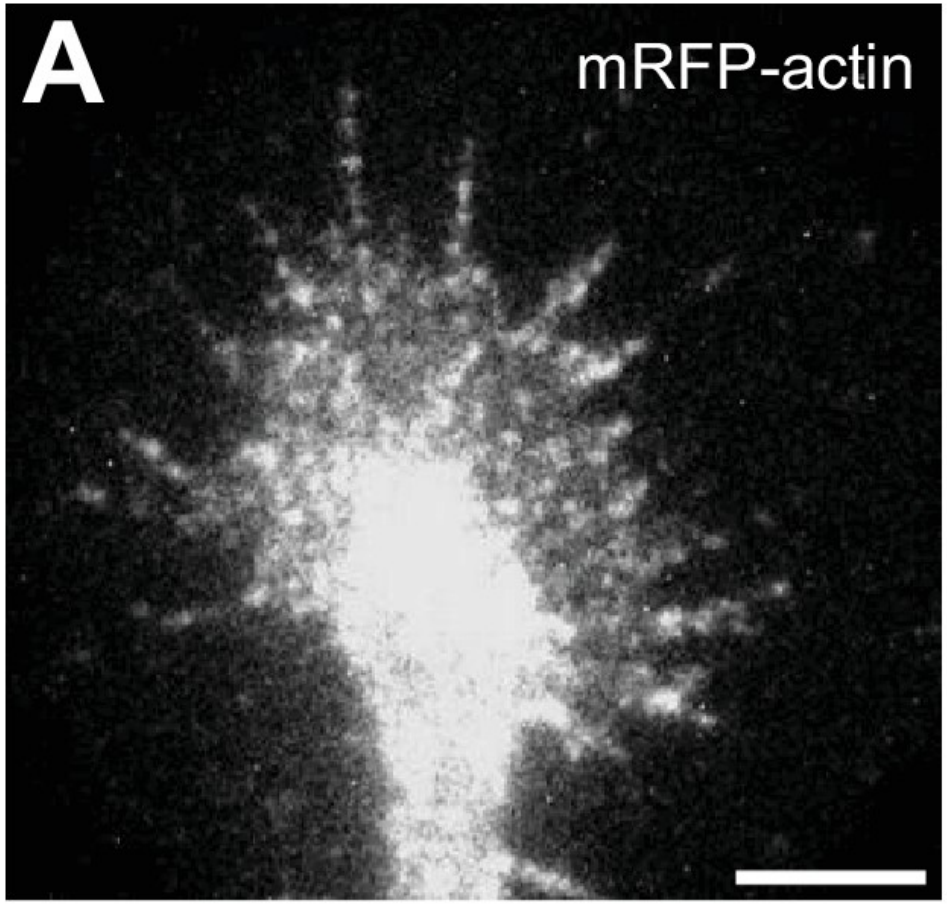
Heng et al. (2017), media1.

Growth cones turn local cues into movement

An axon grows through the activity of its growth cone, a motile structure at the advancing tip. The growth cone extends actin rich protrusions that explore nearby surfaces and signals. Receptors on its membrane detect chemical cues, while adhesive interactions allow the cell's internal forces to act against the surrounding material and move the tip forward.

Actin filaments continually assemble and move within the growth cone. Myosin motors generate tension, and molecular linkages couple the moving actin network to adhesion proteins in the membrane. Without effective coupling, internal movement can occur without equivalent forward advance. Directional growth therefore requires both sensing where a cue is stronger and generating force asymmetrically across the cone.

Guidance cues can attract or repel an axon depending on its receptors and developmental state. Netrin is one such cue, and its spatial distribution can influence the direction of extension. A shallow chemical gradient presents a demanding problem: the growth cone must amplify a small difference across its width into a sufficiently unequal mechanical response to turn.



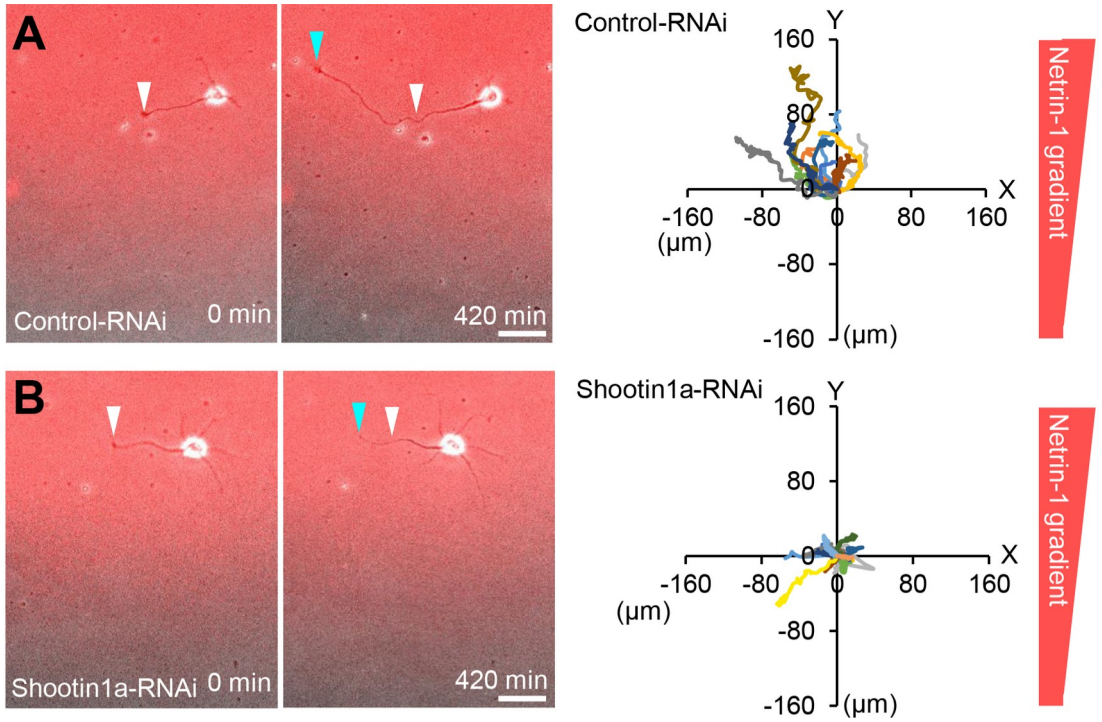
Baba et al. (2018), Fig. 6.

Adhesion enables directed axon extension

An axon cannot reach its target merely by detecting the presence of an attractive molecule. It must extend through tissue, maintain useful contacts, and release attachments that would prevent advance. Adhesion proteins connect the growth cone to other cells or extracellular material, providing a surface against which the cytoskeleton can generate traction and change the axon's direction.

Inside the growth cone, signaling molecules regulate how strongly actin is coupled to those contacts. Increasing coupling on one side can produce more traction there, biasing extension. A developmental mutation can therefore impair guidance even when the cue and receptor remain present, because the cell cannot translate sensory information into the forces required for a turn.

The outcome is a pathway, not simply an increase in neurite length. A longer axon that grows in the wrong direction can still fail to establish the appropriate circuit. Experiments consequently measure orientation, turning, and target entry as well as growth, distinguishing the capacity to extend a process from the ability to navigate it toward a useful connection.



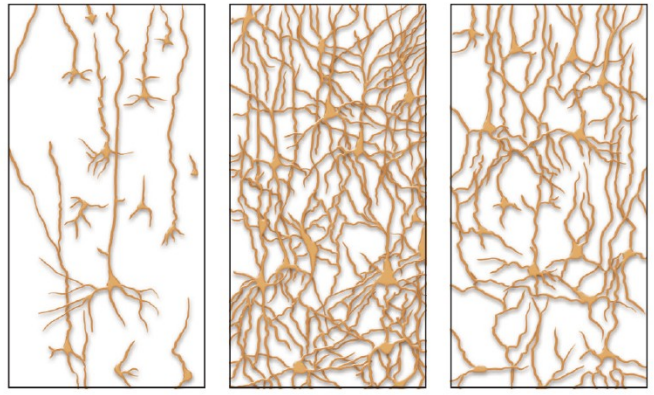
Baba et al. (2018), Fig. 7.

Synapses require two coordinated specializations

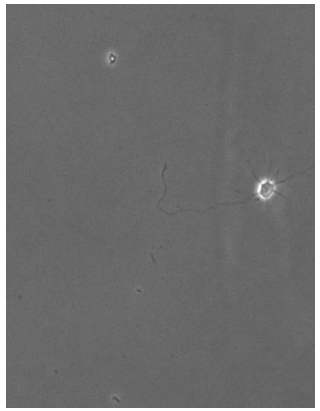
A synapse requires coordinated changes on both sides of a cellular contact. The presynaptic axon terminal accumulates vesicles and release machinery, while the postsynaptic cell organizes receptors and signaling proteins. Adhesion molecules and local signals help the two cells align these components so that transmitter release can influence the receiving cell effectively.

An initial contact is not necessarily a stable mature synapse. Contacts can strengthen, change, or disappear as cells compete for targets and their activity patterns develop. Electrical activity and trophic signals influence which connections persist. A trophic signal supports cellular growth or survival, linking the success of a developing connection to the state of the cells it reaches.

Circuit refinement involves both synaptic elimination and, in some populations, programmed cell death. These are different processes: removing a synapse does not require killing its entire neuron. Keeping them separate explains how a surviving neuron can retain some branches while losing others and why a reduction in synapse number does not automatically imply loss of the same number of cells.



Carlson and Birkett (2022), Fig. 15.4, p. 496.



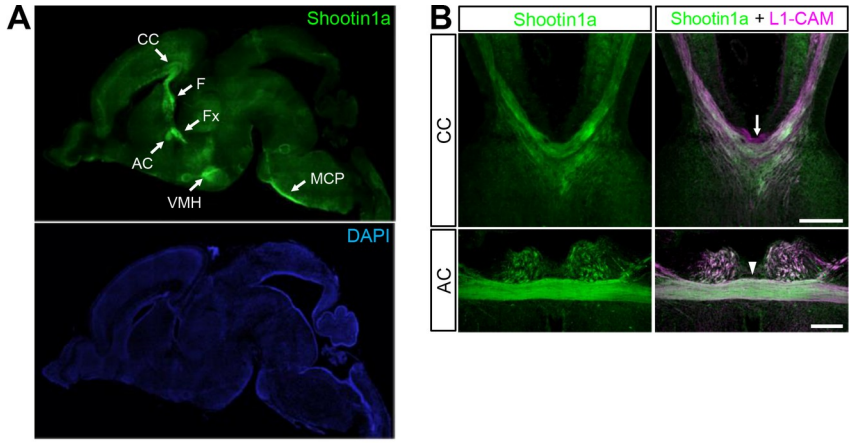
Baba et al. (2018), Video 4.

Netrin gradients bias force within growth cones

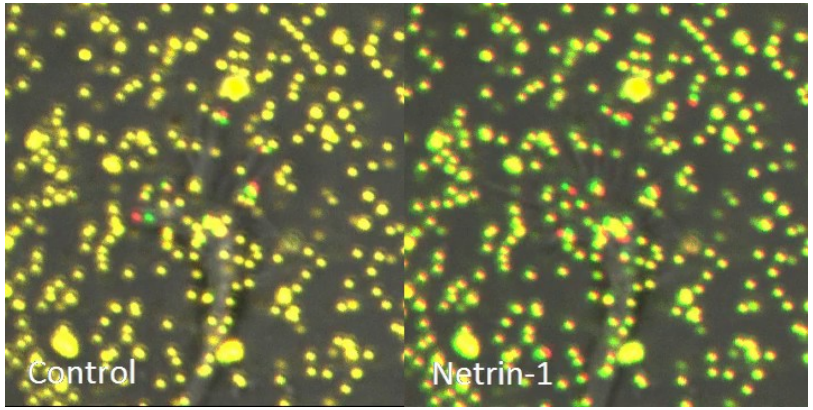
A study investigated how the guidance molecule netrin directs an axonal growth cone toward a stronger signal. The researchers used cultured neurons and experiments in mice lacking shootin1, a protein that helps couple internal actin movement to membrane adhesion. They asked how a chemical difference across the growth cone becomes a physical difference in traction.

Netrin increased phosphorylation of shootin1 through a kinase called Pak1. A kinase transfers phosphate groups to target proteins, changing their activity or interactions. Phosphorylated shootin1 bound more effectively to L1-CAM, a cell adhesion molecule. This strengthened the link between the moving actin network inside the growth cone and contacts that could transmit force to the outside.

The mechanism supplies an intermediate step between sensing a gradient and turning an axon. Stronger coupling allows actin movement to generate more traction rather than slipping without advancing the membrane. The study also found abnormal forebrain commissural projections in shootin1 deficient mice, connecting the cellular force mechanism to the organization of axons that normally cross between brain regions.



Baba et al. (2018), Fig. 1.



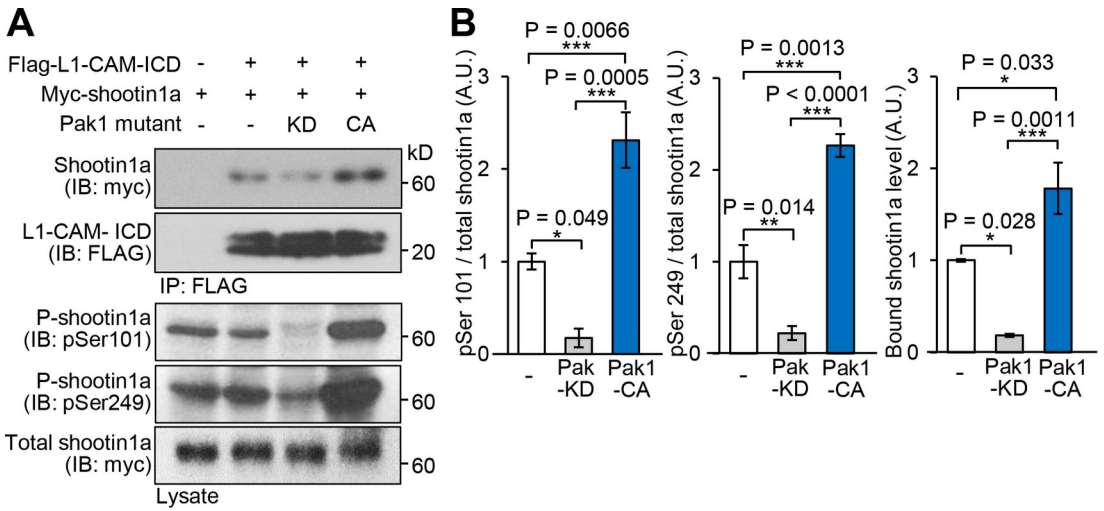
Baba et al. (2018), Video 3.

Asymmetric adhesion coupling steers the growing axon

The netrin study found that a shallow gradient produced a strongly polarized phosphorylation response across the growth cone. The side facing the stronger cue acquired more effective coupling between actin and adhesion. This spatially unequal response generated a mechanical bias, allowing a small difference in extracellular signal to produce a measurable change in the direction of axon extension.

When shootin1 function was reduced or its relevant interactions were disrupted, growth cones failed to turn normally toward netrin. Restoring appropriate protein function recovered guidance in the tested cellular conditions. These manipulations distinguished the ability to sense or encounter the cue from the ability to convert that information into the forces required for a directional response.

The published recordings include real growth cone movement and measurements of traction related bead displacement. The beads report forces transmitted through the substrate rather than electrical activity in the neuron. Reading the images with that measurement in mind identifies the physical output of the signaling pathway: unequal coupling creates unequal traction, which changes where the axon grows.



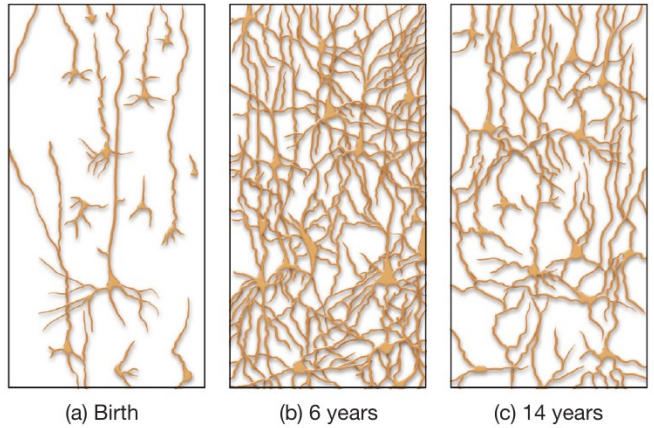
Baba et al. (2018), Fig. 4.

Cell death and synapse elimination are distinct

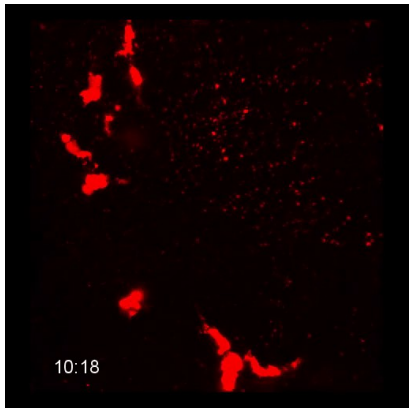
Programmed cell death, or apoptosis, removes cells through a regulated sequence of molecular events. Developing neural populations can initially contain more cells than will survive. Access to trophic support and appropriate targets influences survival, allowing the developing system to adjust population size as cells establish connections and compete within the available tissue environment.

Synaptic pruning instead removes selected contacts or branches while the neuron can remain alive. A neuron may lose some inputs, retain others, and extend new processes during the same developmental period. Reductions in synapse density therefore cannot automatically be interpreted as equivalent reductions in neuron number, because cells and their individual connections are different biological units.

Neural activity helps determine which connections are stabilized, but glial cells also participate actively in remodeling tissue. Microglia can engulf cellular material, and other glial populations can remove synaptic elements. These processes can refine a circuit during development without representing an immune attack on the entire brain or a failure to generate enough neurons in the first place.



Carlson and Birkett (2022), Fig. 15.4, p. 496.



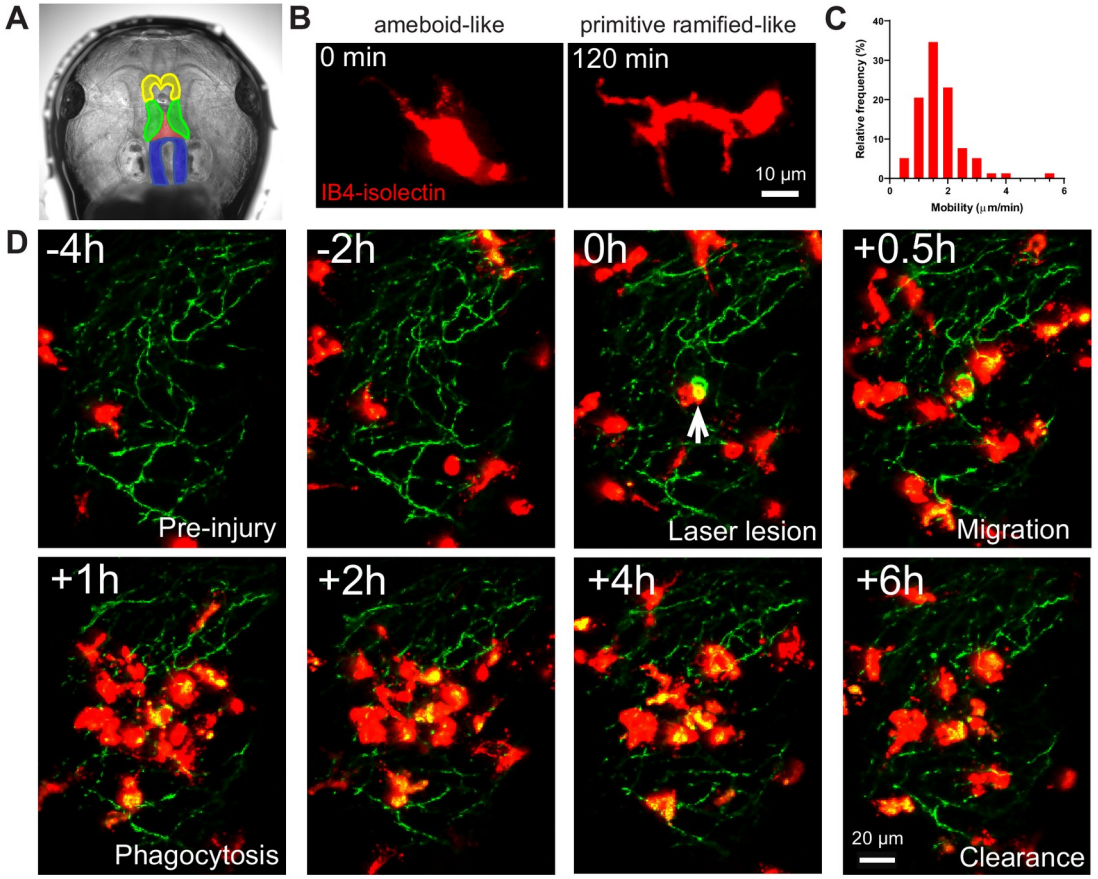
Lim et al. (2021), Video 1.

Microglia can remove portions of developing axons

A study imaged microglia interacting with retinal axons in the developing Xenopus tadpole brain. Retinal ganglion cells send axons to the tectum, a visual processing region. The researchers asked whether microglia removed pieces of those axons inside an intact developing animal rather than only interacting with damaged tissue in a dish.

They directly observed trogocytosis, a process in which a cell takes small portions of material from another cell instead of engulfing it completely. Axonal material entered microglia during live imaging. The event therefore differed from the death and complete removal of the retinal neuron, providing a mechanism for selectively remodeling its projection while the neuron remained present.

Microglial manipulation changed retinal axon pruning and behavioral responses to looming visual stimuli. Looming stimuli expand as an approaching object would, allowing the animal's visually guided response to be measured. The study connected a cellular remodeling process with developing circuit function through an actual behavioral task rather than assuming that any change in axonal appearance had the same functional consequence.



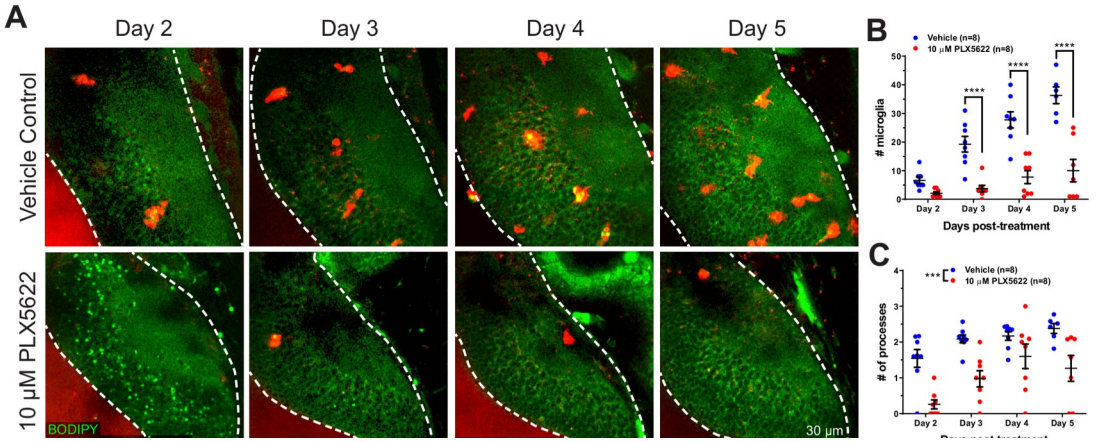
Lim et al. (2021), Fig. 1.

Complement signals regulate microglial pruning

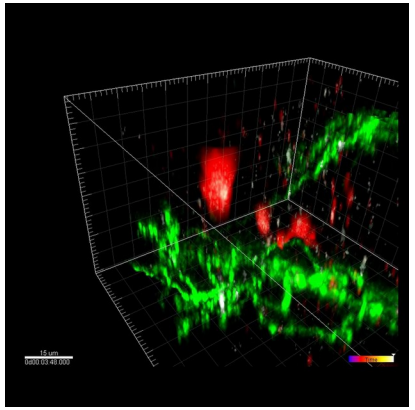
The tadpole pruning study examined complement, a group of proteins that can mark material for interaction with immune cells. The researchers identified a neuronal complement regulatory molecule related to human CD46. By changing complement regulation on neurons, they tested whether the neuron could influence how strongly microglia removed portions of its developing axonal arbor.

Increasing a complement inhibiting molecule reduced trogocytosis and pruning, while enhancing complement activity with a membrane associated C3 construct increased pruning. These opposite manipulations linked a defined extracellular signaling system to the amount of axonal material removed. The neuron was therefore an active participant in regulating its interaction with microglia, not merely a passive target for indiscriminate engulfment.

The original live movie shows transfer of labeled axonal material during an interaction, while the experiments quantify how signaling changes alter the broader arbor. Together they distinguish physical contact from actual removal. A microglial process touching an axon is not sufficient evidence of pruning unless the experiment also demonstrates material uptake or an associated structural change.



Lim et al. (2021), Fig. 4.



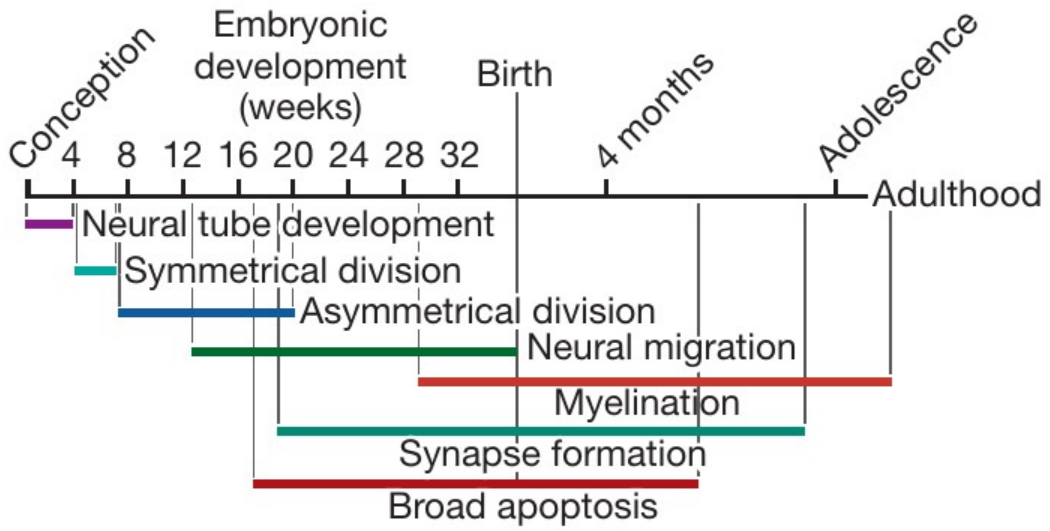
Lim et al. (2021), Video 4.

Critical periods regulate sensitivity to experience

A critical period is a developmental interval during which a particular kind of experience has an especially strong or lasting effect on a circuit. The relevant interval differs across functions and brain systems. It is not a single date when the entire brain becomes fixed, because learning and plasticity continue after many early developmental windows have closed.

Experience changes activity patterns in developing pathways, influencing which synapses are strengthened, stabilized, or removed. In binocular vision, coordinated input from the two eyes contributes to circuits that compare their information. If the eyes provide persistently mismatched input during a sensitive stage, normal stereoscopic processing may fail to develop even if the peripheral alignment is corrected later.

The cellular mechanisms that open and close developmental sensitivity include changes in inhibition, glial activity, extracellular structure, and synaptic signaling. Experimental model systems allow these mechanisms to be manipulated directly. A critical period effect is established by comparing exposure at different times and testing whether the resulting circuit changes persist beyond the immediate period of stimulation.



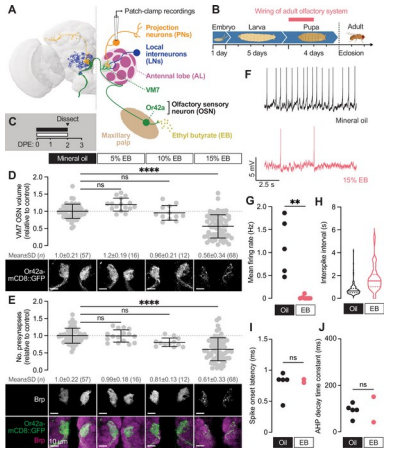
Carlson and Birkett (2022), Fig. 15.5, p. 497.

Early odor experience removes selected fly synapses

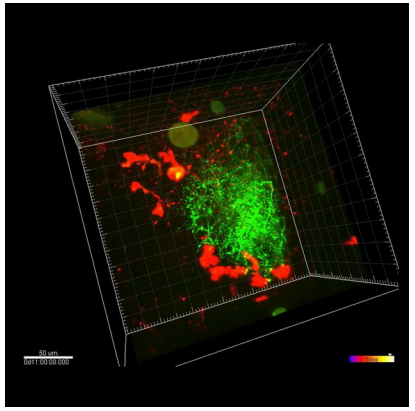
A study examined an early adult critical period in the Drosophila olfactory system. Sensory neurons detecting the odor ethyl butyrate project into a defined glomerulus, a compact region where they contact downstream projection neurons. The researchers exposed young adult flies to the odor and measured how the structure and function of those contacts changed.

Exposure during the first two days after emergence reduced the volume of the responsive glomerulus, the number of presynaptic sites, and activity in the downstream projection neurons. The response therefore included physical remodeling and altered signaling. It was not simply a temporary reduction in movement caused by the animal being surrounded by a strong odor during measurement.

The reduced synapse number and altered spontaneous activity persisted after the early exposure. This established a lasting consequence of developmental experience. The experiment provided a tractable circuit in which the odor, receptor neuron population, target region, and timing of exposure could all be specified, allowing cellular mechanisms of a critical period to be tested directly.



Leier et al. (2025), Fig. 1.



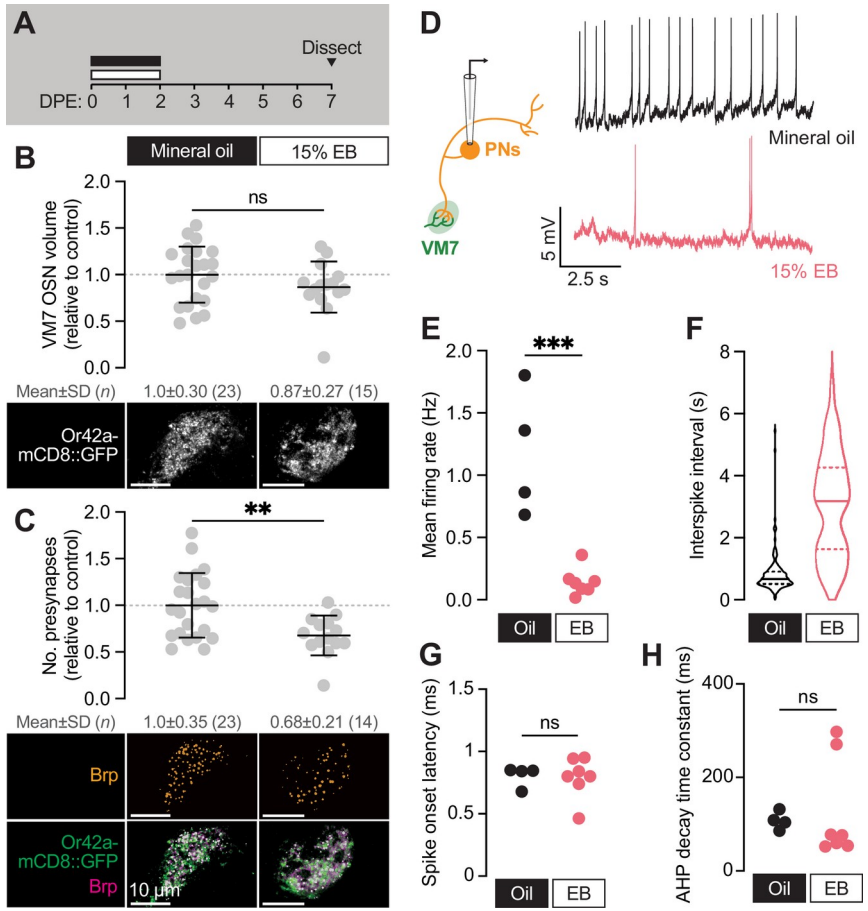
Lim et al. (2021), Video 3.

Glial engulfment is required for odor driven pruning

The fly olfactory study identified ensheathing glia as active participants in the experience dependent change. After early odor exposure, these glial cells increased expression of Draper, an engulfment receptor, entered the affected glomerulus, and removed sensory neuron presynaptic terminals. The circuit changed through a defined cellular interaction rather than an unexplained reduction in regional size.

Loss of Draper prevented the structural and physiological consequences of the critical period odor exposure. Presynaptic terminals and downstream activity no longer showed the same reductions under the tested conditions. This manipulation established that the engulfment pathway was required for the observed form of plasticity, connecting glial receptor function to both synapse number and neural signaling.

The comparison also distinguished an early exposure effect from ordinary aging of the circuit. The timing of the odor experience determined whether persistent remodeling occurred, while the glial manipulation identified machinery that executed it. A developmental window can therefore depend on when supporting cells become able to remove selected connections in response to neural activity.



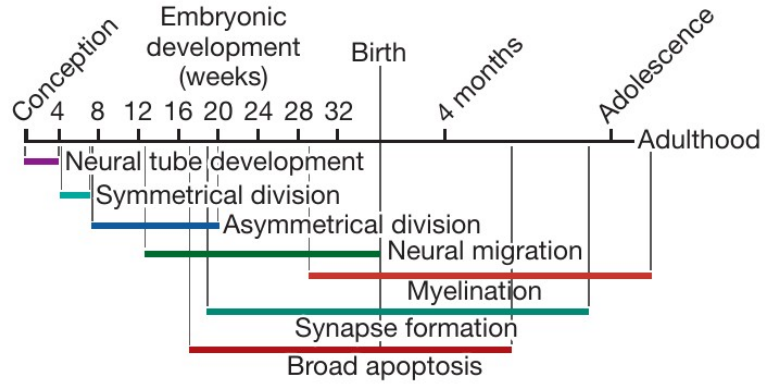
Leier et al. (2025), Fig. 3.

Myelination changes the timing of communication

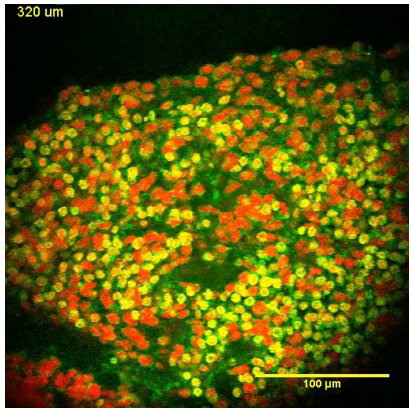
Myelin is a multilayered membrane sheath wrapped around axons by specialized glial cells. In the central nervous system, oligodendrocytes form these sheaths. Myelin reduces current loss across the axonal membrane and allows action potentials to propagate efficiently between nodes, the short gaps where voltage dependent channels regenerate the electrical signal.

Myelination begins on different schedules in different pathways and continues after birth. Changing conduction speed alters when signals arrive at their targets, affecting coordination among distributed regions. Development therefore involves not only adding synapses but also refining the timing with which established pathways communicate, especially as sensory, motor, and cognitive abilities become more reliable.

White matter imaging reflects several tissue properties, including aspects of myelin and axonal organization, but an imaging contrast is not a direct count of sheaths. A developmental change in a scan must be interpreted alongside the measurement method. Similarly, continued maturation of white matter does not mean that all children pass through the same behavioral transition at one universal age.



Carlson and Birkett (2022), Fig. 15.5, p. 497.



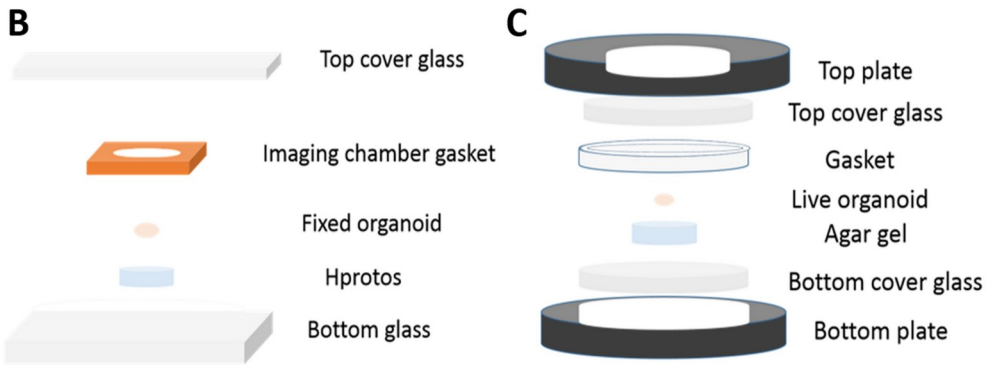
Yildirim et al. (2022), fig3-Video 1.

Deep imaging reveals organoid architecture

A study developed three photon imaging to examine intact human cerebral organoids, including organoids modeling Rett syndrome. Conventional imaging has difficulty reaching deeply into scattering tissue. The method used optical signals that allowed ventricular like structures and migrating cells to be examined with less reliance on physically sectioning the entire organoid into thin slices.

The investigators compared mutant organoids with genetically related control organoids. The controls helped distinguish effects associated with the disease relevant mutation from differences between unrelated genetic backgrounds. Imaging revealed changes in the size and organization of progenitor rich ventricular regions, providing a structural measure of early developmental architecture within the tissue model.

The organoids reproduced selected features of early cortical development rather than an entire human brain. Their cavities and cell populations allowed questions about progenitor organization and migration to be tested. The source images and movies identify fixed structural stacks separately from live recordings, preventing a sweep through tissue depth from being mistaken for the movement of cells over developmental time.



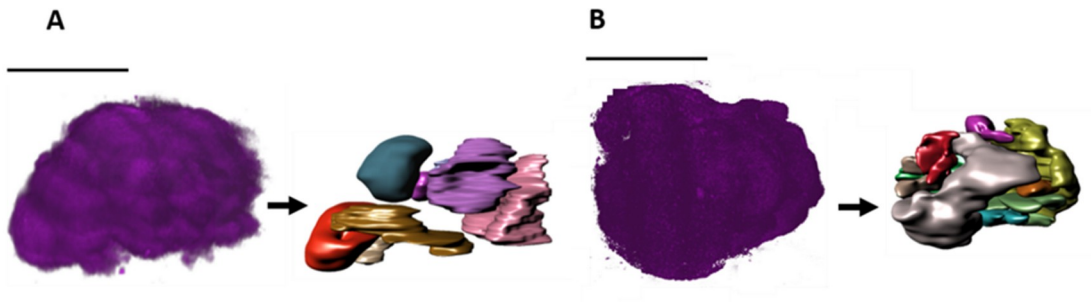
Yildirim et al. (2022), Fig. 1.

Rett model neurons follow altered migration paths

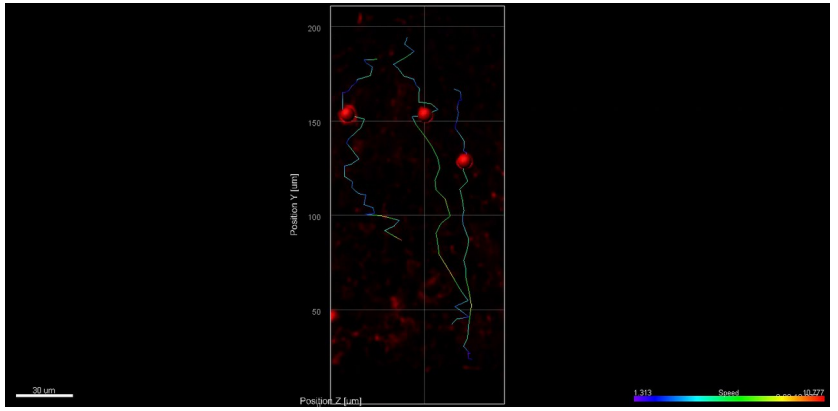
Live imaging in the organoid study followed neurons migrating over hours in mutant and matched control tissue. Mutant neurons traveled shorter distances and moved more slowly, with more tortuous trajectories. Tortuosity describes how indirect a path is, so a cell can spend time moving without making the same progress toward its destination as a more directly migrating cell.

The researchers could measure these features in intact organoids rather than inferring every migration defect from a final tissue section. Tracking the same cells separated path geometry from endpoint location. This is useful because two neurons might finish near one another after following different routes, or one might reach a shorter net displacement despite substantial local movement.

The cellular phenotype concerns migration in a model of Rett syndrome, not a direct observation of every neuron in an affected child. The experiment specifies the preparation, mutation comparison, and measured trajectory. Its contribution is a way to visualize how a developmental genetic change alters cellular movement within three dimensional human derived neural tissue.



Yildirim et al. (2022), Fig. 4.



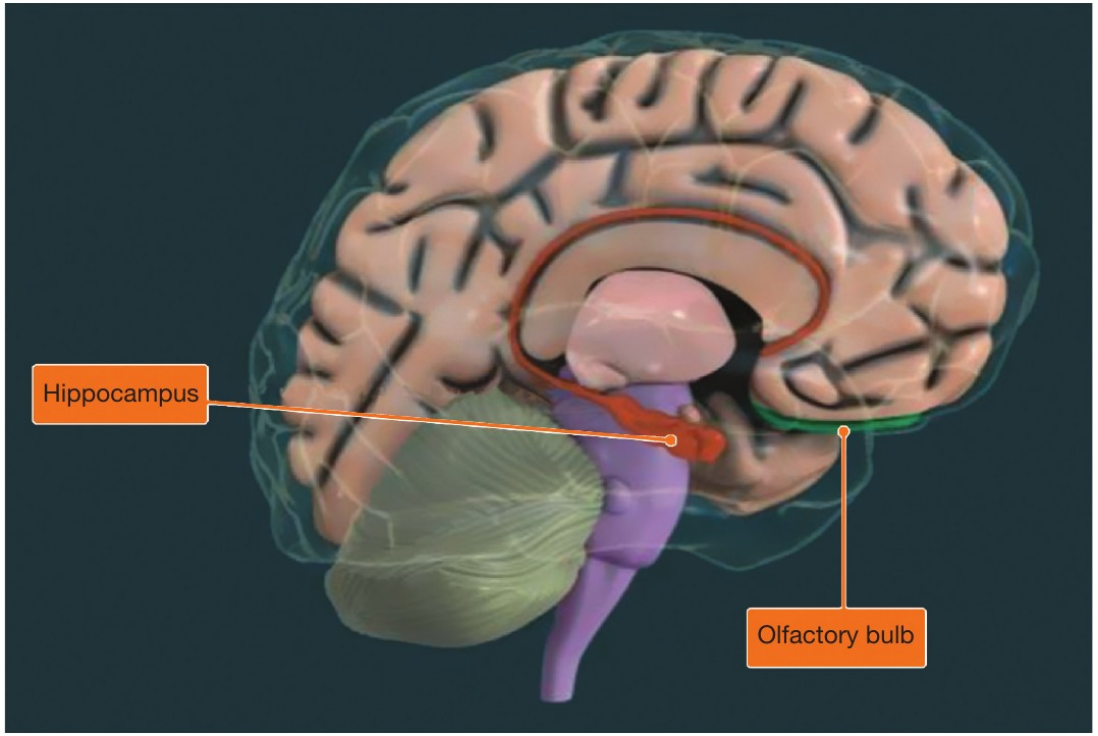
Yildirim et al. (2022), fig5-Video 2.

New neurons require integration as well as birth

Neurogenesis is the production of new neurons from dividing precursor cells. Demonstrating it requires more than detecting cell division, because many dividing brain cells become glia rather than neurons. Researchers combine lineage labels, neuronal identity markers, and developmental follow up to determine whether a newly produced cell survives and acquires the properties of a neuron.

Adult neurogenesis is well established in particular regions of several laboratory animals, including the rodent hippocampus and olfactory system. The extent and persistence of neurogenesis in adult humans remain debated and vary with the region, age, and measurement method. The animal findings should not be converted into a claim that every adult human brain routinely replaces neurons throughout the cortex.

A new neuron must also extend processes, receive appropriate inputs, and develop useful physiological responses. Birth alone does not establish successful circuit repair. Experiments that follow its synapses and activity can identify whether the surrounding tissue supports integration, separating production of a cell from its eventual contribution to sensory processing, memory, or behavior.



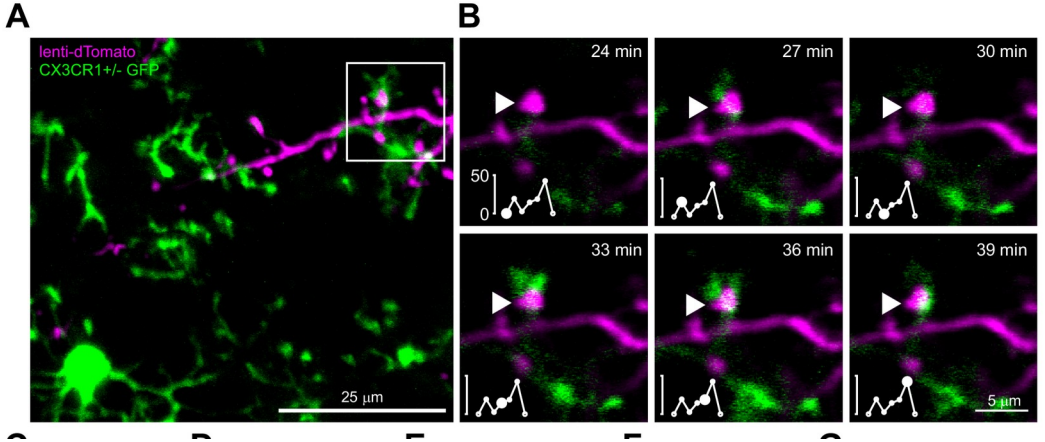
Carlson and Birkett (2022), Fig. 15.6, p. 498.

Microglia support adult born neuron maturation

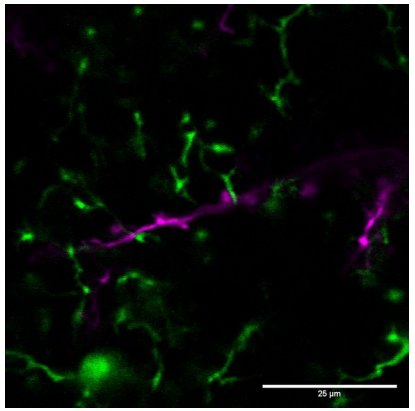
A mouse study investigated newly generated granule cells in the adult olfactory bulb. These neurons become incorporated into odor processing circuits after birth. The researchers temporarily depleted microglia and asked whether the new neurons developed normally, testing a supportive role for these cells in addition to their established capacity to remove cellular material.

Microglial depletion reduced odor responses in developing adult born granule cells but did not produce the same effect in preexisting granule cells. The manipulation therefore affected a developmental stage within the adult circuit rather than uniformly suppressing every neuron. Newly integrating cells had different requirements from neighbors that had already established their synaptic organization.

The source microscopy shows microglial processes near dendrites and spines of the developing cells. A dendritic spine is a small protrusion that commonly receives an excitatory synapse. By combining structural imaging with functional recordings, the study tested whether altered glial availability changed both the contacts being formed and the signals those contacts could convey.



Wallace et al. (2020), Fig. 1.



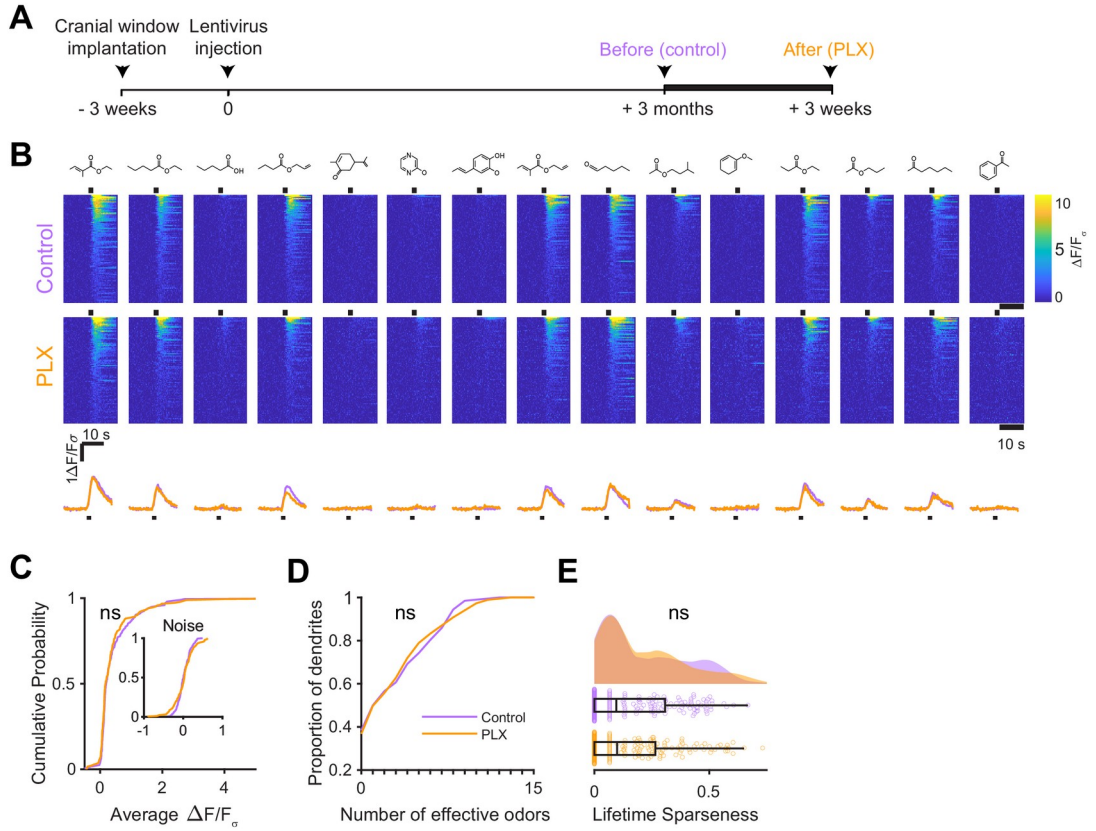
Wallace et al. (2020), Video 2.

Weaker synaptic inputs accompany microglial loss

When microglia were absent, adult born olfactory granule cells developed smaller dendritic spines and received weaker excitatory synaptic inputs. These changes supplied a cellular explanation for their reduced odor responses. The problem was not merely that an imaging marker became dimmer; the receiving neuron's physiological response to its inputs was also altered.

Microglial processes preferentially interacted with particular spine types in the developing cells. Live imaging captured these contacts, while complementary recordings assessed the strength of incoming excitation. Contact alone did not prove that the microglial cell built a synapse, but the depletion experiment showed that normal microglial presence was required for the usual structural and functional maturation.

The findings show that glial involvement is not synonymous with destruction. In one developmental context, microglia help remove axonal material; in another, their absence leaves new neurons with weaker inputs. The direction of the effect depends on the cell population and stage being studied, so removing all microglia cannot be assumed to improve every form of neural plasticity.



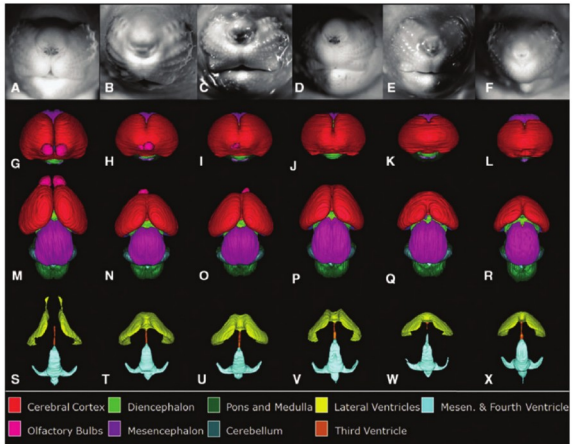
Wallace et al. (2020), Fig. 4.

Prenatal exposures act at particular stages

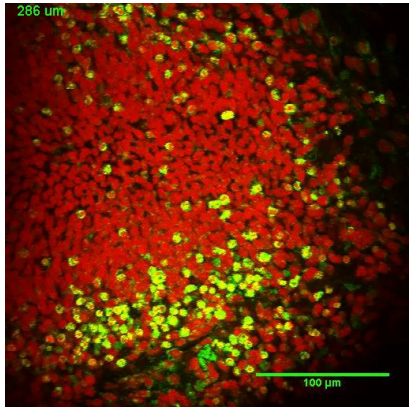
A teratogenic exposure disrupts development during prenatal life. Its effects depend on the agent, dose, timing, and the biological processes occurring when exposure takes place. An early disturbance of tissue patterning differs from a later effect on synapse formation, so the same substance can produce different outcomes across developmental windows and different tissues.

Alcohol can interfere with several processes involved in nervous system development, including cellular signaling, growth, adhesion, and survival. Prenatal alcohol exposure can produce a range of cognitive and behavioral effects, with or without the facial features associated with a particular fetal alcohol syndrome diagnosis. Facial appearance therefore cannot determine whether every relevant developmental effect is present.

Experimental models allow the exposure to be timed precisely and its cellular consequences measured. These models also reveal interactions with genetic background. A mutation that is tolerated under one condition can make a developing pathway more vulnerable to an environmental disturbance, producing an outcome that cannot be predicted by considering the gene or exposure separately.



Carlson and Birkett (2022), Fig. 15.7, p. 500.



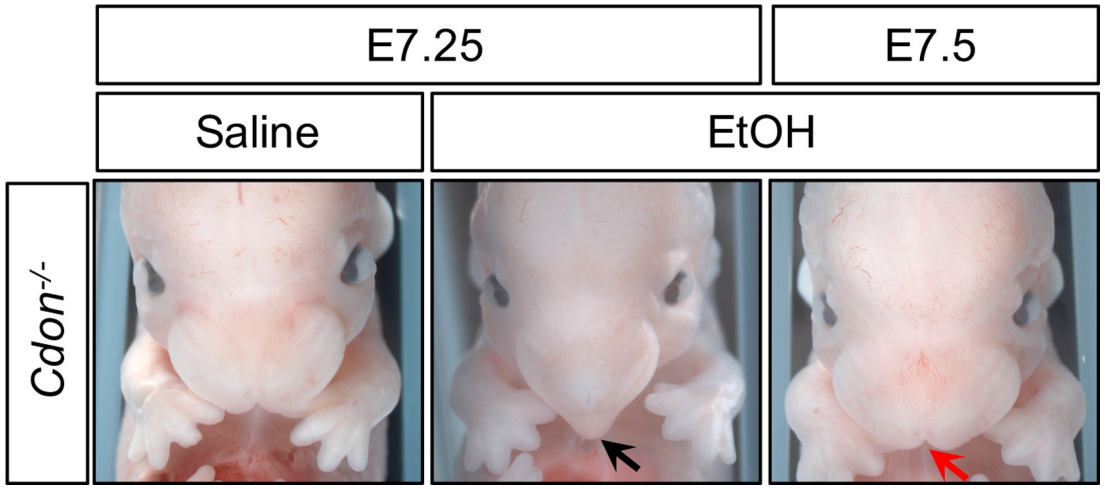
Yildirim et al. (2022), fig3-Video 2.

Alcohol and Cdon changes suppress Nodal

A mouse study examined how fetal alcohol exposure interacted with mutation of Cdon, a gene encoding a cell surface signaling protein. The combination produced holoprosencephaly, a defect in which early forebrain midline development is disrupted. The investigators asked which signaling pathway mediated the interaction rather than assuming that the exposure caused a nonspecific reduction in brain growth.

The experiments identified Nodal signaling as an important point of convergence. Nodal is a developmental signaling system involved in early tissue patterning. Brief ethanol exposure during a sensitive stage transiently reduced this pathway's activity in the susceptible genetic background, with later consequences for the organization of the forebrain midline.

The model therefore related an early molecular disturbance to a later anatomical phenotype. Measuring only the final malformed brain would not identify when the critical interaction occurred. The timed exposure and pathway analysis showed how a short developmental event can change the conditions under which subsequent brain structures are assembled.



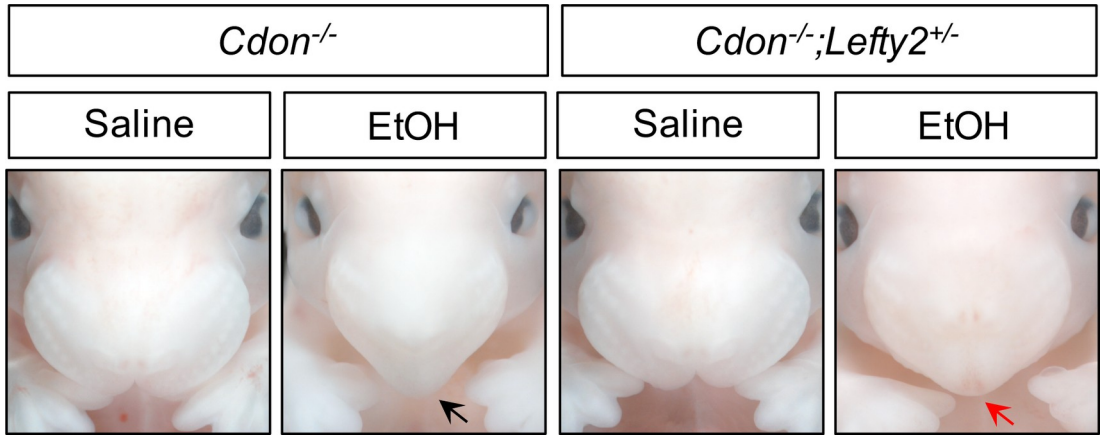
Hong et al. (2020), Fig. 1.

Genes alter vulnerability to developmental exposure

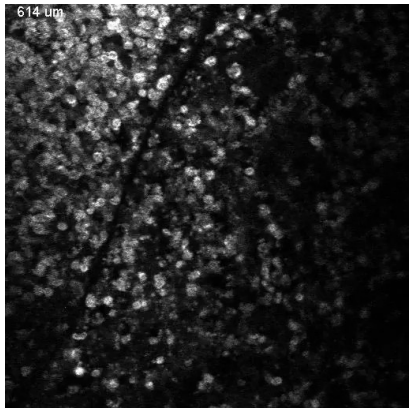
The Cdon and alcohol study varied exposure timing and used genetic and cellular experiments to test the proposed mechanism. The strongest interaction occurred within a particular early window consistent with the role of Nodal in patterning. A later visible brain abnormality therefore originated from an earlier disruption in the signals that organized tissue development.

The findings distinguish susceptibility from inevitability. A genetic change can alter how strongly a pathway responds to an environmental challenge, while the challenge supplies an additional disturbance. The combined outcome can be much more severe than either condition considered alone, providing a concrete example of why developmental effects are not always additive or easily assigned to one isolated cause.

This experimental interaction should not be generalized into a claim that one variant explains all effects of prenatal alcohol exposure. Alcohol influences multiple developmental processes, and human backgrounds vary widely. The study's specific contribution is to identify Nodal suppression during a defined embryonic interval as a mechanistic route through which this particular gene exposure combination disrupted forebrain development.



Hong et al. (2020), Fig. 4.



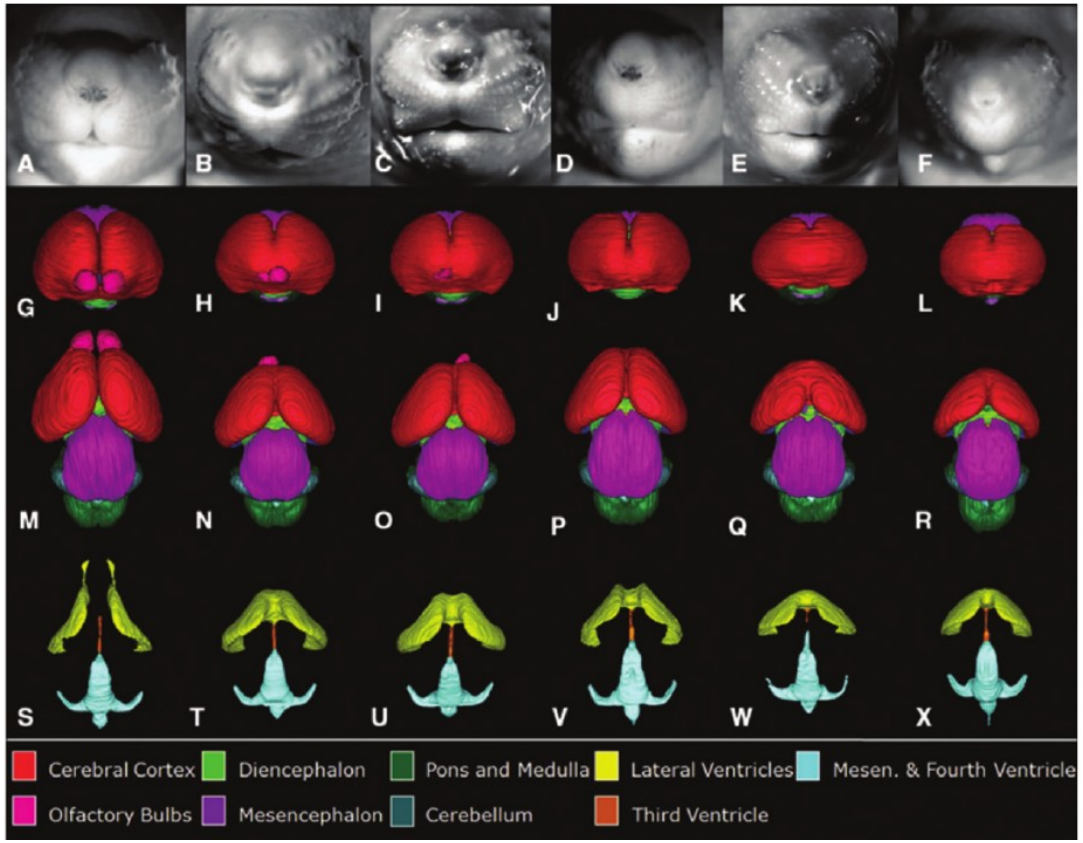
Yildirim et al. (2022), fig3-Video 3.

Different prenatal injuries can disrupt different processes

Prenatal brain injury can arise through chemical exposures, infection, inflammation, or reduced oxygen, each of which can affect several developmental stages. Alcohol related structural changes in mouse fetuses illustrate one exposure, while viral infections such as congenital Zika can alter neural progenitor survival and tissue growth through different biological mechanisms. Similar outcomes do not make the initiating causes identical.

A smaller brain can reflect reduced production of cells, increased death, or altered growth of tissue components. A migration defect can instead place surviving neurons in inappropriate locations. These possibilities require different measurements, so a visible anatomical difference alone does not identify whether the primary disturbance involved proliferation, survival, movement, or connection formation.

Timing helps distinguish those routes. Neural tube closure, cortical neuron generation, interneuron migration, and synaptic refinement occur across different intervals. An exposure can influence whichever processes are active and vulnerable at that time. Experimental interpretation therefore links the observed anatomy to the actual preparation and developmental window rather than treating every prenatal insult as the same generalized damage.



Carlson and Birkett (2022), Fig. 15.7, p. 500.

Metabolic defects alter the developing brain

An inherited metabolic disorder can change the ability to synthesize, transport, or break down a necessary compound. Enzymes catalyze chemical reactions, allowing them to proceed efficiently under cellular conditions. A mutation that reduces an enzyme's function can cause its substrate to accumulate or prevent production of a molecule required for normal neural development and physiology.

Phenylketonuria commonly involves reduced activity of phenylalanine hydroxylase, which converts phenylalanine to tyrosine. Without adequate control, phenylalanine accumulates and disrupts brain function and development. Newborn screening identifies the biochemical problem early, allowing treatment before severe effects emerge rather than waiting for intellectual or behavioral difficulties to become obvious.

Management reduces excessive phenylalanine exposure while preserving adequate nutrition, and continued metabolic monitoring remains important beyond infancy. Other disorders require different interventions because the missing step differs. A diet that helps one pathway cannot automatically correct a transport defect, a vitamin dependent enzyme problem, or a lysosomal storage disorder with a different accumulating substrate.

Causes	Description	Symptoms
Exposure to Teratogens	Prenatal exposure to teratogens (such as alcohol) impairs brain development.	Intellectual disability
Inherited Metabolic Disorders (PKU, pyridoxine dependency, galactosemia, Tay-Sachs disease)	Genetic abnormalities in the codes for specific enzymes. If untreated, leads to impaired brain development.	Intellectual disability; brain swelling in Tay-Sachs
Down Syndrome	Three copies of the 21st chromosome contribute to impaired brain development.	Intellectual disability; brain degeneration in adulthood

Carlson and Birkett (2022), Table 15.1, p. 502.



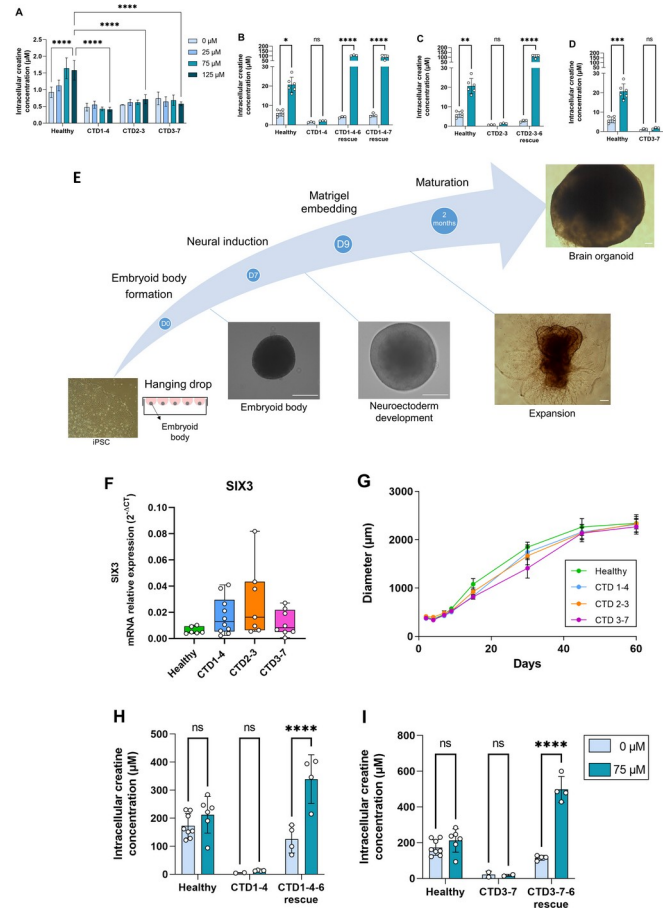
Sun et al. (2022), fig1-Video 1.

Creatine transport shapes organoid development

A study generated brain organoids from people with creatine transporter deficiency and healthy donors. The condition results from changes in SLC6A8, which encodes a protein that transports creatine into cells. Creatine participates in a chemical energy buffering system, helping cells maintain ATP availability when energy demand changes rapidly.

Patient derived organoids showed reduced creatine uptake and altered expression of proteins involved in neural progenitor identity and development. SOX2 and PAX6, proteins associated with progenitor programs, were reduced, while GSK3 beta was increased. GSK3 beta is a kinase that regulates multiple cellular processes, so its change supplied a measurable molecular phenotype rather than a complete explanation by itself.

The researchers also profiled many proteins to identify broader changes associated with the developmental condition. An organoid permits measurements in human derived neural tissue that would be difficult to perform repeatedly in a living child. Its findings concern cellular development and metabolism in the model, while language delay, seizures, and other clinical features require assessment in the person.



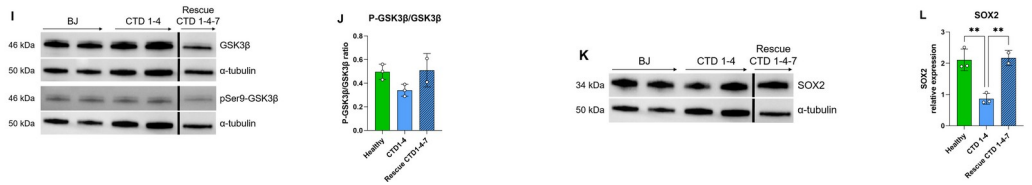
Broca-Brisson et al. (2023), Fig. 1.

Restoring a transporter rescues cellular phenotypes

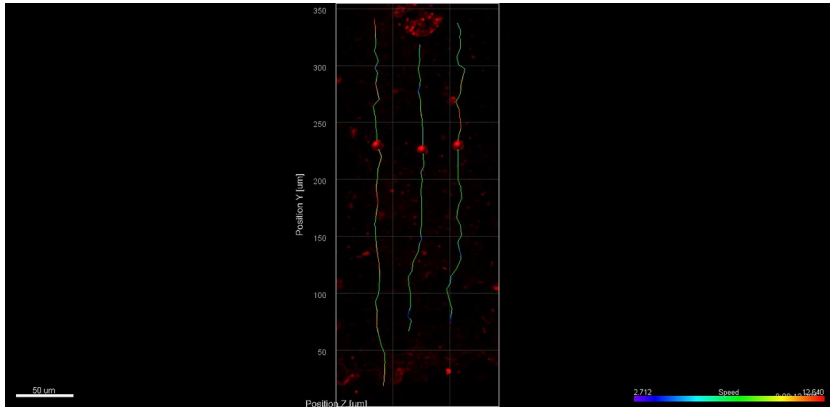
The creatine transporter study reintroduced a functional SLC6A8 into patient derived organoids. This restored creatine uptake and normalized several altered proteins, including markers associated with progenitors and developmental signaling. The manipulation directly tested whether the transport defect contributed to the measured cellular changes rather than leaving the patient control difference as an unexplained association.

The result separates a transport problem from an inadequate supply in the surrounding medium. Adding more of a molecule outside a cell may not solve a defect in the protein required to move it across the membrane. Restoring the transporter changes access to the molecule and therefore can alter intracellular metabolism and downstream developmental processes.

The experiment demonstrated rescue in a tissue model, where the gene could be delivered and the biochemical response measured. Delivering an effective treatment throughout a living human brain presents additional biological requirements. The source supported claim is that reinstating transport corrected specific organoid phenotypes, identifying a causal cellular target for further therapeutic research.



Broca-Brisson et al. (2023), Fig. 6.



Yildirim et al. (2022), fig5-Video 1.

Lysosomal defects prevent molecular breakdown

Lysosomes are membrane enclosed compartments containing enzymes that break down selected cellular materials. The products can then be reused or removed. In a lysosomal storage disorder, a missing or ineffective enzyme allows particular substrates to accumulate, altering cellular organization and function. The biological problem is a specific failure of processing, not simply the presence of an unusual gene label.

Tay Sachs disease commonly results from reduced hexosaminidase A function, leading to accumulation of GM2 ganglioside, a membrane associated lipid. Neurons are especially vulnerable to the resulting cellular disturbance. Infantile disease causes progressive neurological loss, while variants retaining more enzyme function can produce different ages of onset and clinical courses.

Treatment strategies must address the missing biochemical function or its consequences. Possible approaches include supplying an enzyme, restoring its gene, or reducing production of the accumulating substrate, but delivery to neural tissue is a major requirement. These mechanisms differ from dietary management of phenylalanine and from restoring a creatine transporter because the defective step occurs in a different cellular compartment.

Causes	Description	Symptoms
Exposure to Teratogens	Prenatal exposure to teratogens (such as alcohol) impairs brain development.	Intellectual disability
Inherited Metabolic Disorders (PKU, pyridoxine dependency, galactosemia, Tay-Sachs disease)	Genetic abnormalities in the codes for specific enzymes. If untreated, leads to impaired brain development.	Intellectual disability; brain swelling in Tay-Sachs
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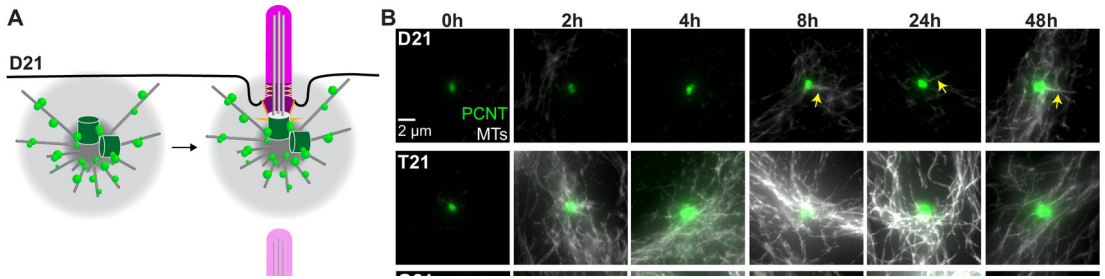
Carlson and Birkett (2022), Table 15.1, p. 502.

Chromosome dosage can alter primary cilium assembly

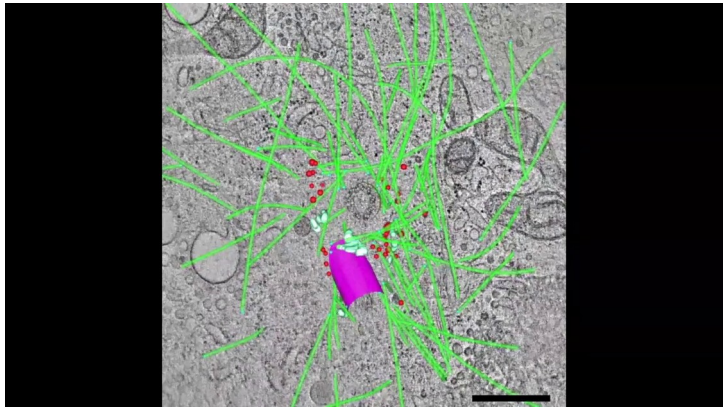
A trisomy 21 study examined how extra chromosome dosage affects primary cilia. A primary cilium is a small membrane covered cellular projection that serves as a signaling compartment rather than propelling the cell through fluid. Its assembly depends on coordinated delivery of membrane and proteins near the centrosome, a structure that also organizes microtubules.

Pericentrin, a centrosomal protein encoded on chromosome 21, was elevated in cells with extra chromosome copies. The researchers found increased material and microtubules around the centrosome that interfered with early steps of cilium construction. This pericentrosomal crowding delayed membrane delivery and the removal of a cap that normally permits the cilium to grow.

The original study combined cultured cell experiments, electron tomography, and a Down syndrome mouse model. These preparations addressed different parts of the mechanism. A three dimensional reconstruction of a centrosome shows physical organization at a cellular scale, while measurements in developing cerebellum test whether the signaling problem is also relevant in neural tissue.



Jewett et al. (2023), Fig. 1.



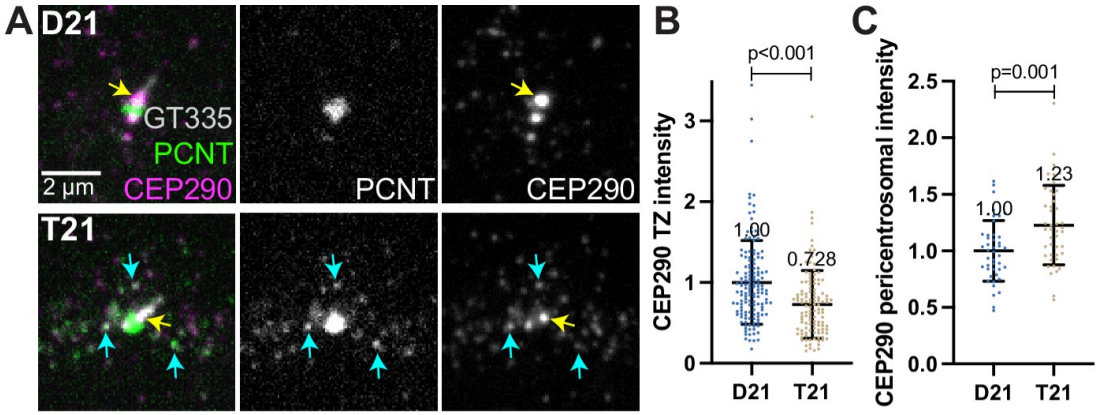
Jewett et al. (2023), fig2-Video 1.

Ciliary trafficking defects persist after delayed assembly

Given sufficient time, many trisomy 21 cells in the cilium study eventually formed cilia. Their signaling compartment nevertheless remained abnormal because trafficking defects reduced the localization of transition zone proteins. The transition zone regulates entry into the cilium, so forming a visible projection did not guarantee that its molecular contents and signaling properties had become normal.

The cells showed reduced sonic hedgehog signaling associated with elevated pericentrin. Sonic hedgehog is a developmental signal that influences proliferation and patterning through a cilium dependent pathway in many cells. A trafficking defect can therefore alter the cell's response to a signal even when the signaling molecule is present in the surrounding environment.

In the mouse model, developing cerebellar granule neuron progenitors had fewer primary cilia and a thinner external granule layer at the tested age. This connected a chromosome dosage related cellular disturbance with a neural progenitor population. It did not identify one mechanism that explains every cognitive or medical feature associated with Down syndrome.



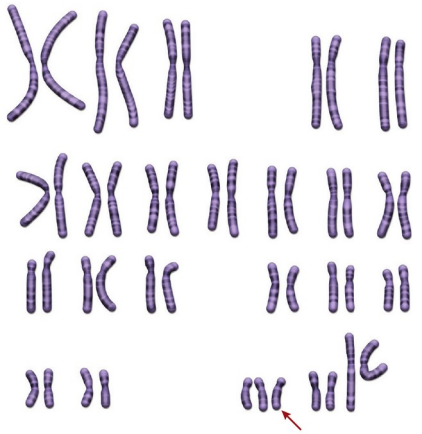
Jewett et al. (2023), Fig. 5.

Trisomy 21 changes gene dosage across many systems

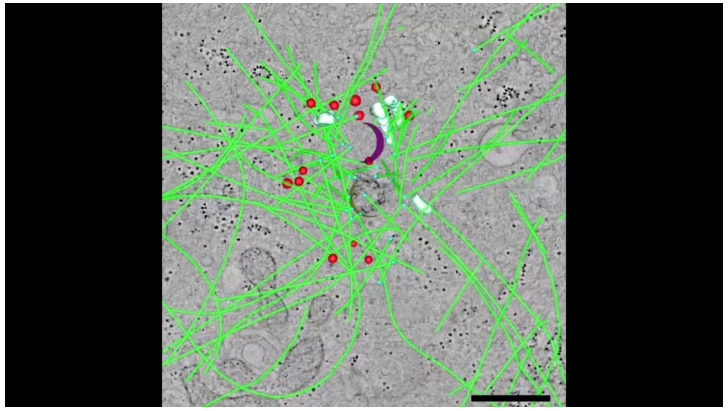
Down syndrome usually results from an extra copy of chromosome 21. The additional genetic material changes the dosage of many genes, influencing development across multiple tissues. A karyotype displays chromosomes and can reveal the extra copy, but it does not show how each gene changes the behavior of a developing neuron or predict one person's abilities.

People with Down syndrome vary in learning profiles, language, health conditions, and support needs. Development includes both strengths and challenges, with individual experience and access to appropriate services influencing participation. The chromosomal cause identifies a broad biological condition while leaving substantial variation in the specific neural processes that contribute to cognition and behavior.

Experimental models isolate candidate mechanisms such as progenitor signaling, synaptic inhibition, or cellular trafficking. A mouse model often contains an extra copy of only a particular set of corresponding genes rather than a complete human chromosome arrangement. The model's genotype must therefore be specified when interpreting what a cellular or behavioral result says about trisomy 21 in people.



Carlson and Birkett (2022), Fig. 15.8, p. 501.



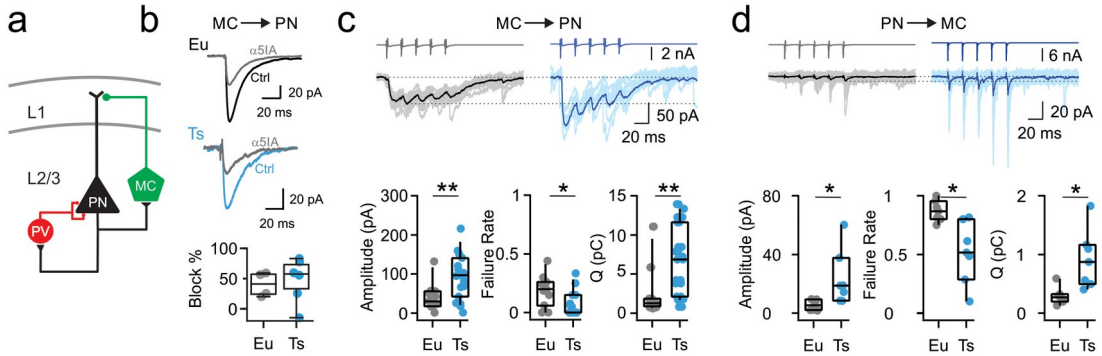
Jewett et al. (2023), fig2-Video 2.

One inhibitory pathway is enhanced in Ts65Dn mice

A study of Ts65Dn mice examined inhibition in prefrontal cortical microcircuits. Inhibitory interneurons release GABA onto other cells, reducing or shaping their activity under mature physiological conditions. The investigators compared distinct interneuron populations because a general claim of excess inhibition would not explain which synapses, receiving compartments, or cellular properties had changed.

The inhibitory loop between somatostatin positive Martinotti cells and pyramidal neurons was enhanced. Martinotti cells contact dendritic regions, where inhibition can regulate how incoming excitation influences the neuron. The increased synaptic inhibition occurred without a corresponding change in the basic excitability of those Martinotti cells, separating synaptic transmission from how readily the sending cell fired.

Paired recordings allowed the response in a receiving neuron to be measured when a connected interneuron was activated. This method identifies the functional strength of a particular connection. It provides a more specific account than inferring inhibition solely from the amount of GABA in a large piece of tissue or from the appearance of a regional brain scan.



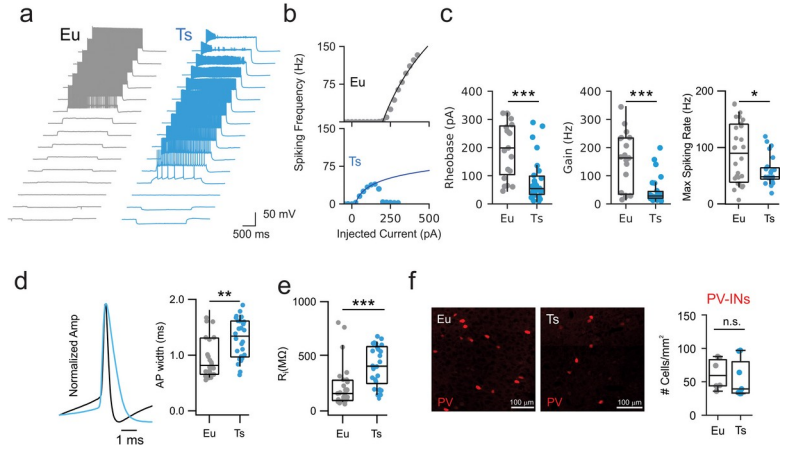
Zorrilla de San Martin et al. (2020), Fig. 1.

Interneuron changes depend on cell type

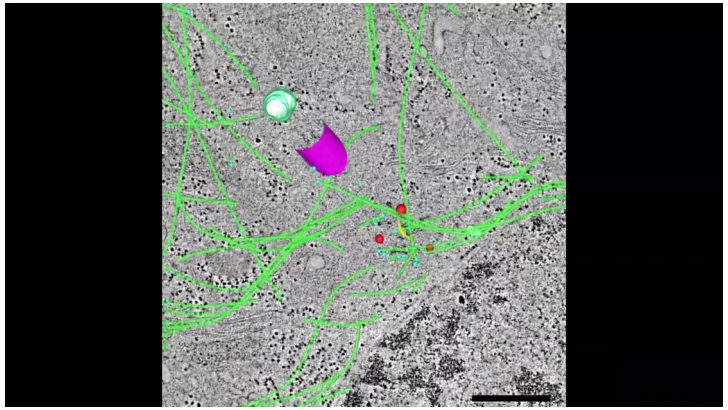
The Down syndrome model study also examined parvalbumin interneurons, which commonly provide inhibition near the cell body of pyramidal neurons. Their perisomatic synaptic inhibition was not enhanced in the same way as the Martinotti cell pathway. Instead, the parvalbumin cells lost aspects of their usual fast spiking behavior and showed increased excitability.

Excitability describes how a cell converts input current into action potentials, whereas synaptic strength describes how one cell's output influences its target. The two properties can change independently. Comparing them explained why an altered interneuron population did not necessarily produce the same increase at every inhibitory synapse in the circuit.

The combined microcircuit changes were associated with reduced pyramidal neuron firing and altered locking of activity to network rhythms in vivo. These results identify particular routes by which a genetic model changes cortical communication. They also explain why globally reducing inhibition is an imprecise therapeutic idea: the affected circuit contains several inhibitory populations with different physiological changes.



Zorrilla de San Martin et al. (2020), Fig. 4.



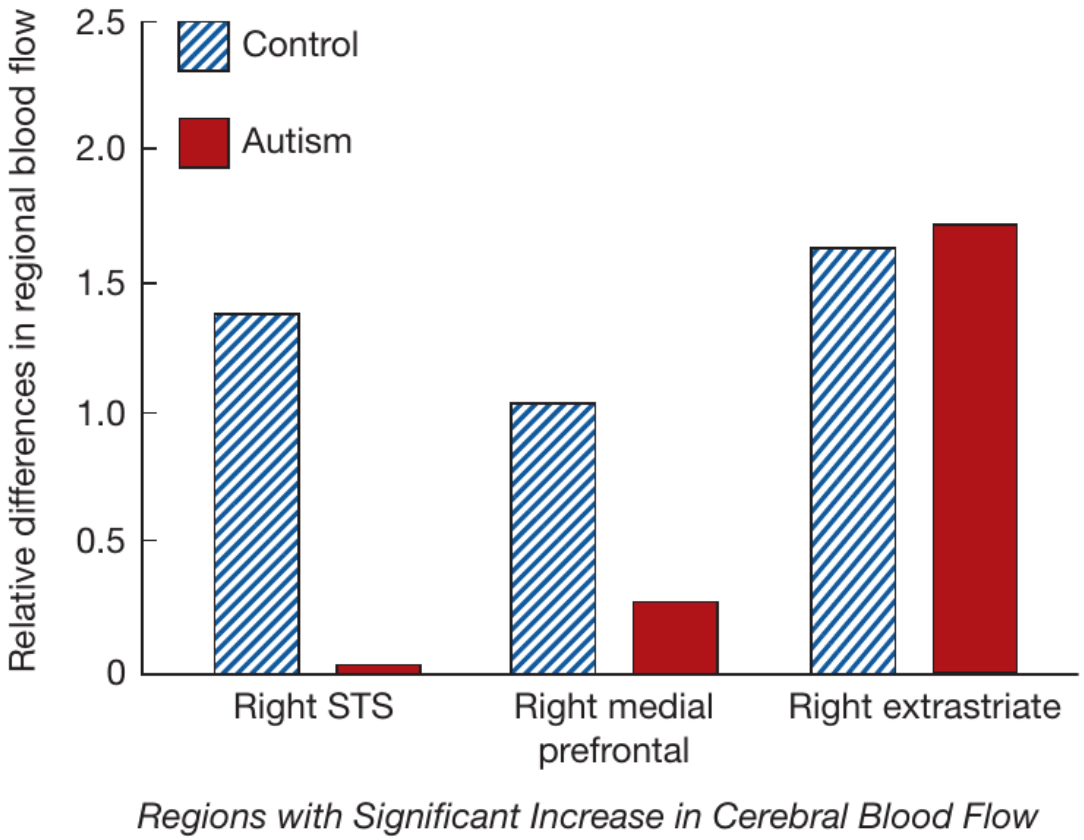
Jewett et al. (2023), fig2-Video 3.

Autism includes diverse developmental profiles

Autism is a neurodevelopmental condition involving differences in social communication together with restricted or repetitive patterns of behavior, interests, or activities. Sensory processing differences are also common. The presentation varies greatly, including differences in spoken language, intellectual abilities, movement, coexisting conditions, and the kind and amount of support a person needs in daily life.

A spectrum is not a single line from little to much autism. A person can have strong abilities in one domain and substantial difficulty in another, and their needs can change with context and development. Spoken fluency alone does not capture adaptive functioning, sensory burden, or the effort required to navigate social situations.

Biological research examines many levels, including genetic variation, precursor cell behavior, circuit physiology, perception, and interaction. No one cellular phenotype or brain activation pattern describes every autistic person. A finding in a defined genetic model can identify a mechanism worth studying while remaining distinct from a complete explanation of autism or an assessment of an individual's capabilities.



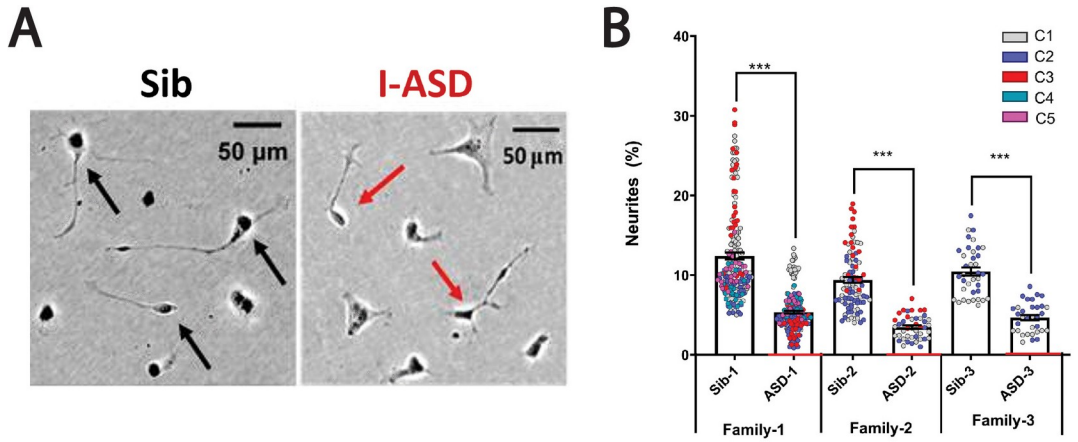
Carlson and Birkett (2022), Fig. 15.9, p. 505.

Autism derived cells can share growth changes

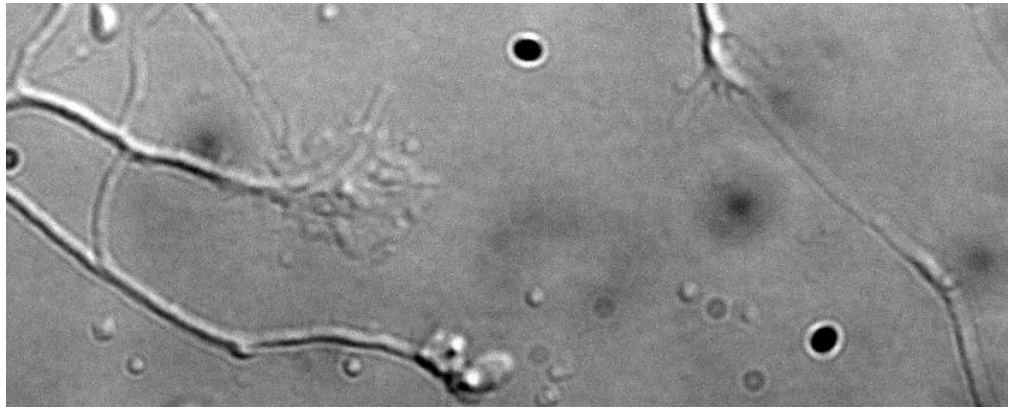
A study generated neural precursor cells from people with idiopathic autism and from people with a deletion at chromosome region 16p11.2. Idiopathic indicates that no single known cause explains the condition in those participants. The investigators asked whether genetically distinct groups could nevertheless show shared cellular changes under the same experimental culture conditions.

Both groups showed reduced neurite outgrowth and migration compared with their respective controls. A neurite is an extending cellular process that can later develop into an axon or dendrite. Measuring the fraction of cells with processes and their movement provided concrete developmental phenotypes rather than assigning a broad behavioral diagnosis directly to a cultured cell.

Genetic analysis did not identify a shared causative variant linking the two sampled groups. The common cellular behavior therefore motivated a search for convergent signaling pathways. Similar outcomes can arise through different upstream causes, and an experiment can test convergence without claiming that all people who share a diagnosis have identical genetic or cellular biology.



Prem et al. (2024), Fig. 1.



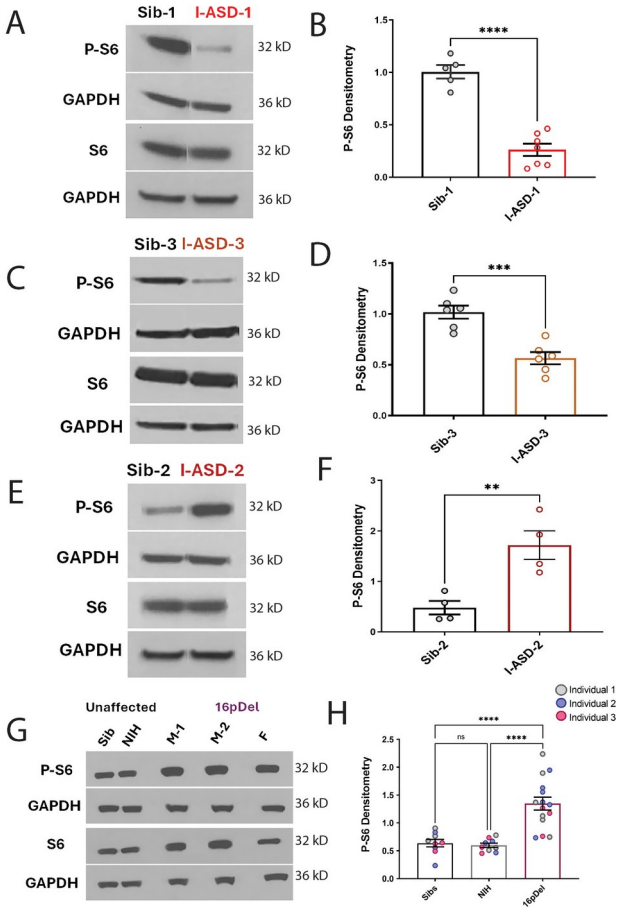
Dos-Santos Carvalho et al. (2020), Video 1.

mTOR modulation alters precursor development

The autism precursor study profiled protein phosphorylation to identify signaling changes shared across the sampled groups. Phosphorylation changes how proteins act or interact, so the measurements supplied information about pathway state beyond the amount of each protein present. The analysis implicated mTOR related signaling, a system already involved in cellular growth and cytoskeletal regulation.

The researchers then altered mTOR signaling experimentally. Modulation rescued the tested outgrowth and migration phenotypes in the affected cells, while related manipulations produced similar abnormalities in control cells. These experiments connected pathway dysregulation to the measured developmental behavior more directly than a comparison of phosphorylation levels alone could establish.

The findings concern the sampled precursor cells and the tested conditions, not a prescription for treating every autistic person by uniformly increasing or decreasing one pathway. Two idiopathic lines had reduced S6 phosphorylation, while another idiopathic line and the deletion lines had increased phosphorylation. The experiment identifies convergence on a regulatable cellular mechanism while preserving the distinction between cellular rescue and an established clinical benefit.



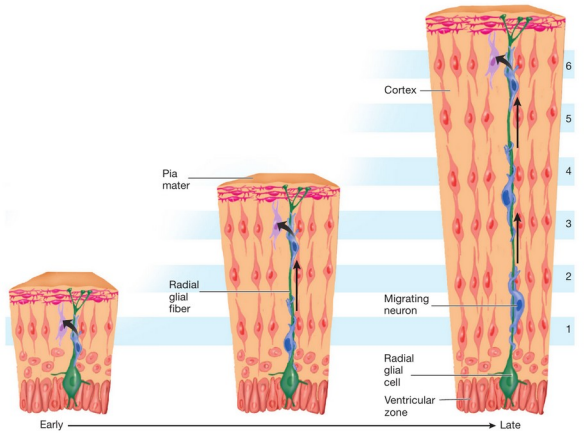
Prem et al. (2024), Fig. 4.

Autism genetics includes many routes to development

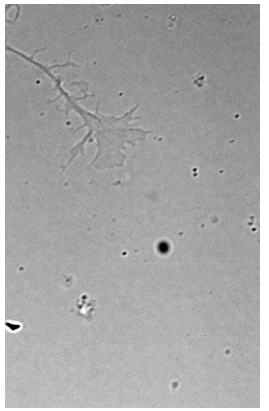
Inherited and newly arising genetic variants contribute to the probability of autism, with many different genes and genomic regions involved. Some variants have relatively large effects, while many common variants each contribute small influences. A genetic contribution does not mean that one test predicts a person's complete developmental trajectory or explains every feature of their experience.

Genes influence processes such as transcription, cell growth, synapse formation, and signaling, often in multiple tissues and developmental stages. A variant can therefore change the conditions under which circuits form without encoding a particular social behavior directly. The route from DNA to behavior passes through cellular and network processes that can be examined experimentally.

Environmental and experiential factors interact with those developmental processes, but interaction is not evidence for parental blame. Everyday parenting differences do not supply a biological explanation for autism. Research must identify an actual exposure, developmental interval, and mechanism rather than converting a broad statistical association into an unsupported story about how a family caused the condition.



Carlson and Birkett (2022), Fig. 15.3, p. 495.



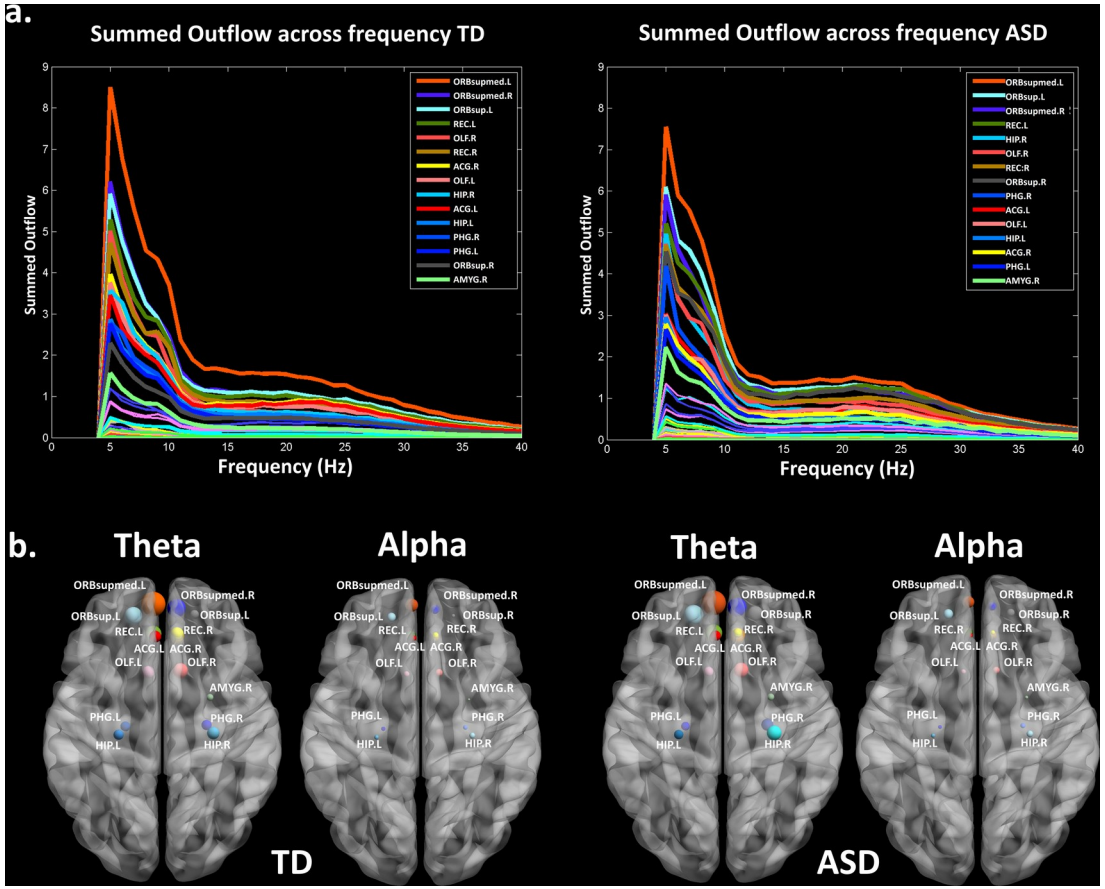
Dos-Santos Carvalho et al. (2020), Video 2.

Social scenes reveal early network differences in autism

A study recorded gaze and electrical brain activity in toddlers with autism and typically developing comparison children while they watched dynamic social scenes. The investigators examined directed functional interactions among estimated cortical sources. This approach asked how regions coordinated during viewing, rather than looking only for one brain area that responded more or less strongly to a face.

The groups differed in frequency specific network patterns, including theta and alpha bands. Some regions also differed in their estimated influence on other nodes. The finding concerned the organization of activity while children viewed the scenes, with gaze measurements documenting how their visual sampling of the available information varied during the same period.

Gaze is a behavior, while an estimated cortical source is a model based interpretation of electrical signals at the scalp. Neither directly measures a child's internal interest or empathy. Combining them can identify relationships between visual attention and neural coordination without assuming that reduced looking at one feature means that the child lacks an emotional response to other people.



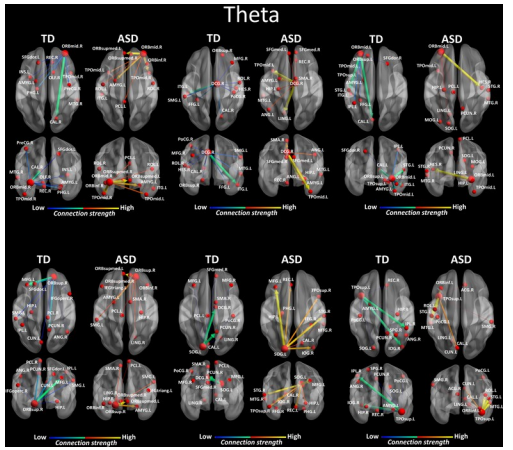
Sperdin et al. (2018), Fig. 1.

Network variation relates to gaze and clinical measures

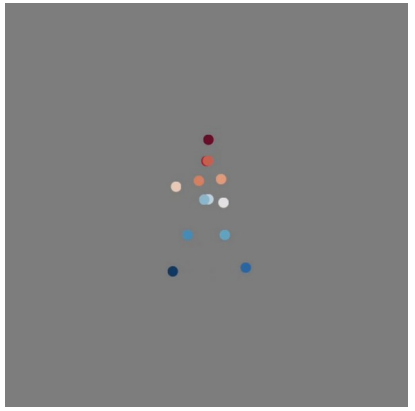
Within the autistic group, the social scene study related network measurements to gaze patterns and clinical measures. Activity involving dorsomedial frontal, inferior temporal, and insular regions was associated with less atypical gaze patterns and lower measured impairment. The result suggested that some patterns of coordination could contribute to adaptation within the sampled children rather than simply indexing a uniform deficit.

The analysis estimated directional functional connectivity, not direct anatomical axons or experimentally proven causal transmission between every pair of regions. Electrical source models and the chosen connectivity method influence the resulting network. The study therefore interpreted the pattern alongside observed behavior instead of treating a larger arrow in a network diagram as a newly discovered physical pathway.

The children varied within groups, and the measured relationships concerned a specific viewing situation. A social scene supplies information through motion, faces, voices, and context, so several routes can support engagement. Comparing network organization with actual gaze helps describe those routes while leaving the child's communication needs and abilities to be assessed through broader developmental observation.



Sperdin et al. (2018), Fig. 3.



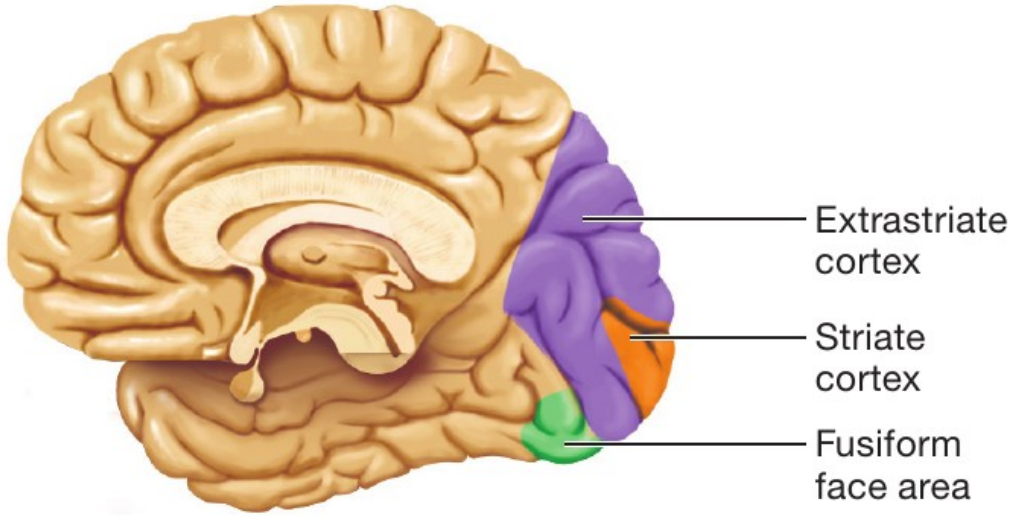
Tian et al. (2024), Video 1.

Context changes measured sensory responses

Sensory differences can influence the experience of sound, touch, light, and movement in autism and other developmental conditions. A stronger response in one situation can coexist with reduced discrimination or reduced tolerance in another. The measured behavior depends on stimulus intensity, predictability, attention, prior experience, and the decision required from the person or animal.

An experiment may ask whether a stimulus was detected, whether two stimuli differed, or which learned category a stimulus belongs to. These tasks involve different computations even when they use the same sensory channel. A change in choice behavior can therefore arise from sensory encoding, the weighting of recent information, or how the response rule is applied.

Model systems permit those components to be varied independently. A genetic mouse model can perform a calibrated tactile discrimination task while researchers manipulate stimulus strength and cognitive load. The relevant result is the specific pattern across conditions, rather than a blanket claim that a developmental condition makes all perception uniformly more sensitive or uniformly less accurate.



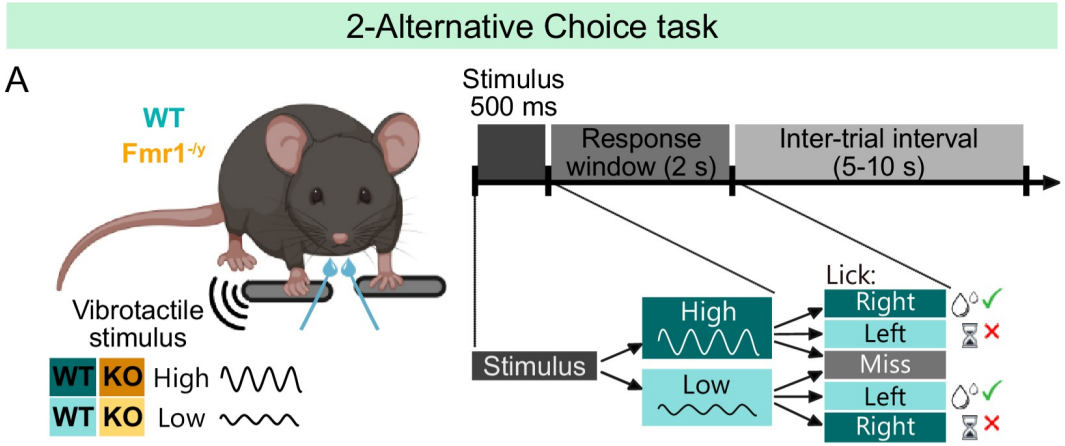
Carlson and Birkett (2022), Fig. 15.10, p. 506.

A tactile task separates bias from sensitivity

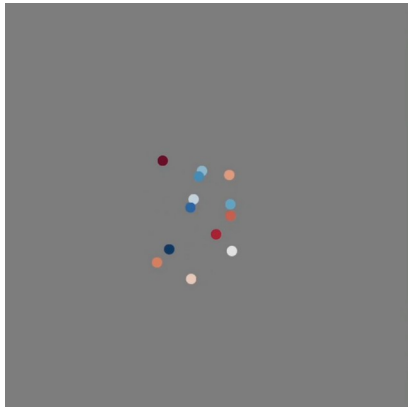
A study used male mice lacking Fmr1, a gene whose loss causes fragile X syndrome and is associated with autism related features. The animals learned to classify forepaw vibrations by licking at different response ports. Stimulus strength and the choice rule were controlled so that sensory discrimination could be compared with the effects of prior choices and task demands.

The mutant mice showed a stronger choice consistency bias in low salience trials during training. Salience refers to how noticeable or behaviorally prominent the signal is. A consistency bias means that choices tend to repeat in a particular way, so the resulting accuracy can differ even when the physical sensory input is held comparable.

The preparation measures a defined behavior in a genetic animal model rather than asking a mouse to report a human sensory experience verbally. Its value lies in separating controllable task components. Researchers can then determine whether performance changes with signal intensity, recent history, or the amount of information the animal must retain and use.



Semelidou et al. (2026), Fig. 1.



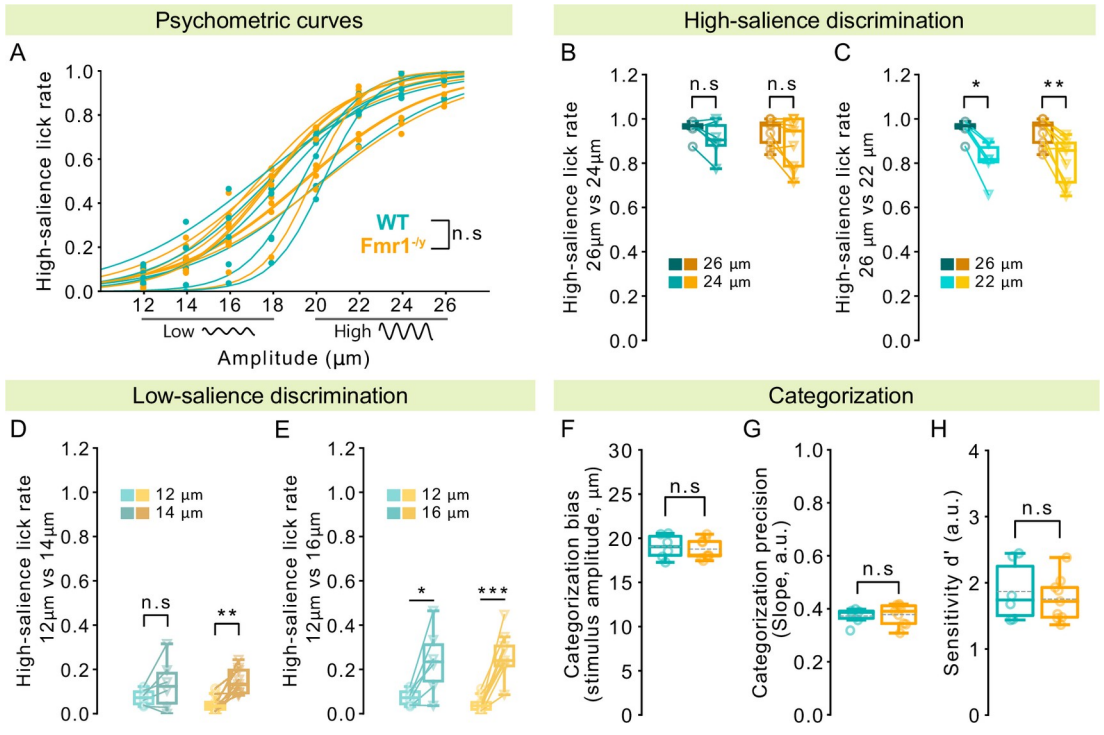
Tian et al. (2024), Video 2.

Tactile discrimination changes with category boundaries

The Fmr1 study tested intermediate vibration intensities after the animals learned the initial categories. Mutant mice discriminated some low salience stimuli more effectively but showed less facilitation for comparisons crossing the learned category boundary. The phenotype therefore combined differences that would be missed by describing the animals as simply more sensitive or generally impaired.

The mice also integrated sensory history differently, and attention related difficulties emerged when cognitive load increased. These effects were distinguishable from the basic discrimination pattern. A response to the present stimulus can therefore reflect both current sensory evidence and how strongly earlier events or task rules influence the decision made from that evidence.

The source psychometric curves show how response probability changes across stimulus intensity, while additional panels compare learning and categorization measures. The experiment identifies context dependent weighting of information in this genetic model. It does not establish that the same perceptual profile occurs in every autistic person or that one threshold summarizes the full sensory experience of the condition.



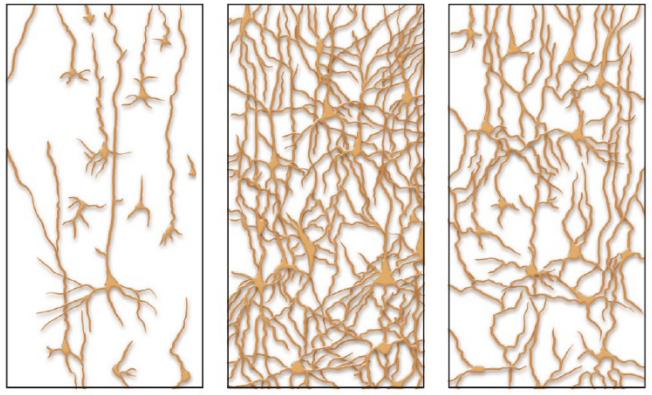
Semelidou et al. (2026), Fig. 3.

Excitation and inhibition develop within specific circuits

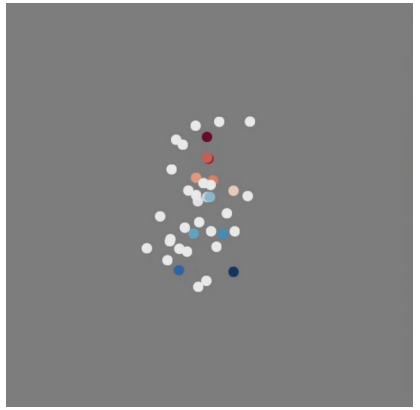
Excitatory and inhibitory neurons develop through partly different origins and schedules before becoming integrated into shared circuits. Excitatory synapses often use glutamate, while many inhibitory interneurons use GABA. Their effects depend on receptors, cellular chloride regulation, connection location, and developmental stage rather than transmitter identity alone determining the same response throughout life.

Circuit balance is therefore not one fixed ratio measured across the whole brain. Inhibition near a cell body can shape spike timing differently from inhibition on a distant dendrite. Different interneuron classes can also change independently, as shown by microcircuit recordings in developmental genetic models. A regional average can conceal those cell type and compartment specific effects.

Behavioral consequences depend on the computation performed by the affected circuit. A cerebellar pathway that learns the timing of a blink differs from a cortical pathway that maintains a task rule. Studying a defined learning or sensory behavior allows an altered cellular property to be connected to a particular circuit operation rather than being used as a general explanation for every developmental symptom.



Carlson and Birkett (2022), Fig. 15.4, p. 496.



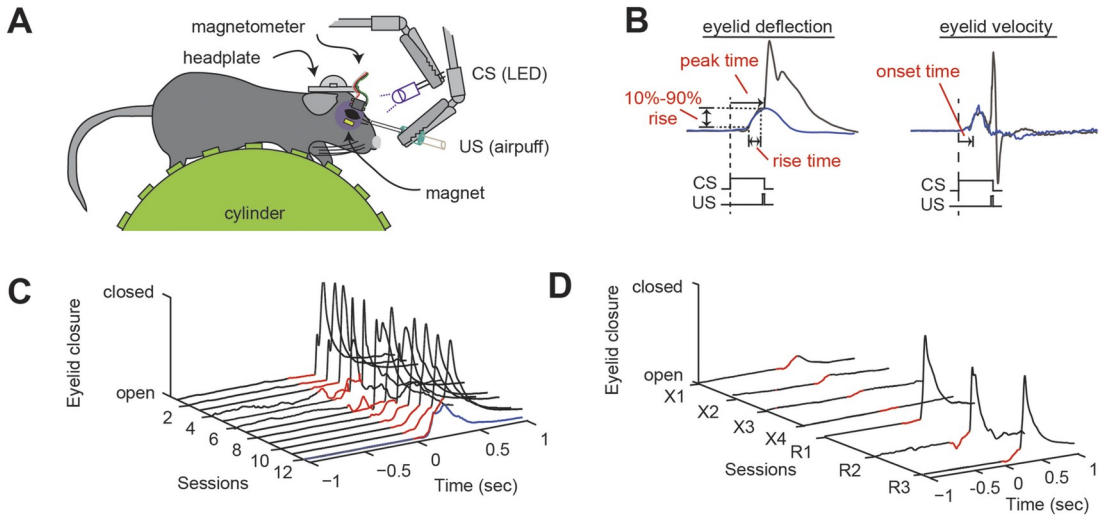
Tian et al. (2024), Video 3.

Eyeblink conditioning tests cerebellar sensory learning

A study compared five mouse models carrying genetic changes associated with autism related biology. The animals underwent delay eyeblink conditioning, in which a neutral cue overlaps with a later air puff to the eye. With training, the cue can evoke a blink timed to precede the puff, providing a measurable form of associative sensory learning.

The cerebellum contributes to learning and timing this response through interacting pathways that convey sensory input and teaching signals. The experiment measured whether a blink occurred, its size, and its timing. These are distinct response properties: an animal might learn to blink on many trials but produce a smaller or poorly timed closure.

Head fixation and a movement platform allowed the eyelid response to be measured consistently while the animal could still move its body. The preparation therefore tested a specific learned response rather than unrestricted social behavior. Comparing several genetic models asked which components of the same cerebellar learning task were shared and which varied with the biological alteration.



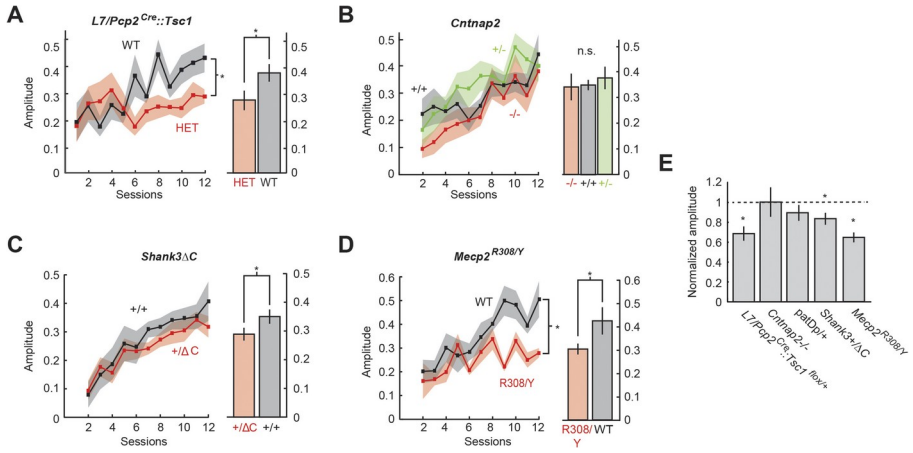
Kloth et al. (2015), Fig. 1.

Genetic models alter different components of learning

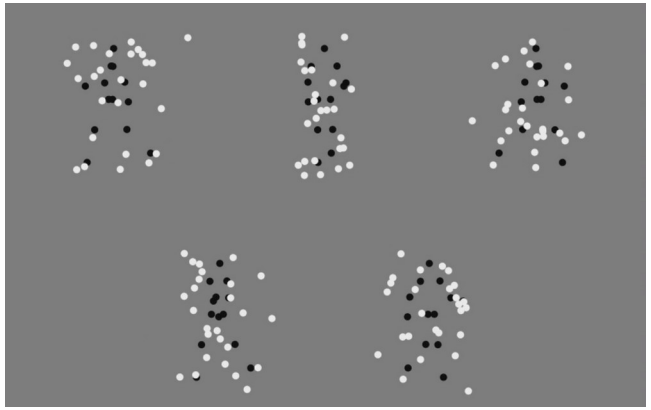
The eyeblink study found that the tested models differed in the probability, amplitude, and timing of learned responses. Several showed reduced response probability, while a partly overlapping set showed smaller blinks. Shank3 and Mecp2 related models also showed abnormal timing and reduced dendritic spine density in Purkinje cells, major output neurons of cerebellar cortex.

These differences were interpreted in relation to the expression of the affected genes in cerebellar pathways. A change in the teaching loop can influence whether learning develops, while a change in the representation of sensory input can affect the size or timing of the response. The experiment therefore separated components of performance rather than treating all deviations as one identical learning defect.

The original plots preserve the individual model comparisons and response measures. The study shows that related developmental phenotypes can converge on cerebellar learning while retaining different circuit signatures. It does not mean that a mouse's blink response reproduces the complete social or sensory profile of a person with the corresponding genetic condition.



Kloth et al. (2015), Fig. 4.



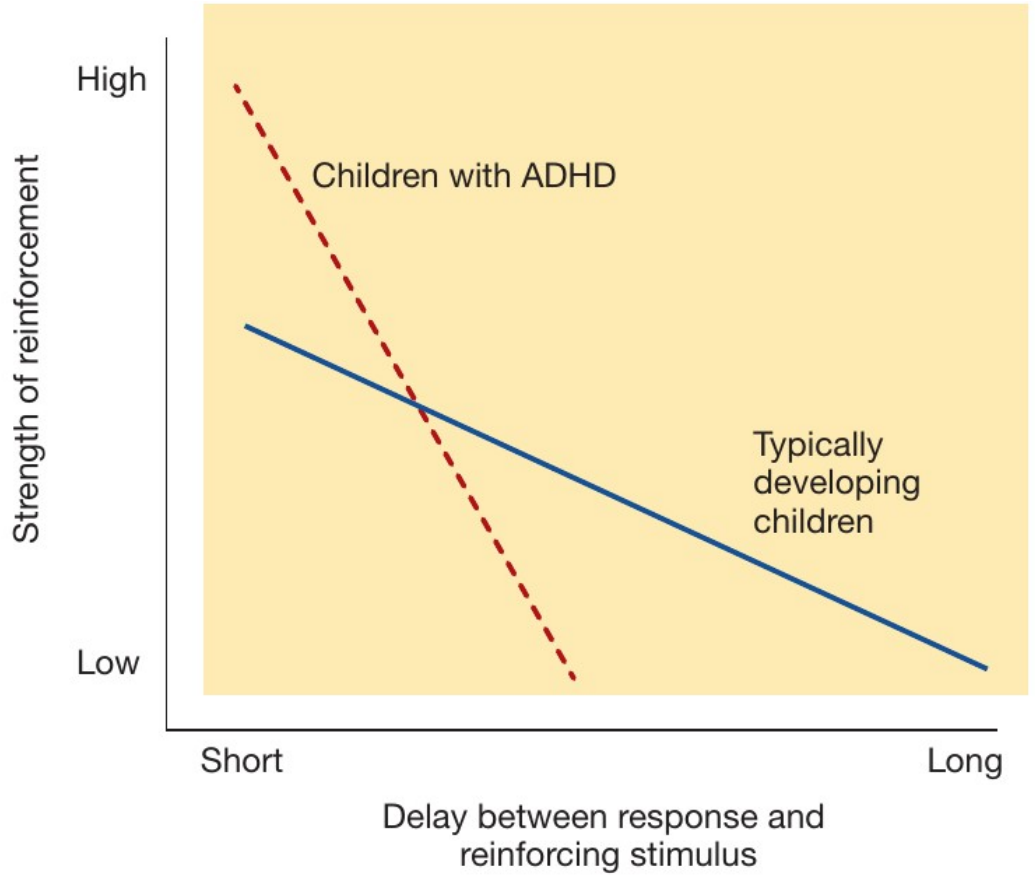
Tian et al. (2024), Video 4.

ADHD affects attention activity and impulse regulation

Attention deficit hyperactivity disorder is a neurodevelopmental condition involving persistent patterns of inattention and or hyperactivity and impulsivity that interfere with functioning. Presentation varies across people and settings. Difficulty sustaining attention to a required task can coexist with prolonged engagement in an activity that provides strong interest or immediate feedback.

Attention requires selecting useful information, maintaining a goal, monitoring errors, and adjusting behavior as conditions change. Hyperactivity and impulsivity concern regulation of movement and responses rather than an absence of knowledge about what should be done. The same person can understand an instruction while having difficulty keeping it active and applying it consistently over time.

Assessment considers developmental history, impairment, and behavior across situations rather than relying on one brain scan or one moment of restlessness. Sleep, anxiety, learning difficulties, and other conditions can influence similar behaviors. Biological studies examine the systems contributing to regulation while clinical assessment determines whether the person's pattern meets the developmental and functional requirements of the diagnosis.



Carlson and Birkett (2022), Fig. 15.12, p. 509.

Context changes anatomical symptom associations

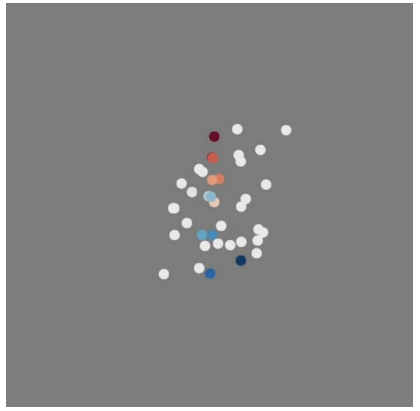
A study examined relationships between ADHD symptoms and cortical anatomy in two large child cohorts. The researchers measured cortical surface area, thickness, and volume from magnetic resonance images. They then asked how the relationships changed when analyses accounted for additional developmental and family characteristics that could influence both reported symptoms and brain measurements.

Initial analyses adjusting mainly for demographic variables identified widespread associations. Adding socioeconomic and maternal behavioral factors reduced many of those relationships. The measured association between symptoms and a cortical property therefore depended partly on other variables included in the analysis, rather than identifying one anatomical pattern that remained equally strong under every comparison.

The result concerns population level relationships, not whether a particular child has ADHD based on the appearance of a scan. Cortical measurements vary widely among children with and without symptoms. Understanding how those measurements relate to behavior requires identifying the developmental and environmental context rather than treating every correlated image feature as a direct cause.



Dall'Aglio et al. (2022), Fig. 1.



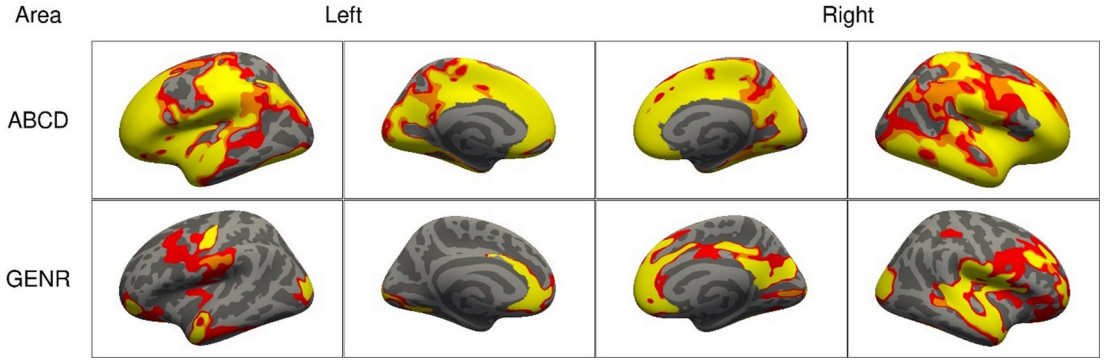
Tian et al. (2024), Video 5.

Adjustment can change the apparent anatomical pattern

The ADHD morphology study compared maps produced by several sets of statistical adjustments. Some previously widespread surface area associations became smaller after additional factors were considered. This does not mean that the brain is unrelated to attention; it means that the observed map combined multiple sources of developmental variation that needed to be separated more carefully.

Head motion and measured intelligence posed additional interpretation problems because they can be related to both symptoms and the imaging outcome. Correcting for a variable is not automatically neutral if that variable lies along a relevant developmental pathway. The study therefore examined how different choices changed the result instead of presenting one adjusted map as a complete biological explanation.

The anatomical lesson is specific: a group association depends on the comparison being made and the factors that could create it. Direct behavioral and clinical assessment remains necessary for the child. Imaging research contributes by testing reproducible relationships with development, while the maps describe those relationships rather than a unique boundary between diagnostic groups.



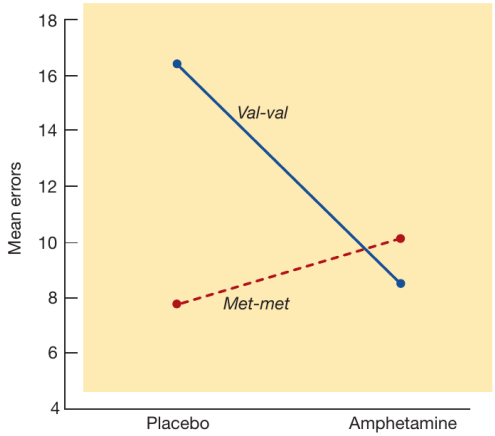
Dall'Aglio et al. (2022), Fig. 3.

Prefrontal circuits help maintain goals across delays

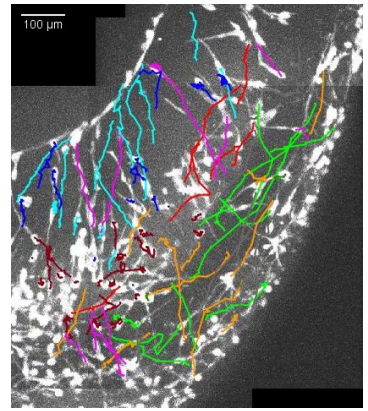
Working memory temporarily maintains and uses information needed for the current task. Prefrontal circuits contribute to keeping a rule or goal available when it is not continuously visible. This allows behavior to depend on a remembered instruction rather than only on the most recent or most immediately rewarding stimulus in the environment.

The prefrontal cortex interacts with parietal, sensory, and striatal systems to select information and regulate responses. A failure can involve losing the goal, being distracted by competing input, or acting before the relevant information has been evaluated. These are related but separable operations, so one attention test does not measure every function involved in everyday planning.

Dopamine and norepinephrine modulate these circuits through receptors that change cellular responsiveness and network dynamics. Their effects depend on the amount of stimulation and the receiving cell population. A developmental difference in regulation cannot be reduced to the claim that the entire brain simply contains too little of one transmitter at every time and location.



Carlson and Birkett (2022), Fig. 15.13, p. 509.



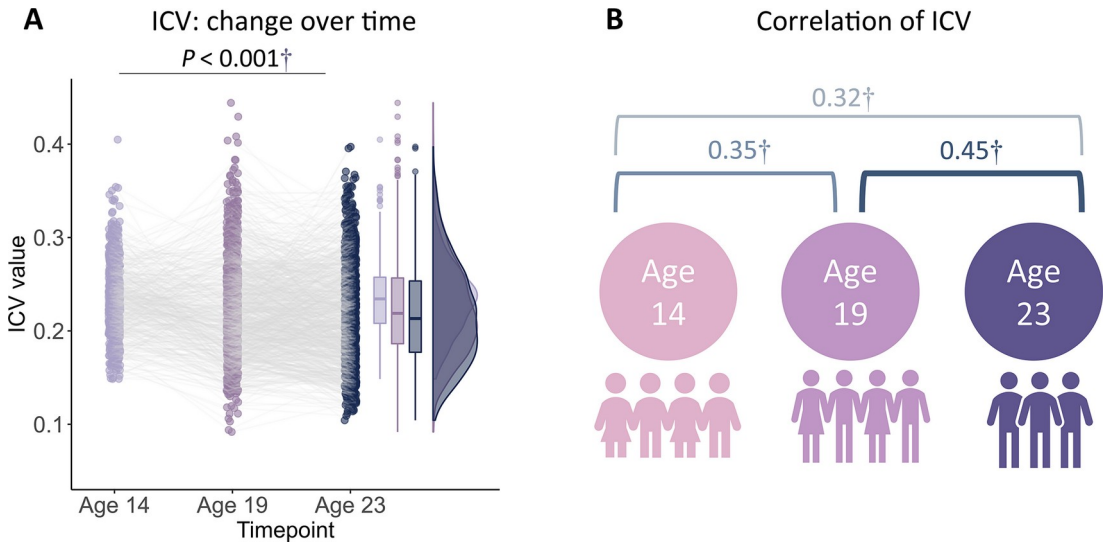
Pedraza et al. (2025), fig6-Video 2.

Sustained attention changes across adolescence

A longitudinal study measured sustained attention and brain connectivity from adolescence into early adulthood. Sustained attention is the ability to maintain consistent task engagement over time. The researchers followed participants across multiple ages rather than inferring the whole developmental trajectory from separate groups tested only once at different stages of life.

The behavioral measure included variability in response timing within a person. Greater variability can indicate that performance fluctuates across trials, even if the average response speed looks acceptable. Measuring consistency therefore captures an aspect of attention that would be lost by considering only the mean reaction time or total number of correct responses.

Individual differences showed continuity across developmental assessments alongside broader changes with age. This means that maturation and stable variation can coexist: performance can improve across the group while some people remain relatively more variable than others. The study's age related findings concerned a measured task, not a fixed birthday on which attention becomes fully mature.



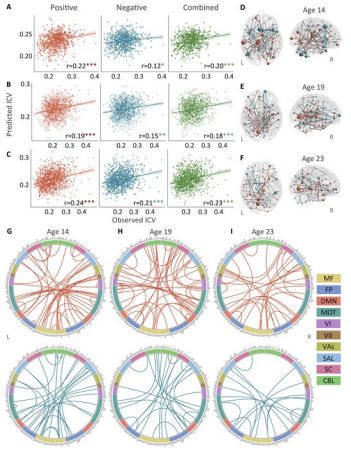
Weng et al. (2024), Fig. 1.

Distributed connectivity predicts attention variability

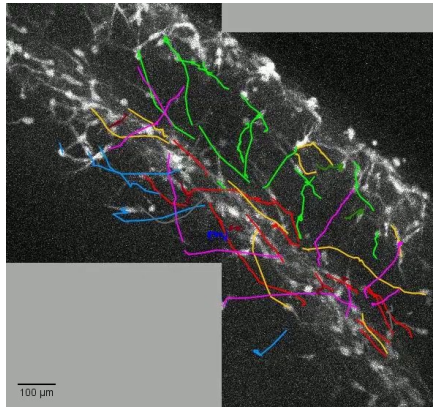
The sustained attention study used patterns of connectivity across the brain to predict individual differences in response timing variability. A predictive network combines information from multiple region pairs rather than searching for one uniquely responsible attention center. The reported network generalized across developmental assessments and to an external dataset, testing whether its information extended beyond one particular sample.

Poorer sustained attention and related network measures in adolescence also predicted later increases in some substance use behaviors. A prospective relationship establishes that the predictor was measured before the later outcome, but it does not prove that the attention difference alone caused that outcome. Developmental behavior reflects additional social, environmental, and biological influences.

The finding supplies a network level complement to cell and synapse studies. Consistency of attention can be related to coordinated activity distributed across regions, while neurotransmitter systems influence how those regions operate. These levels describe different measurements of the developing system and should not be collapsed into a claim that one connectivity value reveals a person’s future behavior.



Weng et al. (2024), Fig. 3.



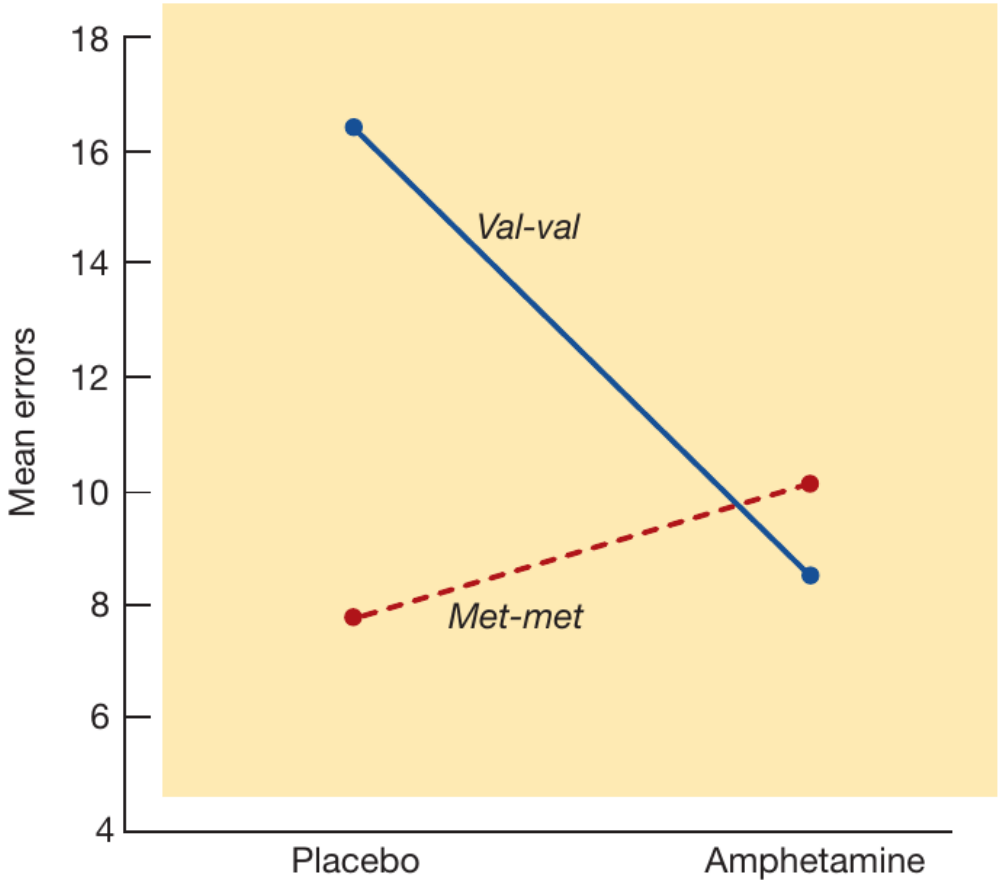
Pedraza et al. (2025), fig6-Video 3.

Catecholamines modulate control circuits

Dopamine and norepinephrine are catecholamine transmitters that regulate activity in several brain systems. In prefrontal circuits, their receptors influence how neurons respond to inputs and maintain useful patterns of activity. The effect depends on receptor type, location, and ongoing state, so increasing transmitter action can improve performance under some conditions and disrupt it under others.

Methylphenidate blocks transporters that normally remove dopamine and norepinephrine from extracellular space. This prolongs aspects of transmitter availability after release. Amphetamine acts through additional mechanisms affecting catecholamine handling. These drugs can reduce ADHD symptoms in appropriate clinical use, but their effectiveness does not prove that every person with ADHD shares one uniform transmitter deficiency.

The original COMT comparison illustrates how baseline biology can change a drug's effect on a working memory task. COMT is an enzyme that contributes to catecholamine breakdown. Different genetic variants can influence the starting condition, so an intervention that moves one system toward useful signaling levels can move another system away from its effective range.



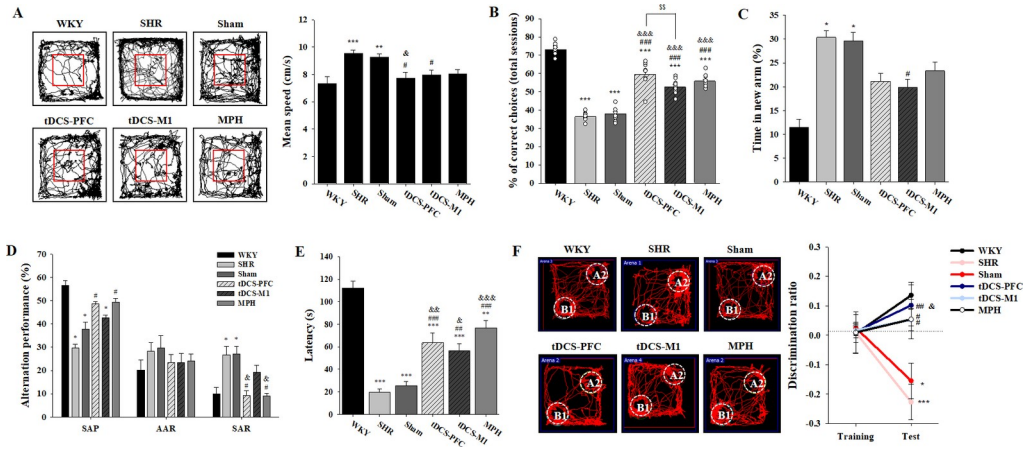
Carlson and Birkett (2022), Fig. 15.13, p. 509.

Prefrontal stimulation changes behavior in a rat model

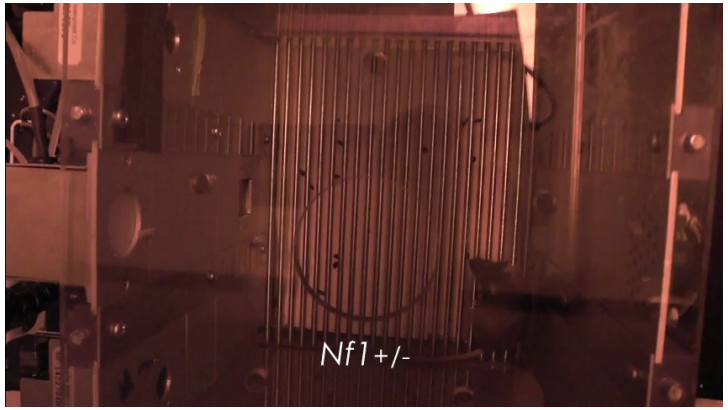
A study used spontaneously hypertensive rats as a model for selected ADHD related behavioral features. The animals received repeated anodal high definition transcranial direct current stimulation targeted over prefrontal cortex. This method applies a weak electrical current through external electrodes to alter neural excitability and related plasticity, rather than implanting an electrode to trigger each individual action potential.

The treated rats improved on tested cognitive tasks and showed reduced hyperactivity in the open field comparison. A sham condition controlled for handling and the stimulation procedure without the same effective current. Measuring both performance and movement helped distinguish a change in the required task from a general increase in the amount the animal moved.

The model represents selected traits and a particular rat strain, not the full human diagnosis. The experiment's concrete claim is that this stimulation schedule changed specified behaviors under its tested conditions. It does not establish an equivalent effective dose or clinical recommendation for children, whose anatomy, development, and treatment evaluation differ from the animal preparation.



Jung et al. (2020), Fig. 1.



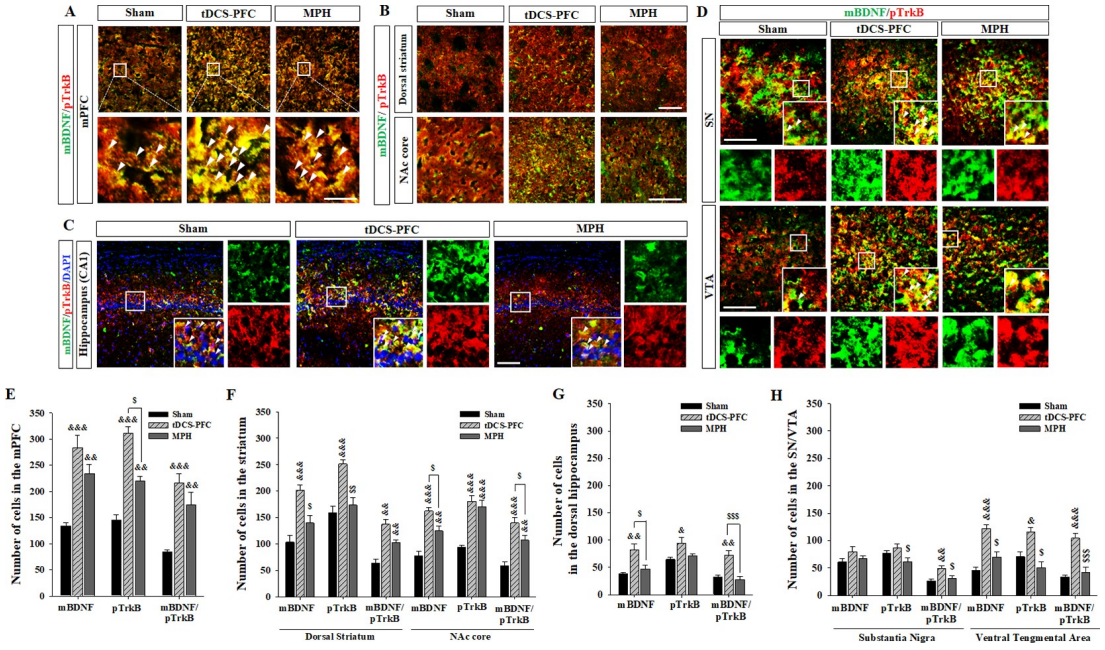
Robinson et al. (2019), Video 1.

Stimulation alters dopamine related and trophic proteins

The rat stimulation study measured molecular changes alongside behavior. Prefrontal stimulation increased markers related to dopamine synthesis and vesicular storage while reducing the dopamine transporter in the reported comparisons. These proteins influence different stages: synthesis produces transmitter, vesicles store it for release, and the transporter contributes to removing it from extracellular space.

The investigators also found changes in brain derived neurotrophic factor, or BDNF, and its receptor related signaling, together with increased hippocampal neurogenesis. BDNF is a signaling protein that can support neuronal development and plasticity through receptor activation. The source micrographs show measured protein labeling in defined regions rather than a direct movie of transmitter molecules crossing a synapse.

These changes supplied candidate mechanisms associated with the behavioral response, but observing several markers after treatment does not identify which one was indispensable for improvement. A necessity test would selectively block a proposed step while evaluating the outcome. The reported experiment established a coordinated response to stimulation in the model and identified specific biological processes for further testing.



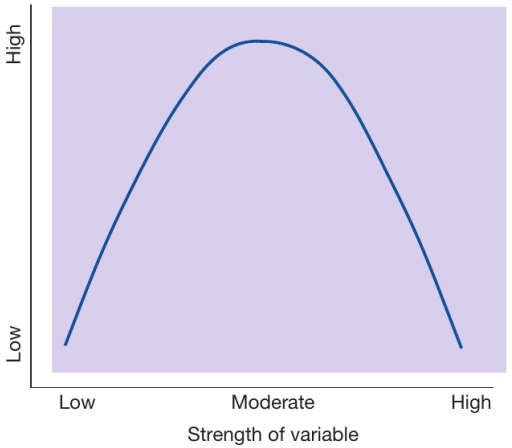
Jung et al. (2020), Fig. 4.

Modulation has an effective intermediate range

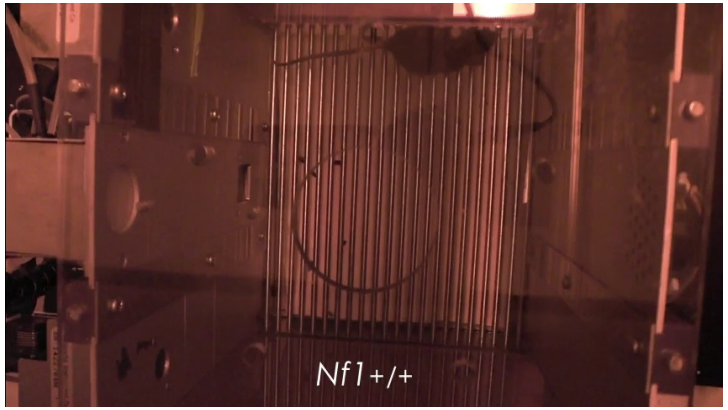
An inverted U relationship means that performance is best within an intermediate range of a variable and poorer at lower and higher values. Catecholamine effects on some prefrontal tasks can follow this pattern. Insufficient receptor stimulation can leave task relevant activity weak, while excessive stimulation can destabilize or suppress the patterns needed for effective control.

The useful range depends on the receptor, circuit, task, and starting state. A dose that improves one person's attention need not produce the same effect in another person or on a different task. This explains why treatment response is measured and adjusted clinically rather than inferred from the rule that more transmitter must always produce better concentration.

The curve is a conceptual description, not a measurement of one universal brain dopamine level. It also does not mean that every behavior has the same optimal point. Its biological value is to make the direction of an intervention conditional on the system's current state, replacing a simple excess or deficiency account with a testable dose and response relationship.



Carlson and Birkett (2022), Fig. 15.14, p. 510.



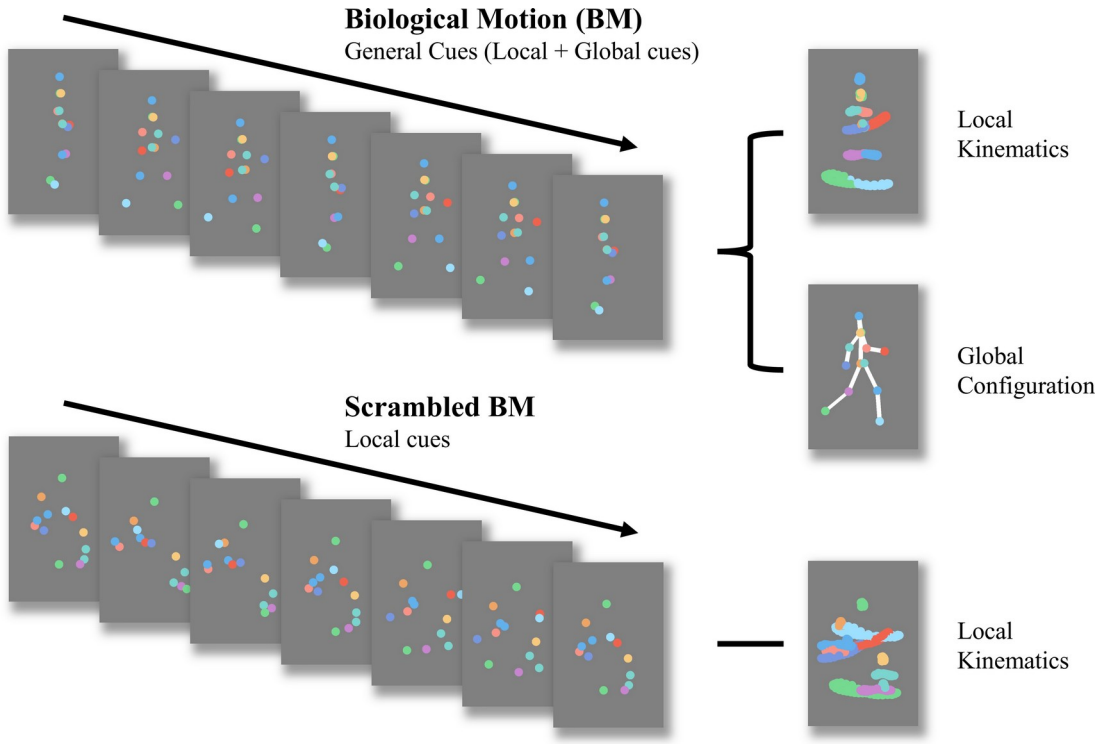
Robinson et al. (2019), Video 2.

Biological motion combines movement and form

A study compared biological motion perception in children with ADHD and typically developing peers. Point light walkers depicted movement using dots at major body joints. An intact arrangement preserves the configuration of a walking body, while scrambling the positions disrupts the overall body form but can retain information in the movement trajectories of individual dots.

The researchers designed tasks to separate local kinematics from global configuration and from performance when distracting dots were present. Kinematics describes movement through space and time. These comparisons asked whether difficulty concerned the paths of individual joints, their organization into a coherent body, or the selection of relevant motion from competing visual information.

The published videos are the study's original stimulus demonstrations, not recordings of participating children. Their point light patterns show exactly which information the tasks made available. This distinction keeps the visual example tied to the experimental manipulation without presenting a designed stimulus as a direct observation of a child's neural activity or everyday social behavior.



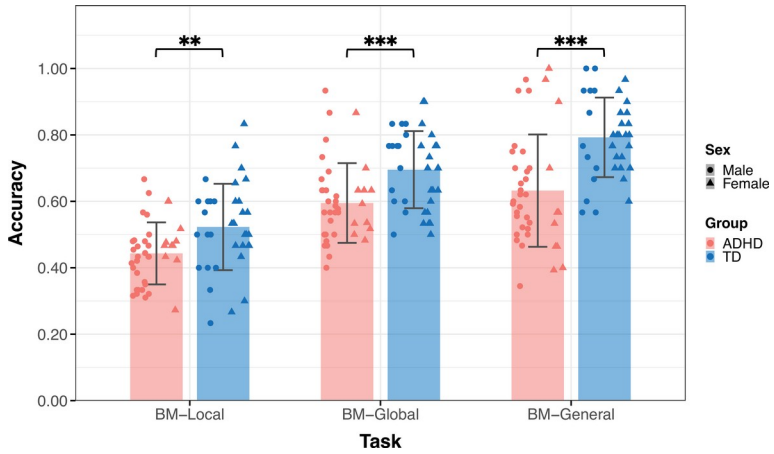
Tian et al. (2024), Fig. 1.

Motion perception reflects attention and development

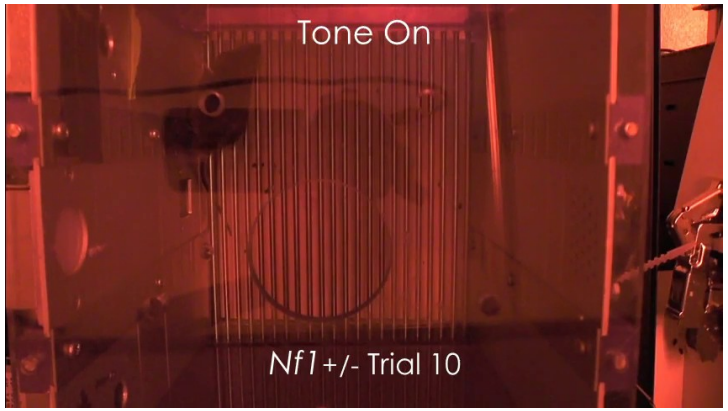
Children with ADHD performed less accurately across the reported biological motion tasks, but local and global processing had different relationships with other measures. Local processing related to social interaction measures, whereas global processing improved with age. The findings therefore separated components that would be obscured by reporting one combined score for social perception.

Sustained attention also related to overall biological motion performance, with reasoning ability involved in the reported relationship. A child must maintain the task rule and select relevant dots as well as perceive their movement. Performance consequently reflects interactions among sensory analysis, attention, and decision demands rather than a pure readout of one isolated visual detector.

The original task videos make the differences among intact, scrambled, and masked displays visible. Comparing those conditions helps specify what information a participant used. The results concern group patterns on controlled tasks and do not imply that every child with ADHD has the same difficulty interpreting people's actions in all real world situations.



Tian et al. (2024), Fig. 3.



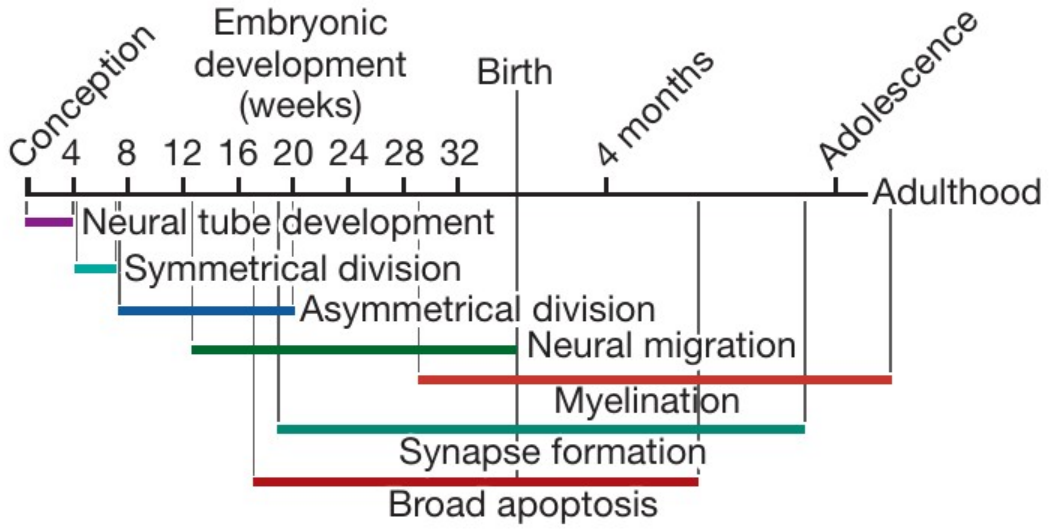
Robinson et al. (2019), Video 3.

Development reflects interacting probabilities

Genetic inheritance, prenatal conditions, experience, health, and social context can influence development through different biological routes. Some changes strongly alter a particular pathway, while others shift probability by a smaller amount. A risk factor is therefore not equivalent to an inevitable outcome, and the absence of a recognized risk factor does not establish that development will follow one uniform course.

The same molecular pathway can participate in several developmental conditions because it regulates a basic process such as cell division, migration, or synapse formation. Conversely, people with the same diagnosis can reach a similar behavioral difficulty through different cellular mechanisms. These two patterns explain why diagnostic categories and molecular mechanisms do not map onto each other one to one.

Research tests these possibilities by measuring intermediate steps and comparing defined conditions. A manipulated receptor can change cell movement, an altered circuit can change learning, and a developmental history can change measured perception. The explanation becomes stronger when the actual sequence from biological change to measured outcome is demonstrated rather than supplied by a broad label.



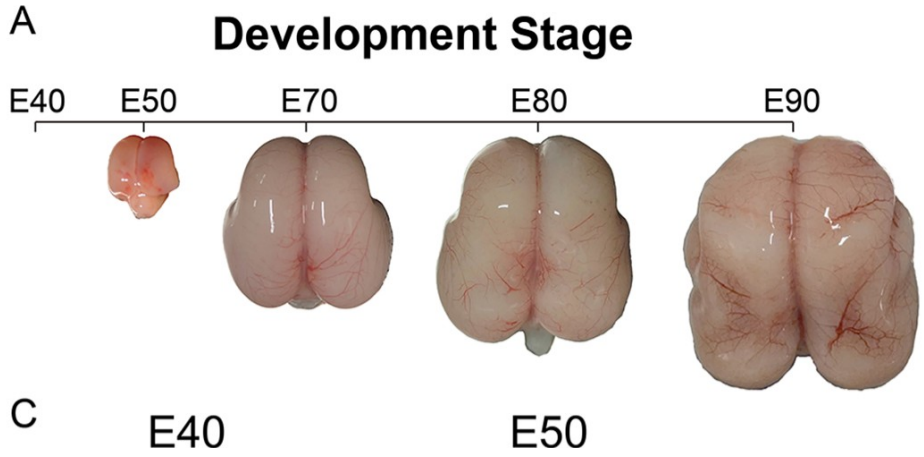
Carlson and Birkett (2022), Fig. 15.5, p. 497.

Single cell profiles reveal changing cortical populations

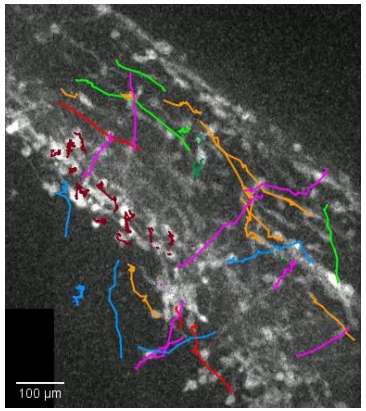
A study sampled developing macaque parietal cortex across prenatal stages and measured gene expression in individual cells. Single cell RNA sequencing estimates which genes are being transcribed in each sampled cell. By grouping related expression profiles, the researchers identified progenitor and neuronal populations and examined how their proportions and molecular states changed during cortical neurogenesis.

The samples spanned the period when deep and superficial cortical layers are being generated. Deep layer neurons appeared earlier, while upper layer neurons became prominent later, consistent with the developmental sequence of cortical construction. The analysis therefore provided molecular information alongside a well defined anatomical and temporal pattern rather than treating all fetal cortical cells as one undifferentiated population.

The published cell clusters are computational summaries of measured RNA profiles, not literal islands of cells floating in the brain. Their positions represent similarities under the analysis used to organize the data. Interpreting the clusters requires identifying the cell markers and sampled developmental stages, while tissue anatomy supplies the location in which those populations actually developed.



Xu et al. (2024), Fig. 1.



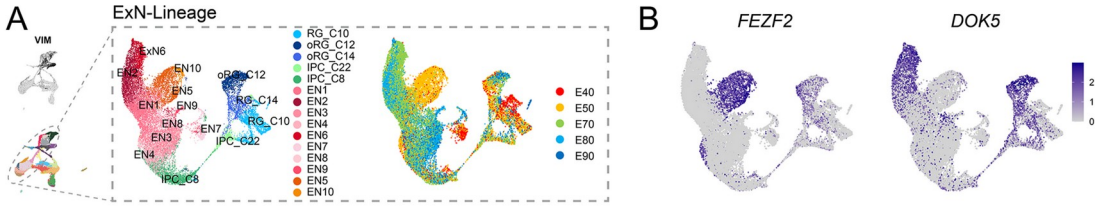
Pedraza et al. (2025), fig6-Video 4.

Progenitor programs diverge during development

The macaque transcriptomic study reconstructed developmental trajectories from progenitors toward neuronal identities. These trajectories summarize ordered changes in gene expression across sampled cells. They are not time lapse recordings of every individual cell, but they can identify candidate transitions when supported by the known developmental stages and the expression of appropriate identity markers.

The researchers found overlapping differentiation programs for deep and upper layer lineages and evidence that aspects of fate became determined after the final cell division. They also compared ventricular and outer radial glia, identifying differences in developmental signaling pathways including BMP, FGF, and WNT. These pathways regulate cellular behavior through receptors and changes in gene expression.

The findings add molecular detail to the radial migration diagram: cells produced at different times acquire different programs as they become integrated into layers. Gene expression and location must be interpreted together. A cell's current transcript profile can indicate its developmental state without proving that one expressed gene alone fixed its entire future identity and connectivity.



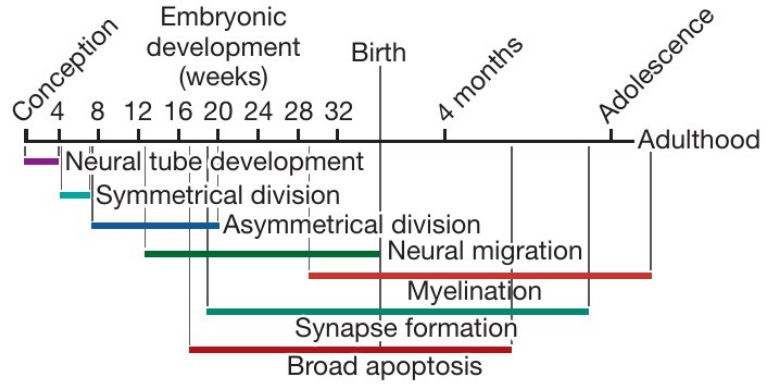
Xu et al. (2024), Fig. 4.

Development continues through adaptation and support

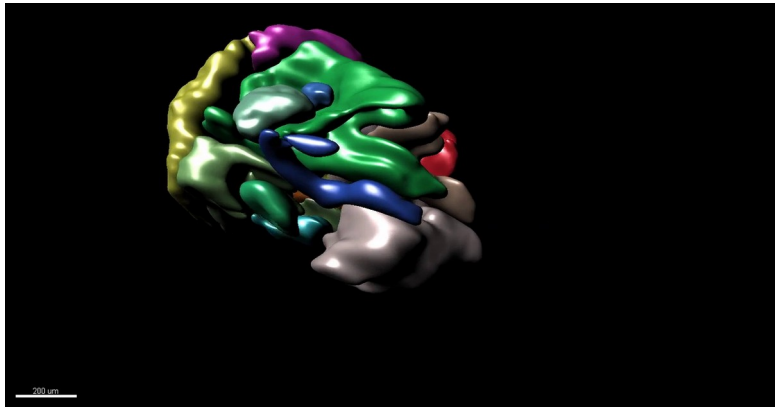
Neural development continues as circuits acquire experience, myelination progresses, and connections are strengthened or removed. The rate and form of change differ across regions and abilities. Continued plasticity provides opportunities for learning and adaptation, while persistent developmental differences can remain important rather than disappearing automatically when a person reaches adulthood.

Support can change the demands placed on a developing system and improve access to communication, education, and everyday activities. A quieter sensory environment, a clearer task structure, or an accessible language route can improve performance without requiring an immediate change in a genetic variant. Functional improvement can arise by changing the interaction between the person's abilities and the environment.

Biological treatments target particular mechanisms when evidence supports them, while developmental services address specific skills and participation. These approaches operate at different levels and can be complementary. Evaluating them requires measuring the outcome that matters, such as metabolic control, task performance, communication access, or daily functioning, rather than assuming that a normalized laboratory marker guarantees a useful change in a person's life.



Carlson and Birkett (2022), Fig. 15.5, p. 497.



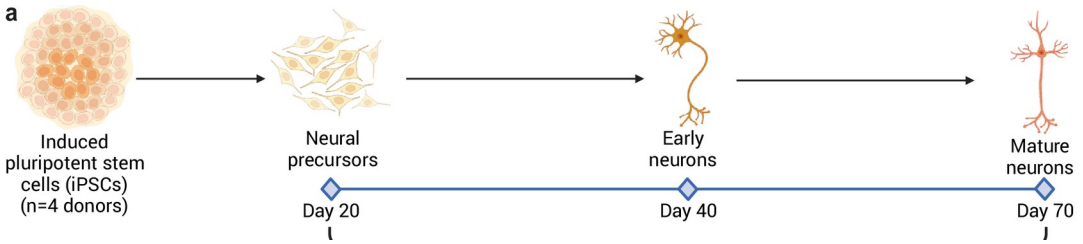
Yildirim et al. (2022), fig4-Video 1.

Cell shape complements gene expression

A study combined detailed cell imaging with gene expression measurements during differentiation of human induced pluripotent stem cells toward cortical neurons. Cell Painting uses several fluorescent stains to describe cellular compartments and morphology. Single cell RNA sequencing measures molecular programs, allowing the researchers to ask how physical appearance and transcriptional state correspond during development.

The methods captured overlapping but different information. Imaging quantified features associated with processes such as cell cycling and mitochondrial organization, while RNA profiles distinguished finer molecular identities. Some cultured cells resembled fetal neural states, while others occupied states more specific to the culture system, demonstrating that a developmental model can contain both biologically relevant and preparation dependent populations.

The investigators also applied the imaging approach to progenitors from people with Kabuki syndrome, identifying patterns consistent with altered cycling and premature differentiation. Combining measurements makes those interpretations more specific than judging a cell by shape alone. A short process, a changed nucleus, and a transcriptional marker can each describe a different part of the same developing cellular state.



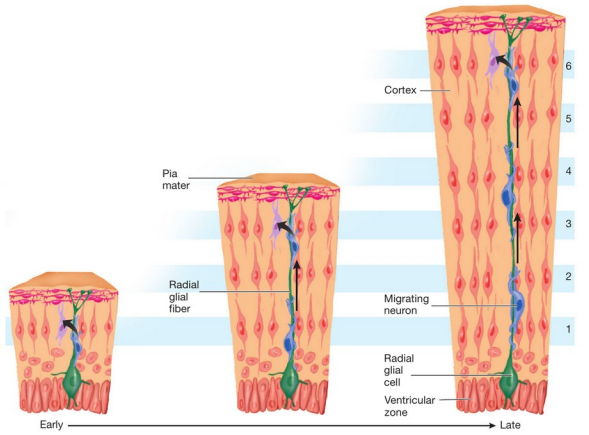
Sundaresh et al. (2025), Fig. 1.

Mechanistic explanations connect changes across scales

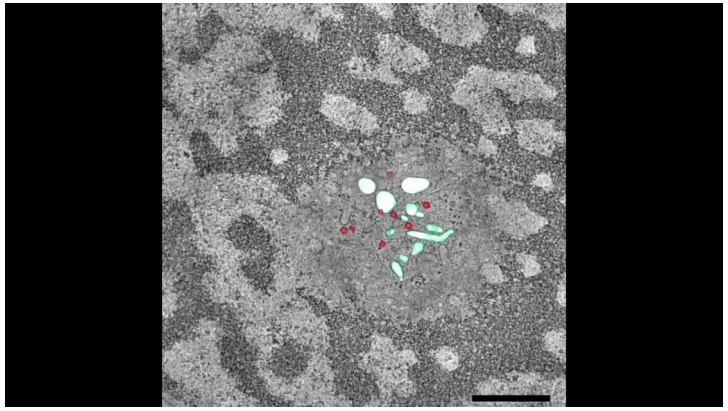
A developmental explanation can begin with a molecular change, but it must specify the cellular action that follows. A transporter changes molecular access, an adhesion system transmits force, and a receptor changes responses to a signal. Those actions influence division, migration, survival, or synapse formation before their consequences appear in larger scale anatomy and behavior.

The sequence is not always linear because developing cells influence one another. A changed glial scaffold alters the route available to a neuron, and altered neuronal activity changes the material that glia remove. These reciprocal interactions help explain why a phenotype can depend on the surrounding tissue, developmental time, and environmental conditions rather than only the identity of one altered molecule.

Experiments become informative when they identify both the preparation and the measured transition. Rescuing migration in a human organoid, changing axonal pruning in a tadpole, and improving a task in a rat each establish different outcomes. Translation requires determining which of those mechanisms also operates in the relevant human developmental context and whether changing it improves a meaningful function.



Carlson and Birkett (2022), Fig. 15.3, p. 495.



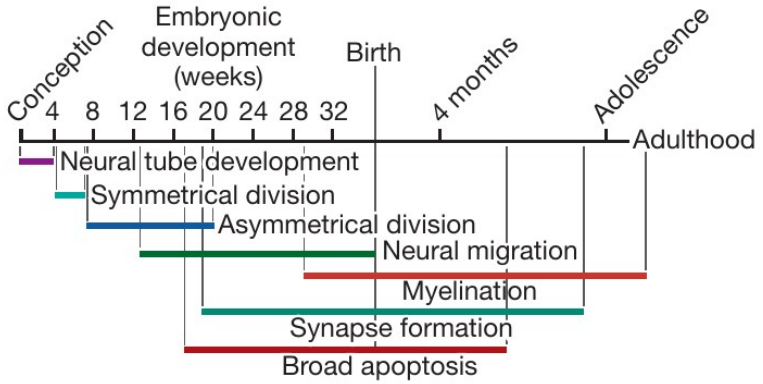
Jewett et al. (2023), fig4-Video 1.

Shared biology produces diverse trajectories

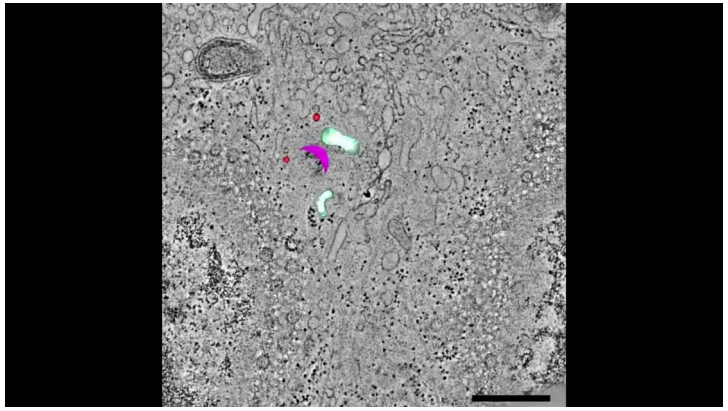
Brain development combines common cellular machinery with variation in timing, genetic background, experience, and health. Neurulation, progenitor expansion, migration, axon guidance, synapse formation, and refinement provide a shared framework. Differences in these processes can alter circuit organization without yielding one inevitable behavioral outcome or erasing the person's other abilities and sources of adaptation.

Developmental conditions must be described at the appropriate level. A chromosome arrangement, a cultured precursor phenotype, a pattern of attention, and a clinical diagnosis are different kinds of evidence. Connecting them requires defined intermediate mechanisms and direct measurement rather than assuming that a shared diagnostic word establishes the same biological change in every individual.

The most useful account also identifies what can change. Early detection can prevent some metabolic injuries, accessible experience can support learning, and targeted interventions can address specific cellular or functional problems. Persistent differences and continued plasticity can coexist, so assessing development means following actual abilities, needs, and responses over time rather than predicting an entire life from one early observation.



Carlson and Birkett (2022), Fig. 15.5, p. 497.



Jewett et al. (2023), fig5-Video 1.

Key takeaways

Development coordinates distinct cellular processes. Neurulation, progenitor division, migration, guidance, synapse formation, myelination, and refinement overlap but require different cellular machinery.

Signals must change a concrete cellular action. Adhesion transmits force, transporters control molecular access, and receptors regulate intracellular pathways; those actions explain how developmental changes occur.

Glia actively build and refine circuits. Radial glia guide migration, oligodendrocytes myelinate axons, and microglia and other glia can support synapses or remove selected cellular material.

Developmental diagnoses contain diverse mechanisms. Trisomy 21, metabolic disorders, autism, and ADHD involve different and sometimes overlapping pathways; a diagnosis does not identify one uniform cellular phenotype.

Models and human findings answer different questions. Specify the species, preparation, mutation, manipulation, and measured outcome; a cellular rescue or animal task improvement is distinct from established clinical benefit.

Development remains responsive while differences persist. Critical periods, continued learning, metabolic management, accessible experience, and targeted supports influence trajectories without making every outcome inevitable.